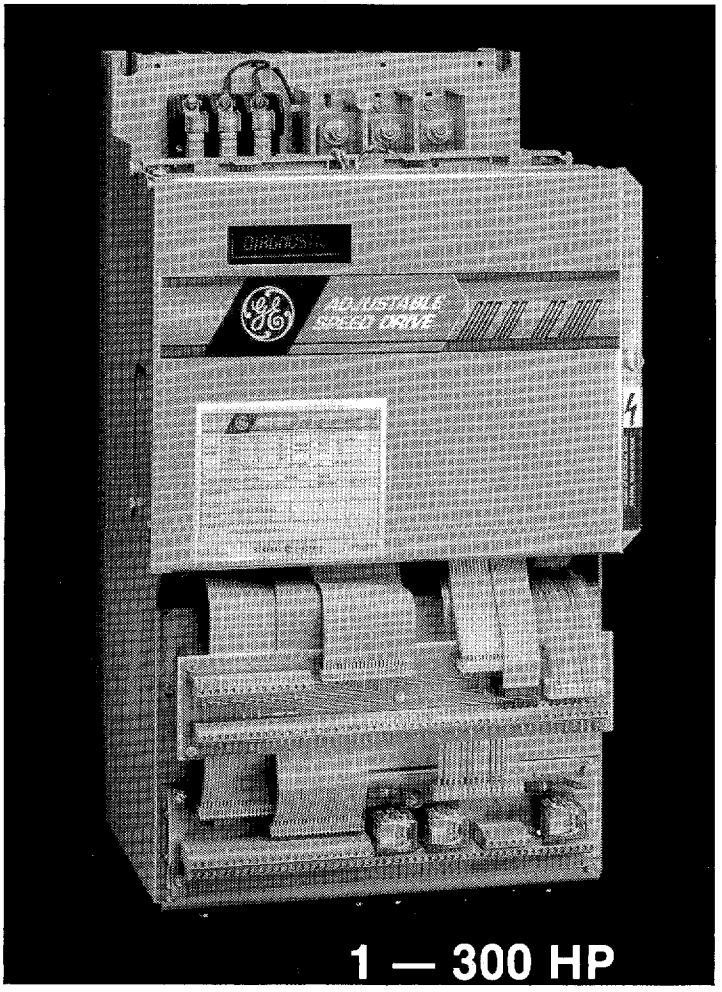
GE Drives

INSTRUCTIONS



AC-300™ Adjustable Speed Drives

TM TRADEMARK OF GENERAL ELECTRIC COMPANY, U.S.A

GE Drive Systems

INSTRUCTIONS

AC-300™ DRIVES MATCHED TRANSISTOR PAIRS ("T" FRAME) AND CONTROL POWER TRANSFORMERS

As of approximately 8/91, AC-300 Drives ("T" Frame Size) of 100 Horsepower and above (230 VAC) and 200 Horsepower and above (460 VAC) changed to the use of **matched pair transistors**. This was done to allow equal load sharing between the transistors. When shipped, these transistors will be packed together in one box that is labeled:

104X170CB032
MATCHED PAIR
DO NOT OPEN UNTIL ASSEMBLY

Each transistor within the box will be labeled with a date code and the following identification:

1/2 of PART #
104X170CB032
MATCHED PAIR

A "**CAUTION**" statement sheet will also be contained in the box and the instructions of it must be adhered to (see below).

CAUTION

THE TRANSISTOR MODULES IN THIS BOX ARE A SET OF MATCHED PARTS. UNDER NO CONDITION SHOULD THESE PARTS BE SEPARATED OR MIXED WITH OTHER PARTS; EVEN OF THE SAME TYPE.

REPLACE BOTH TRANSISTORS EVEN IF ONLY ONE IS SUSPECTED DAMAGED.

STOCK PARTS IN THIS CARTON AND DO NOT REMOVE UNTIL THEY ARE TO BE ASSEMBLED IN THE CONTROLLER.

Refer to the Figures and procedures of this Instruction Book Supplement to make the required change to the matched transistor pairs.

Refer to the following chart instead of the charts on pgs. 13-5 and 13-6 of GEK-85781C for the Catalog # of the new matched transistor pairs.

VAC	HP	Frame	Catalog #
230	75	Q	104X170CB-031*
230	100	T	104X170CB-032**
230	125	T	104X170CB-032**
230	150	T	104X170CB-032**
VAC	**HP**	**Frame**	**Catalog #**
460	150	Q	104X170CB-031*
460	200	T	104X170CB-032**
460	250	T	104X170CB-032**
460	300	T	104X170CB-032**

This is **not a matched pair Catalog #; however, it is a change from the previous Catalog # shown in GEK-85781C.*

***This Catalog # provides a **matched pair of transistors**. The transistors are taped together and packed/shipped as a pair.*

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These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

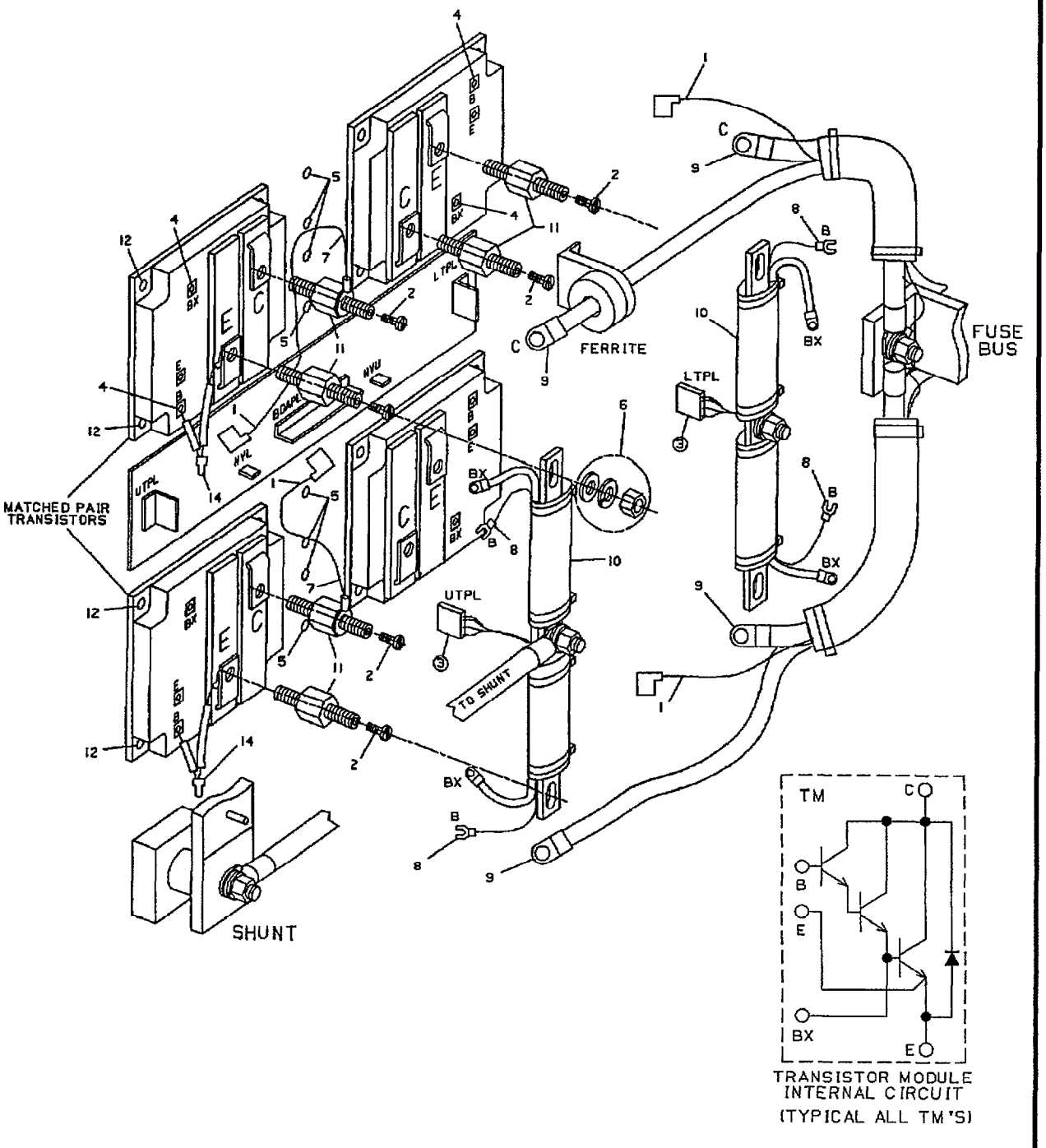


Figure S-1. TRANSISTOR MODULE REPLACEMENT

TRANSISTOR MODULE REPLACEMENT PROCEDURE (See Figures S-1 and S-2)

1. Remove red PI-T faston connectors from the Snubber Card (531X149SNCABG1).
2. Remove Snubber Card mounting screws (4 each) and remove both Snubber Cards.
3. Unplug UTPL and LTPL 4 position plugs from Base Driver Amplifier Card.
4. Loosen phillips head screws at "B" points on transistor modules and remove the screws completely at "BX" points on both modules (see Figure S-1).
5. Loosen all eight #10 screws that retain ferrite supports to allow removal of these supports (don't completely remove screws).
6. Remove all four 3/8"-16 nuts, 3/8" lock washers, and 3/8" flat washers from transistor module connection posts.
7. Remove the ring terminals (#14 black wire) from module connection posts that attach to the Base Driver Amplifier Card.
8. Snap loose the "B" spade terminals from both transistor modules.
9. Remove the power connections from the module connection posts at the "C" and "E" locations.
   -- This can be facilitated by sliding the ferrite supports from under their mounting screws and leaving the ferrites on the wires.
10. Remove bus bar assemblies from the module connection posts and position them out of the way to provide access to the transistor modules (no further disconnection or disassembly on busbar assemblies is required).
11. Loosen all four connection posts on the transistor modules (before loosening the mounting screws that secure the modules to the heatsinks).

CAUTION: Do not remove connection posts at this time or washers under the connection tongues on the modules may easily fall into the drive when posts are removed.

12. Remove the four #10 mounting screws on each transistor module and remove the modules.
13. Remove connection posts now and carefully set aside all hardware (see Figure S-2).
14. Remove and save snubber assemblies.
15. Clean the transistor module mounting surfaces of old lubricant and dirt.

CAUTION: The mounting screws are to be fastened in a certain order and tightness (see chart on following page). Also, care must be taken to achieve maximum contact area (i.e., minimum contact thermal resistance) for the best heat dissipation. Application of thermal compound on the contact surfaces improves its thermal conductivity. Prior to installing the new matched pair transistor modules, apply a thin film of General Electric GL322 Versilube Plus lubricant (Part No. 104X214AB059) to all contacting surfaces.

To replace the transistor modules, follow the preceding procedure in reverse order.

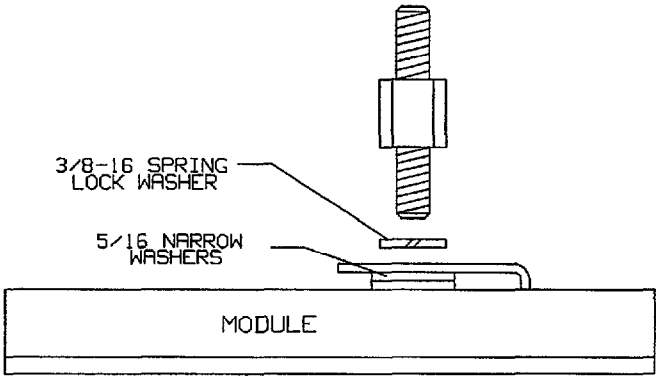
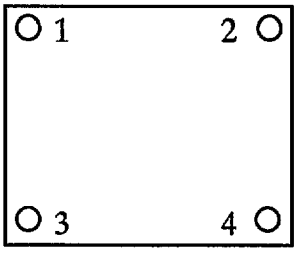


Figure S-2. TRANSISTOR MODULE CONNECTION POST

104X170CB032 Matched Pair Transistor Module Torque Values		
Item	**Torque (in/lbs) Min.**	**Max.**
Mounting Screws*	17.4	26.0
Terminal (B, BX, E)	9.0	12.0
Connection Post (E, C)	65.0	95.0

**Four point tightening method as follows (see sketch):*
Temporary Tightening Sequence: 1-2-3-4
Final Tightening Sequence: 4-3-2-1

CAUTION: When mounting transistor module on a heatsink, certain precations must be taken tp prevent any damage against a sudden torque. If a sudden torque is (one-sided tightening) is applied at only one mounting terminal, the ceramic insulation plate or silicon chip inside the module may be damaged.



CONTROL POWER TRANSFORMERS

The Control Power Transformers shown in the chart on page 13-2 of **GEK-85781C** have been changed. The previous Catalog Numbers of -

230 VAC 36B605538ARG01
460 VAC 36B605538ARG02

have been replaced by those shown in the chart to the right as of approximately 1/92.

CONTROL POWER TRANSFORMERS AS OF APPROXIMATELY 1/92		
Voltage (VAC)	**KVA Rating**	**Catalog Number**
230/460	Up To 60	36B605573BCG01
230/460	75-300	36B605573ACG01

NOTE: *Due to different connection methods between the older Control Power Transformers and the new ones, the new Control Power Transformers are* ***NOT*** *backward compatible (nor the older ones forward compatible). If it is necessary to replace the Control Power Transformer, use the following guidelines:*

- *If the drive was originally supplied with the older Control Power Transformer, replace it with the applicable one from the chart on pg. 13-2 of* ***GEK-85781C*** *(per voltage rating).*
- *If the drive was originally supplied with the new Control Power Transformer, replace it with the applicable one from the chart above (per KVA rating).*

General Electric Company
GE Drive Systems
1100 Lawrence Parkway
Erie, PA 16531

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1. SAFETY PRECAUTIONS

WARNING, CAUTION AND NOTE LABELS PLACED ON THE EQUIPMENT

The following format is used on the safety and informative labels placed on the equipment. Read all labels and follow the directions of them whenever working on the equipment.

WARNING: Denotes operating procedures and practices that may result in personal injury or loss of life if not correctly followed.
WARNING labels will be red in color with black or white lettering.

CAUTION: Denotes operating procedures and practices that, if not strictly observed, may result in damage to, or destruction of the equipment.
CAUTION labels will be amber in color with black lettering.

NOTE: *Notes call attention to information that is especially significant in understanding and operating the equipment.*
NOTE labels will be white in color with black lettering.

WARNING, CAUTION AND NOTE PARAGRAPHS WITHIN THIS INSTRUCTION

The following paragraphs list some general safety reminders and safety recommendations to be followed when installing or operating this equipment. These safety precautions will be repeated throughout this instruction book where applicable.

WARNING - MECHANICAL MOTION HAZARD: Motor drives and control systems cause mechanical motion. It is the responsibility of the user to insure that any such motion does not result in an unsafe condition. Factory provided interlocks and operating limits should not be bypassed or modified.

WARNING - ELECTRICAL SHOCK HAZARD: Electrical shock can cause personal injury or loss of life. Circuit breakers, if supplied as part of the total system, may not disconnect all power to the equipment (see system elementary diagrams). Whether the AC voltage is grounded or not, high voltage to ground will be present at many points.

WARNING - ELECTRICAL SHOCK AND BURN HAZARD: When using instruments such as oscilloscopes to work on live equipment, the oscilloscope's chassis should be grounded and a differential amplifier input should be used. Care should be used in the selection of probes and leads and in the adjustment of the oscilloscope so that accurate readings may be made. See oscilloscope manufacturer's instruction book for proper operation and adjustments to the instrument.

WARNING - ELECTRICAL SHOCK HAZARD: Some controllers are furnished with partial enclosures open at the top and bottom. These are intended only for mounting within another enclosure or in a control room having access by qualified personnel only.

WARNING - STRAIN HAZARD: Improper lifting practices can cause serious or fatal injury. Lift only with adequate equipment and trained personnel.

WARNING - FIRE AND EXPLOSION HAZARD: Fires or explosions might result from mounting drive controllers in hazardous areas such as locations where flammable or combustible vapors or dusts are present. Drive controllers should be installed away from hazardous areas, even if used with motors suitable for use in these locations.

WARNING - ELECTRICAL SHOCK HAZARD: All motor bases and equipment enclosure housings should be grounded in accordance with the National Electric Code or equivalent.

CAUTION: Do not remove connections or printed circuit cards from the controller while power is applied. This can damage the equipment.

CAUTION: Do not remove input power from the drive until it has fully executed a stop sequence as this can damage the drive system.

CAUTION: Installlation wiring must be in accordance with the National Electrical Code and be consistent with all local codes. Secondaries of the 115 volt control transformers typically have one side fused and the other grounded or available for grounding by the user.

CAUTION: The elementary diagrams supplied with the drive controller identify some signal connections as "low level analog and/or digital signals" and indicate special wiring practices. These signals should be isolated from all other wiring (particularly any power wires) and run in a separate conduit or wire run to prevent possible misoperation of the controller.

CAUTION: Meggering can damage electronic components. Do not megger or Hi-Pot the drive system without consulting GE DRIVE PRODUCTS OPERATION, ERIE, PA.

CAUTION: Do not connect any external circuits other than those shown on the elementary diagram. Connection of devices such as ammeters on the shunt or voltmeters on the tachometer may degrade the performance of the drive system.

CAUTION: Do not use power factor correction capacitors with this equipment without consulting GE DRIVE PRODUCTS OPERATION, ERIE, PA. Damage may result from high voltages generated by electrical resonance and when capacitors are switched.

CAUTION: The DB resistor kit is duty cycle rated per NEMA specifications and not intended for continuous operation. A disconnect device should be provided and interlocked to the drive controller to deenergize the drive or disconnect the DB resistor in the event that a continuously high AC input line or DB Transistor failure should cause continuous operation. An optional AC input starter or optional DB contactor (DBC) may serve this function.

NOTE: *Always read the complete instructions prior to applying power or troubleshooting the equipment and follow the start-up procedures step by step.*

NOTE: *The motor chassis should be grounded to earth through a separate ground lead from all other equipment ground leads to prevent noise coupling.*

NOTE: *Read and heed all* ***WARNING, CAUTION,*** *and* ***NOTE*** *labels posted on the equipment.*

2. RECEIVING AND STORAGE

RECEIVING

The controller should be placed under adequate cover immediately upon receipt as packing cases are not suitable for out-door or unprotected storage. Each shipment should be carefully examined upon arrival and checked with the packing list. Any shortage or damage should be reported promptly to the carrier. If required, assistance may be requested from **GE DRIVE PRODUCTS OPERATION, ERIE, PA.** When seeking assistance please use **serial number, requisition number, model number,** and **drive code** to identify the equipment **(Telephone 814-875-3851).**

STORAGE

If this controller is not to be installed immediately, it should be stored in a clean, dry location at ambient temperatures from -20°C (-4°F) to 55°C (131°F). The surrounding air must be free of chemical and electrically conductive or corrosive contaminants.

Precautions should be taken to prevent condensation from forming within the equipment enclosure. If the storage enviroment exceeds a 15°C (27°F) drop in temperature at 50% humidity over a four hour period, a space heater should be installed inside each enclosure to prevent condensation. (A 100 watt lamp can sometimes serve as a substitute source of heat). Higher humidities with smaller temperature changes can also cause condensation.

NOTES:

3. INSTALLATION GUIDELINES

LOCATION

AC-300™ drive controllers are suitable for most areas where industrial equipment is installed. They should be installed in well ventilated areas with ambient temperatures ranging from 0°C to 40°C (32°F to 104°F) and relative humidities up to 90%. It should be recognized, however, that since the life expectancy of any electronic component decreases with increased ambient temperature, reduction of the ambient temperature will bring about extended component life. For example, longer component life should be expected if the ambient temperature is held between 20°C (68°F) and 30°C (87°F). Proper performance and normal operational life can be expected by maintaining a proper environment for the drive system.

WIRING LEVELS

1. Wiring should be run in separate conduits or wireways for signal, control and power wiring levels.
   a. Signal
      -- Low level analog and digital signals
      -- Speed and/or position signals
      -- Power supplies and logic signals
      (See system schematics and notes to identify signal level wires)
   b. Control
      -- AC or DC control circuits
   c. Power
      -- 3-Phase input/output leads
      -- Brakes, 115 VAC circuits
      -- Internal AC and DC circuits

2. Signal wiring and power wiring may cross at right angles with a **minimum** one-inch separation. Avoid parallel runs between signal level wires and power or control level wires. If signal wires must be run in parallel to control or power wires, a **minimum** of a four-inch separation must be maintained between the wires.

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GROUNDING

1. Drive common (PCOMX) should be grounded at **only one** point.
   -- If the reference is supplied by numerical control or process instrument with a grounded common, the drive common should not be grounded separately.
   -- If an external grounded signal is connected to PCOMX, PCOMX should not be grounded.

NOTE: *The Process Follower input is differential; it may be grounded independent of PCOMX.*

2. If an isolation transformer is used, it is recommended that a high resistance, neutral ground be used.

3. Shields for shielded wire must be grounded on one end only. Provision is made to tie shields to chassis ground at the drive I/O. It is preferred to have the shield grounded only at the drive end.

4. Proper case grounding that conforms to UL and CSA standards should utilize wire sizes and crimp-type terminals per the following chart:

Circuit Rating (Amps, Max.)	Copper Ground Wire AWG	AMP Terminal Number	AMP Tool Number
20	14	34120	49900
60	10	33457	49900
100	8	33458	69062
200	6	33465	69062
300	4	31811	69062
500	2	320383	69062
800	1/0	36917	69120
1000	2/0	36923	69120
1200	3/0	36925	69120
1600	4/0	321878	69120

COMMONS

Connection of reference, meter, or any other external commons can only be made at 3TB-PCOMX (unless otherwise indicated by the system elementaries).

SUPPRESSION

Control system relays, solenoids, or brake coils can produce erratic drive behavior due to electrical noise transients. To eliminate this possibility, an RC suppressor should be added in parallel with the coils of these devices. A 220 ohm, 2 watt resistor in series with 0.5 mfd, 600 volt capacitor can be used.

TIGHTEN ALL CONNECTIONS

-- Incoming AC line connections

-- Power connections to motor

-- 1TB, 2TB, 3TB, 4TB, and MATB if furnished

ADDITIONAL CONSIDERATIONS

Attention should be given to the National Electrical Code and any applicable local codes when installing any drive. Wire size and insulation type, conduit sizing, enclosures, grounding, etc., should be in compliance with these codes.

Environments which include excessive amounts of one or more of the following characteristics should be considered hostile to drive performance and life:

1. Dust, dirt, or other foreign matter.
2. Vibration or shock.
3. Moisture or vapors.
4. Rapid temperature excursions or high ambient temperatures.
5. Caustic fumes.
6. Power line fluctuations.
7. Electromagnetic interference or **"noise"** introduced by:
   a. Radio frequency signals (typically from portable transmitters used in the vicinity of the equipment or its wiring).
   b. Stray high voltage or high frequency signals as might be provided by arc welders, or by the operation of unsuppressed relay, contactor, or brake coils as a part of, or in the vicinity of, drive control circuits.

MOUNTING

-- Totally enclosed controllers should be positioned to permit heat radiation from all surfaces.

-- A wall-mounted enclosure may be placed side by side with another enclosure. Clearance at least to the width of the enclosure doors should be available in front so that the doors may be fully opened for easy access.

-- Wall-mounted controllers may be mounted on any firm, reasonably flat, vertical surface.

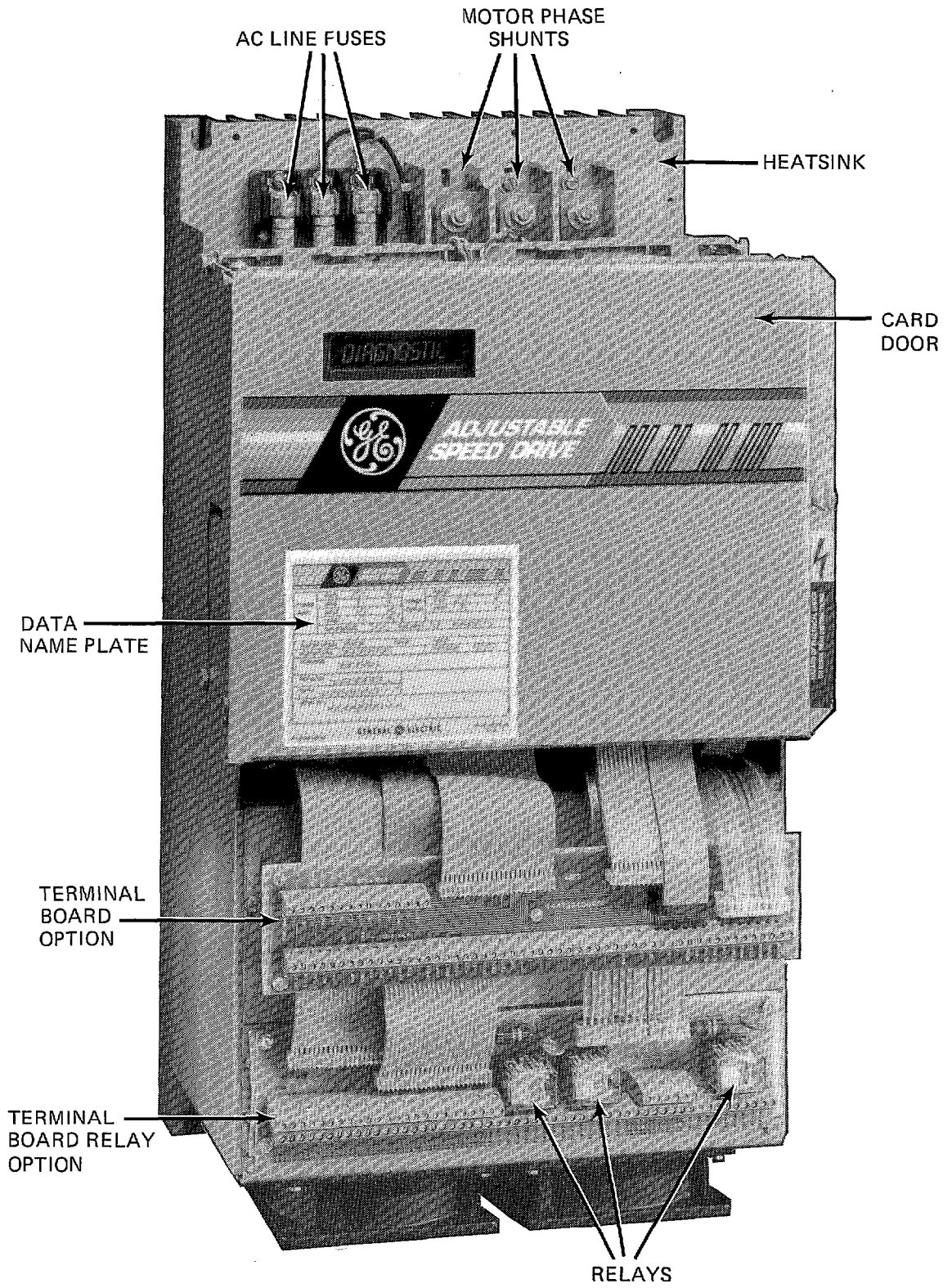


Figure 3-1. AC-300 Controller, Wall-Mounted Open Chassis Construction

NOTES:

4. MAIN CONTROLLER ELEMENTS AND THEIR FUNCTIONS

GENERAL DESCRIPTION

(See Figures 4-1 thru 4-11)

The AC-300 Adjustable Speed Drive provides for control of three phase induction motors. It combines a full wave diode bridge rectifier, pulse width modulated transistor inverter, and microprocessor regulation in a programmable AC drive controller. Drive parameters are stored in an electrically erasable read only memory (EEROM) for drift-free accuracy

There are three versions (or frame sizes) that make up the 1-300 HP range of the AC-300 controllers covered in this Instruction Book. The **P frame** is rated up to 60 HP, the **Q frame** is rated from 75-125 HP, and the **T frame** is rated from 150-300 HP.

This Instruction Book covers these three frame sizes of the AC-300 drive, as they have hardware and software commonality. Attention is drawn to any variations between the frame sizes where necessary.

The AC-300 drive controller provides over 80 diagnostic messages which can be decoded via a LED diagnostic readout or an optional Programmer module with alpha-numeric readout. Diagnostic messages can also be read via an optional coded 8-wire output or over the optional RS422/232C communication link. Drive operating parameters stored in the EEROM may also be examined and changed via the RS422/232C link. In addition, the AC-300 drive controller provides diagnostics which can check the transistors and also provides a power-up self-check down to the printed circuit board level.

The AC-300 drive controller requires only **three** printed circuit cards (Control card, Power Supply card and Base Driver card) for operation and offers optional cards and terminal boards for additional functions. **Optional** cards include the Application card, Process Interface card, Programmer card, and Terminal Board cards (3TB and 4TB). The Programmer can be purchased in a Local or Hand-held version that plugs into the Control card for inspection and adjustment of parameters when tuning up the drive, as well as diagnostic functions.

Three-phase power is input through AC line fuses to a full wave diode bridge rectifier. The rectified DC output is filtered by capacitors. A charge control circuit between the rectifier bridge and filter capacitors limits charge current inrush at power-up. Power transistors, driven in a Pulse Width Modulated (PWM) switching mode, invert the filtered DC Link voltage to provide controlled, three phase AC current to the motor. The resulting current output is a sine wave having a superimposed chopping frequency ripple.

Throughout most of its operating range the AC-300 drive controller uses a field-angle control analogous to the field and armature control used with DC motors. Separate, interactive, regulator loops are provided for speed, torque, flux, and net motor current. Voltage and current signals for all three phases are fedback, digitized, and integrated to resolve flux-producing and torque-producing current vectors. The torque-producing current vector, analogous to the armature current of a DC motor, is regulated in response to the output of the speed regulator. Phase and magnitude of the rotor flux vector are regulated according to the Volts/Hertz rating of the motor. Analogous to the field current of a DC motor, the flux vector may be reduced to achieve the equivalent of "field weakening" when operating above base speed. Outputs of the flux and torque regulators are compensated for stator leakage inductance, then are input to the current regulator and determine the net current vector necessary to develop required torque at commanded speed. In response to the current regulator output, the base drive to the inverter transistors is modulated to produce the three phase AC output necessary to develop the net current vector.

The frequency range of the controller, and therefore speed range of the motor, is dependent on the type of speed feedback used and the characteristics of the motor. Where no external feedback is provided, the speed of the motor is measured as the frequency of the inverter, compensated for slip (CEMF). As an option, provisions may be made to utilize analog or digital tachometers for speed feedback.

AC POWER REQUIREMENTS

The AC-300 drive is available in either 230 VAC +10%, -5% (1 - 30 HP) or 460 VAC +10%, -5% (2 - 300 HP), three-phase power configurations. Nominal line frequency is 50/60 Hz with a tolerance of ±2 Hz.

AC LINE FUSES

AC line fuses are provided on the three incoming power lines to the controller and provide short circuit protection for the power semiconductors and DC link.

REACTORS

The AC-300 controller may require series reactors with a series impedance of at least 2% of the base Power Unit impedance between the controller and its supply transformer to limit the short circuit current available at the controller. The rule governing the need for such reactance is as follows:

An optional three phase reactor is required if the transformer feeding the controller is larger than that shown in the following chart.

Controller HP	Voltage	Transformer (Max. KVA)
2-3 to 15	460	50
20 to 30	460	100
40 to 60	460	200
5 to 125	460	400
150 to 300	460	1000
1 to 7.5	230	25
10 to 15	230	50
20 to 30	230	100
40 to 60	230	200
75 to 150	230	500

The optional three phase reactor should be connected as shown in Figure 4-1.

DC LINK FUSE(S)

A DC link fuse is provided in the DC link circuit to protect against power transistor failure. 75-300 HP drives will have individual inverter phase protection (3 fuses).

CONTROL POWER TRANSFORMER (CPT)

This transformer is energized through its primary fuses (CFU4 and CFU5) on the Power Supply card. The secondary winding provides two isolated voltages:

1. 115 VAC to operate the RUN (K1) Relay and controller cooling fan if required as well as some external control devices.
2. 38 VAC center tapped for the Power Supply card to generate the DC control voltages necessary for controller operation.

RECTIFIER MODULES (DM1, DM2, and DM3)

Three dual-diode modules form the three phase, full wave bridge that rectifies the incoming AC supply voltage to the DC link bus. The modules have electrically isolated base plates and mount to the common heatsink extrusion of the controller.

FILTER CAPACITORS AND DISCHARGE RESISTOR (1C, 2C, AND R2)

A bank of capacitors filter the DC link voltage, providing a low impedance source for the fast-switching inverter transistors and supressing AC supply transients. The electrolytic capacitors are connected in a series-parallel configuration to achieve the proper combination of voltage rating and total capacitance. A single, center-tapped resistor (R2) provides for series voltage matching and capacitor discharge.

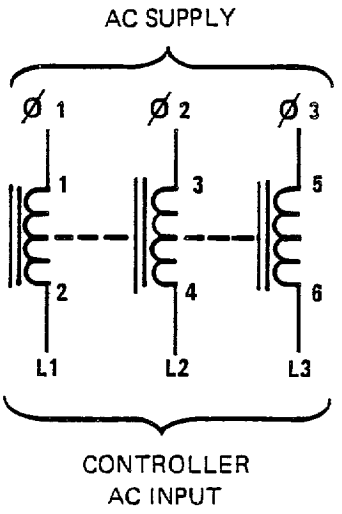


Figure 4-1. Three Phase Reactor Connection

CHARGE CONTROL SCR MODULE AND RESISTOR (SM1 AND R1; also SM2 on 250-300 HP)

At power-up, the charge control resistor (R1), in series with the rectifier output to the DC link, limits the charging current to the filter capacitors. After the capacitors are initially charged, SCR (SM1; also SM2 on 250-300 HP) is gated and latched ON to provide a low impedance shunt around the resistor. SCR conduction is maintained during controller operation and is interrupted only if a DC link undervoltage occurs. The firing signal to the SCR (SM1; also SM2 on 250-300 HP) is provided through the Power Supply card. The SCR module(s) has an electrically isolated base plate and mounts to the common heatsink extrusion of the controller.

INVERTER TRANSISTOR MODULES (TM1 through TM6)

The number of transistor modules varies with the current rating of the drive. Lower power drives utilize three modules (TM1, TM2, and TM3), each with two transistors and anti-parallel diodes. Higher power drives have six modules (TM1 through TM6) or twelve modules (TM1-1 through TM6-2), each with one transistor and diode. The fast-switching transistors are driven by the Base Driver card to provide regulated Pulse Width Modulated (PWM) control. The modules have electrically isolated base plates and mount to the common heatsink extrusion of the controller.

OPTIONAL DYNAMIC BRAKING TRANSISTOR MODULE AND RESISTOR (TM7 and R3)

When the DB Regeneration option is included, transistor module (TM7) and the externally mounted DB resistor kit (R3) are provided to dissipate the energy regenerated to the DC link by the motor. Other factors, such as a high AC line, causing DC link voltage "pump up" may also result in operation of the DB circuit. The DB transistor is driven from the Power Supply card in a pulse width modulated (PWM) mode with the ON and OFF times controlling the regenerative discharge current through the DB resistor. The transistor module has an electrically isolated base plate and mounts to the common heatsink extrusion of the controller. An optional protective DB contactor is used to open the circuit in case of a transistor failure to prevent failure of the DB resistors.

SHUNTS(SH)

The strip shunts in the output leads of the controller provide the current feedback signals to the control circuits.

CONNECTION DIAGRAMS

(See Figures 4-5 thru 4-11)

The connection diagrams for up to 60 HP, 75 - 125 HP and 150 - 300 HP controllers are shown in Figures 4-5 thru 4-11. These graphically describe the internal connections for all plugs and cables. These diagrams aid control hook-up as well as troubleshooting when connection integrity of one of the cables is in question. The functions of the connectors are described in the "Electrical Connector" section of this Instruction Book.

CONTROL CARD

(See Figure 4-12)

This card provides the intelligence for the AC-300 drive controller. It contains three microprocessors and performs most of the regulating functions, sequencing, firing, diagnostic, and protective functions for the drive along with many customer and motor interfacing functions. The functions performed on the card are implemented in hardware and/or software.

The control card (Figure 4-12) contains six Diagnostic LEDs to display the status of the AC-300 drive. The LEDs that are lit are interpreted as a binary number indicative of the fault. See the Troubleshooting section for a listing of Fault Codes. Note that during proper controller operation, the LEDs will blink sequentially right to left when the drive is stopped and sequentially right to left in pairs when the drive is running.

POWER SUPPLY CARD

(See Figure 4-13)

This power supply card provides the necessary +5 VDC, ±15 VDC, and ±24 VDC power used throughout the controller. It also contains the filter capacitor charge control and the optional dynamic braking control, that are at the potential of the negative (N2) DC Link bus. Three relays, "RUN" (K1), "MAX" (K2), and "FAULT" (K3), are also on this card. The latter relay provides a control interlock for operation of an externally supplied output contactor.

BASE DRIVE CARD

(See Figure 4-14)

This card contains the six base drive circuits for the inverter power transistors, that are at different power potentials, as well as the voltage feedback and current feedback isolation circuits. It also contains transistor short circuit detection and protection circuitry.

BASE DRIVE AMPLIFIER CARD (150 - 300 HP Only)

This card contains base drive amplifier circuits for the inverter power transistors. Each card serves a pair of power transistors.

SNUBBER CARDS

These small cards are mounted on the power module bus bars and provide transient switching voltage protection.

POWER CONNECTIONS

The three-phase input AC power connections are made at L1, L2, and L3 on the line side of the AC fuses. The motor connections are made at T1, T2, and T3 terminals at the top of the controller as shown in Figure 4-2.

CONTROL CONNECTIONS

The control connections to the basic controller may be made by ribbon cables to the following connectors on the specific printed circuit boards.

- -- 4PL on the Power Supply card (see Figure 4-13)
- -- 6PL on the Control card (see Figure 4-12)
- -- 10PL & 12PL in the optional Process Interface card (see Figure 4-16) or Encoder Process Interface Card (see Figure 4-18)

DC or AN tach connections are made to the screw terminals of 1TB (located on the Process Interface card).

If the optional Terminal Board card (3TB, see Figure 4-19) is mounted on the controller, all of the control connections can be made on this card through screw terminals identified by nomenclature and numbered points.

If the optional Application card (see Figure 4-15) is mounted in the drive, depending on functions purchased, it has three ribbon cable connectors for the control connections. The optional Terminal Board Relay card (4TB, see Figure 4-20) can be used to provide for screw terminal connections to the Application card and/or to its three relays (K6, K7, K8).

OPTIONAL APPLICATION CARD AND MICROAPPLICATION CARD

(See Figures 4-15 and 4-17)

This optional card provides a combination of analog interfaces between the Control card and external equipment. These interfaces allow the core drive to be expanded to handle optional functions required in applications such as machine tool spindles, material handling cranes, etc. **Two versions** of the Application card are available. The cards, shown in Figures 4-15 and 4-17, contain the following optional functions:

- -- Eight Analog Inputs
- -- Eight Digital Inputs
- -- Sixteen-Bit Parallel Word Input
- -- Five Programmable Signal-Level-Detector Relays (SLDs)
- -- Seven-Bit Digital Fault Status Output
- -- Three Programmable Status Bit Outputs
- -- Two Encoder Interfaces
- -- Two Analog Outputs
- -- Two General Purpose Amplifiers (Version APM card only.

The second type of Appplication card available is the **Microapplication card.** In addition to the above features, it also contains the AC-300 Local Area Network (LAN) feature for communication over a single pair of wires with up to 30 other drives, to a Series Six™ programmable controller. The Microapplication card and LAN are described in the Local Area Network Users Guide, GEK-85789.

™ *Series Six is a trademark of General Electric Company, U.S.A.*

OPTIONAL PROCESS INTERFACE CARD AND ENCODER PROCESS INTERFACE CARD

(See Figures 4-16 and 4-18)

These optional cards provide a mixture of analog and digital interfaces between the Control card and the equipment. The optional functions provided are:

- Analog (AC or DC) tachometer interface.
- Analog speed meter driver (analog tachs only).
- Digital (reluctance or square wave) tachome ter interface.
- Frequency (pulse train) reference interface.
- Process Follower (current or voltage refer ence) interface.
- Digital voltmeter probe.
- RS232C and RS422 serial link interface.

In addition to the above, the **Encoder Process Interface card** also provides for quadrature tachometer inputs.

OPTIONAL TERMINAL BOARD CARDS (3TB and 4TB)

(See Figures 4-19 and 4-20)

These optional cards provide convenient termination points for control interface to the drive. Connections to the Control, Power Supply, and optional Process Interface cards are made thru 3TB, that also contains an optional signal-level-detector (SLD5). Connections to the optional Application card and Microapplication card are made thru 4TB. In addition, three 115 VAC relays are available on 4TB.

OPTIONAL PROGRAMMER

The Programmer is available as either a "Local" Programmer, mounted on the Control card or as a "Hand-Held" Programmer, which is plugged into 18PL on either the Control card or Diagnostic Readout card. The Programmer contains a 10-digit fluorescent alphanumeric display and a keypad for communicating with the drive. Refer to "Use of Optional Programmer Module", Section 11, for further information. The Programmer provides for **three** modes of operation:

- **OPERATE Mode** provides all necessary functions to run the drive and displays the drive status including fault messages.
- **PARAMETER Mode** allows examination and changes, if necessary, of the system operating parameters.
- **DIAGNOSTIC Mode** allows the user to monitor drive values while in operation, self-test the controller, and access diagnostic RUN and STATIC modes. Diagnostic Mode also provides a number of drive tests, including optional Drive Self-Tune features.

SERVICE AND PARTS INSTRUCTION LABELS

Each drive contains, inside the door, a multicolor label. This label provides information on connection point and component locations essential in connecting or troubleshooting the drive. The label contains the following information in color-coded format:

- Card location
- Thyristor module locations
- Fuse information and nomenclature
- Plug locations
- Incoming control connections at 3TB and 4TB
- Power connections
- Factory "hot line" telephone number **(1-814-455-3219)**

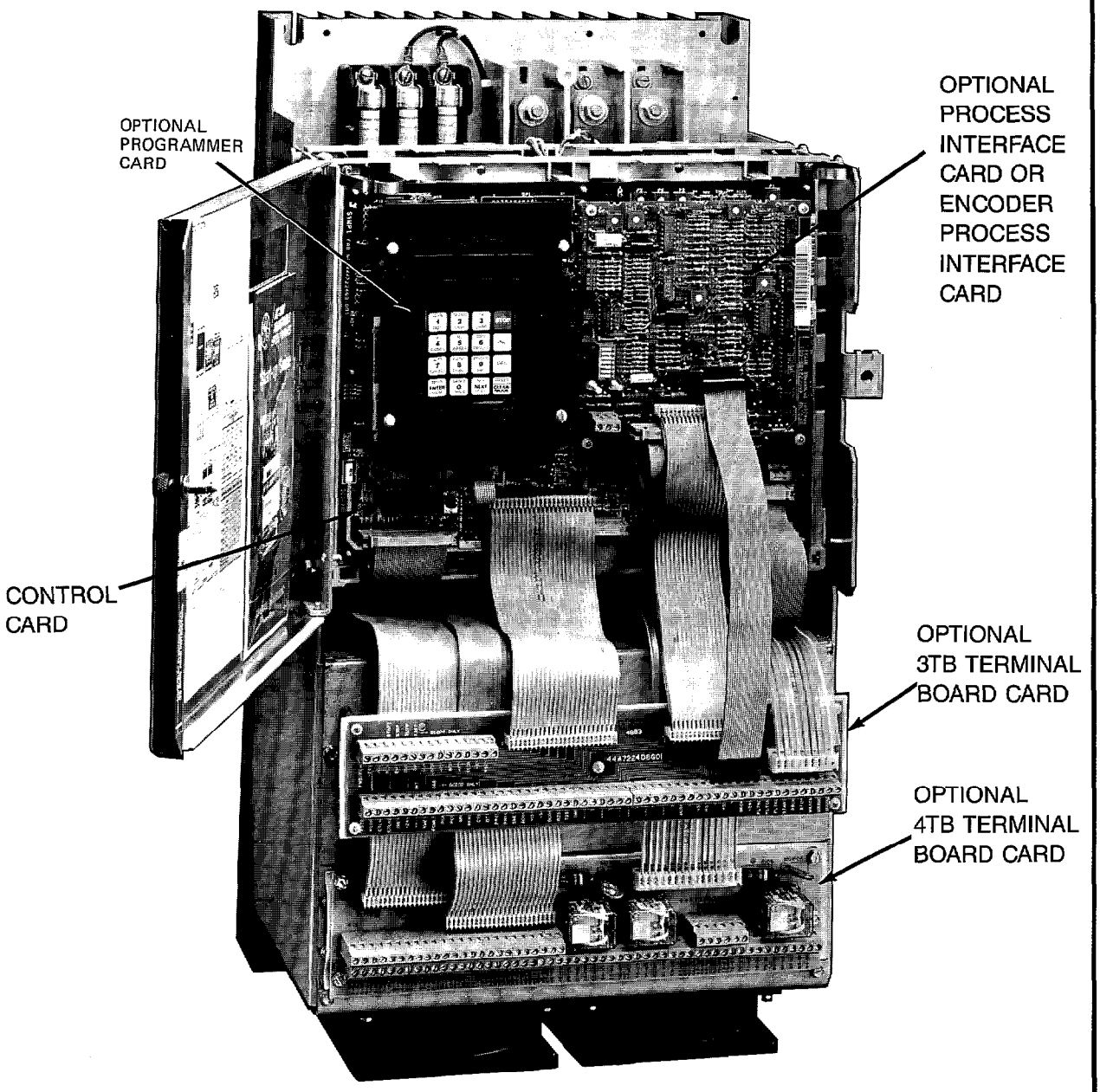


Figure 4-2. AC-300 Controller With Card Door Open

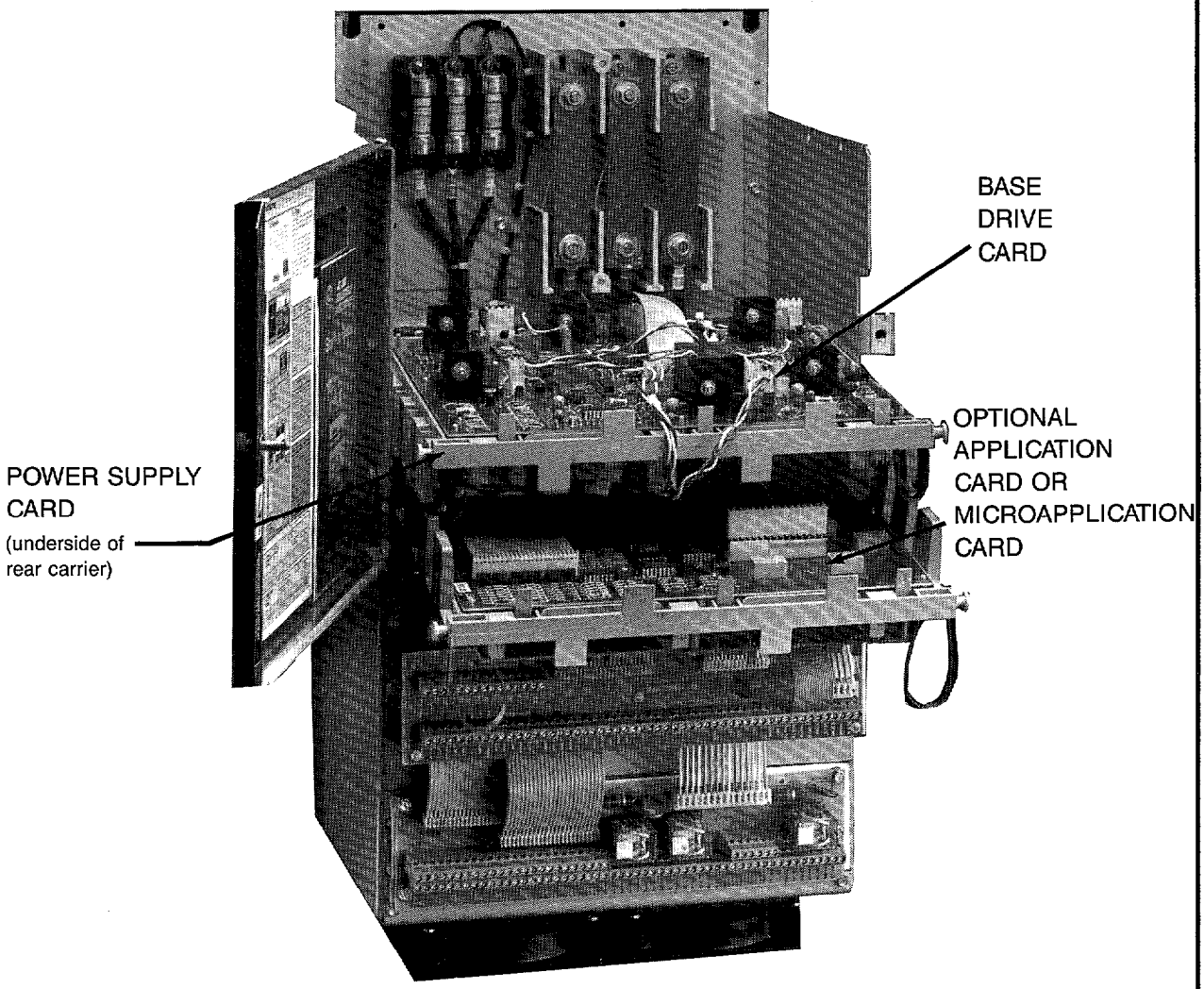


Figure 4-3. AC-300 Controller With Front and Rear Card Carriers lowered

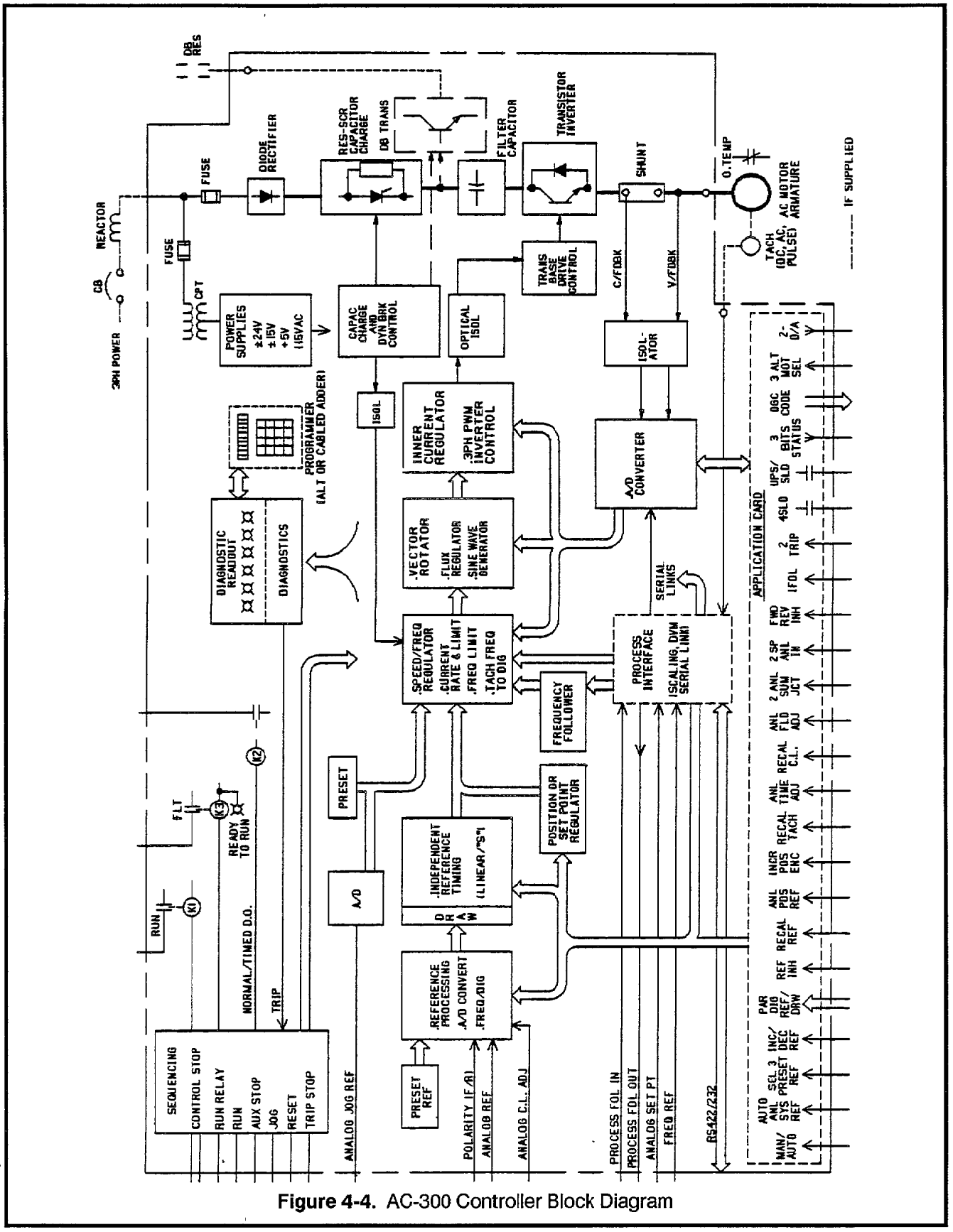


Figure 4-4. AC-300 Controller Block Diagram

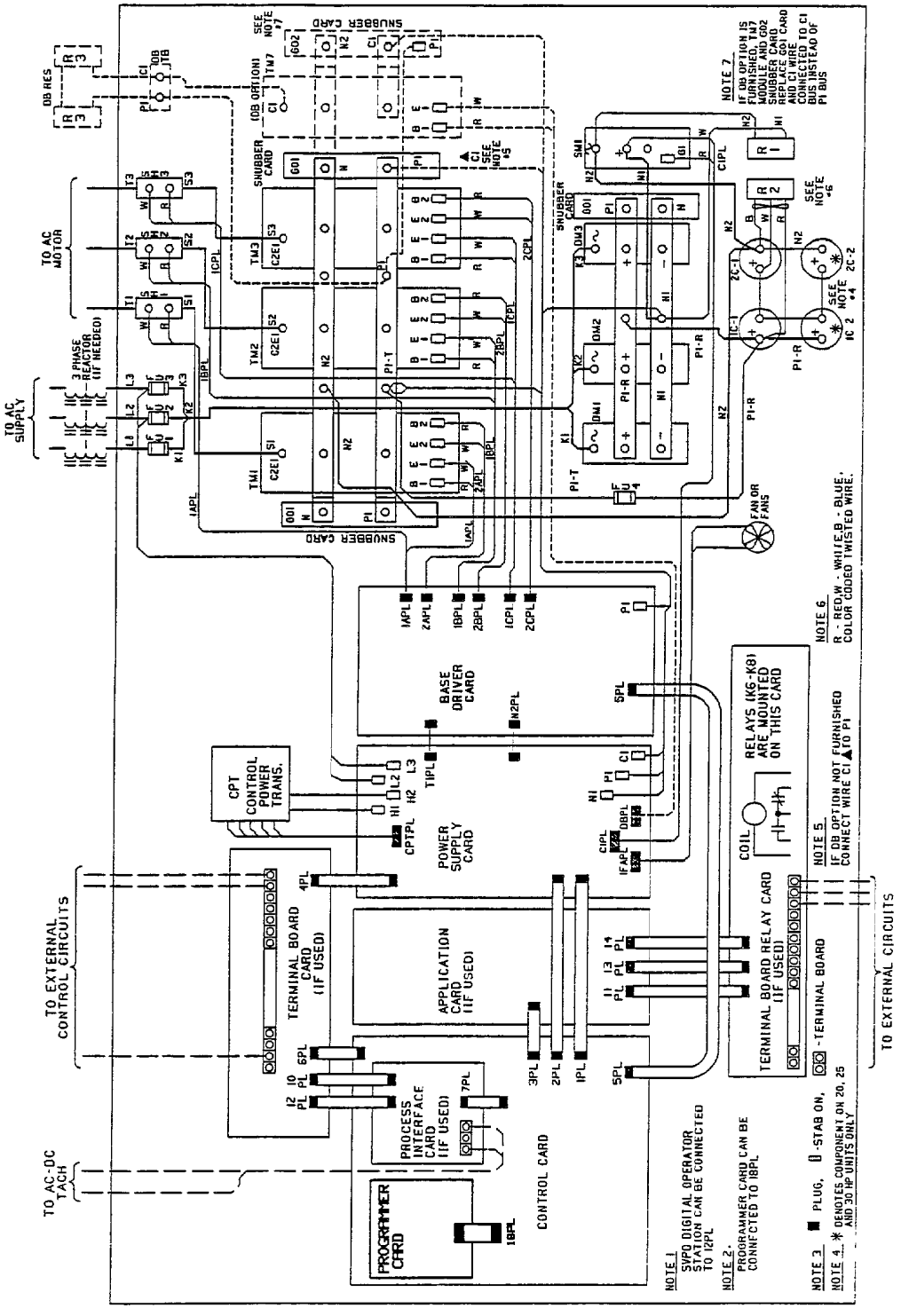


Figure 4-5. Connection Diagram (Up To 30 HP, 460V)

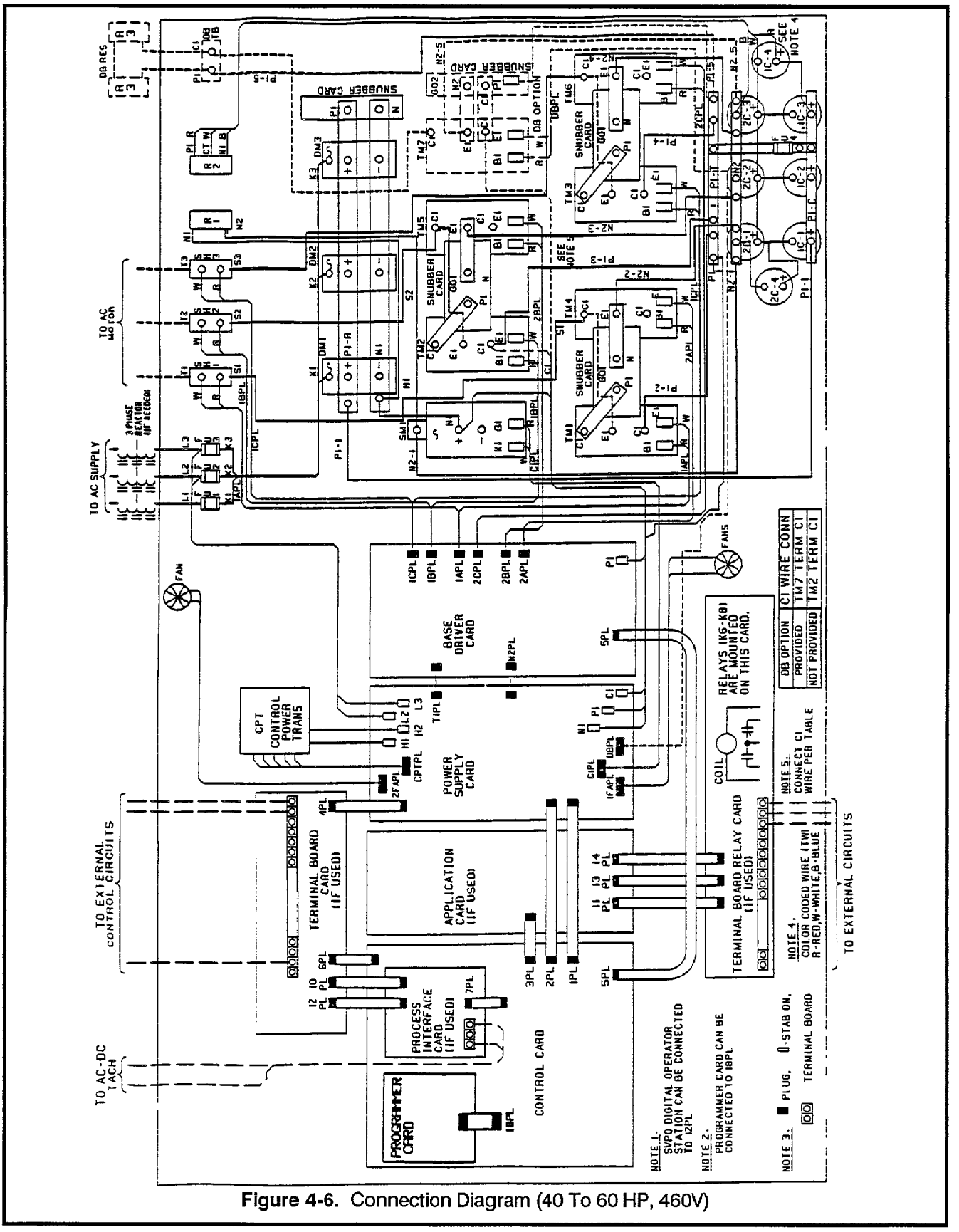


Figure 4-6. Connection Diagram (40 To 60 HP, 460V)

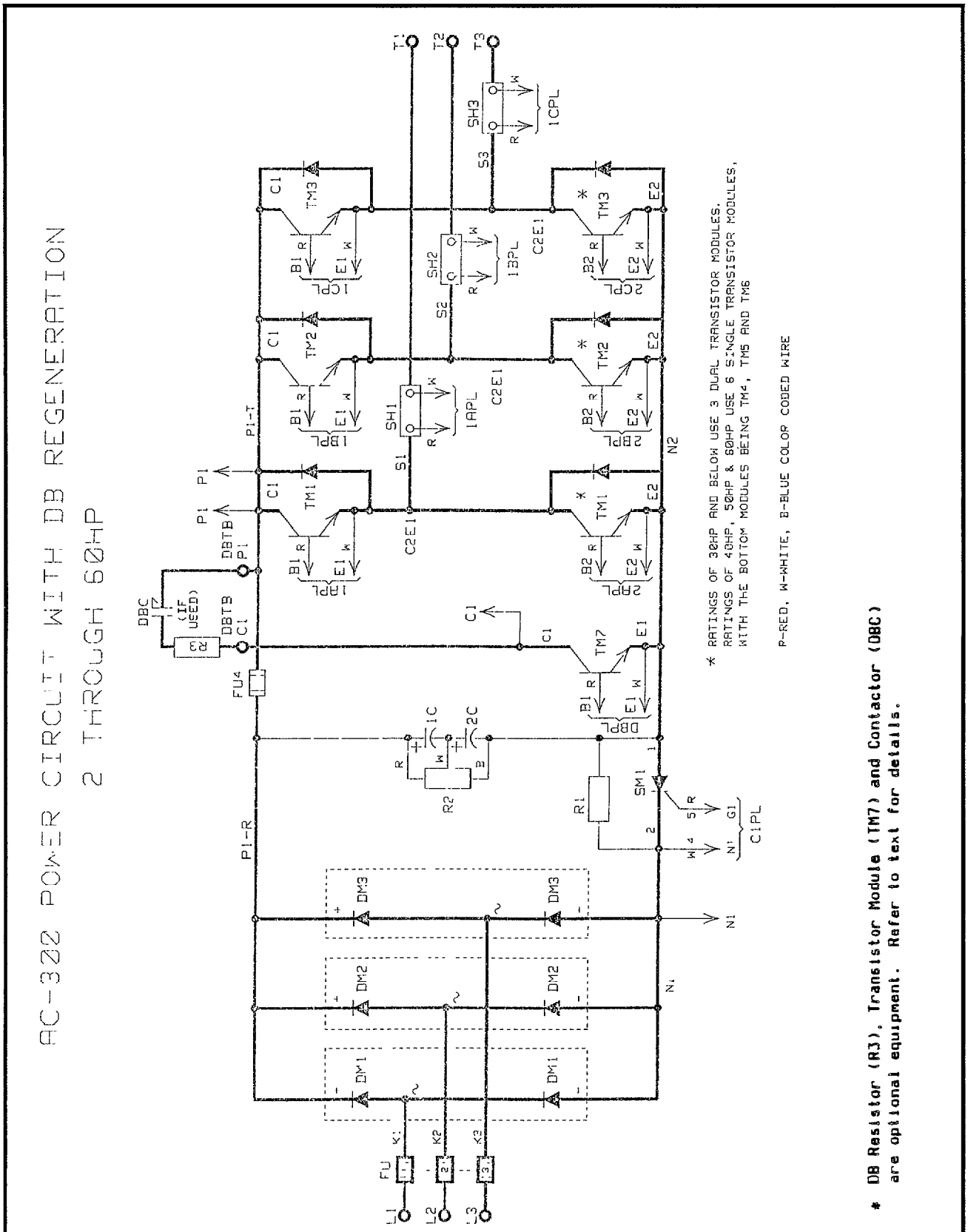


Figure 4-7. Power Circuit With DB Regeneration (2 To 60 HP, 460V)

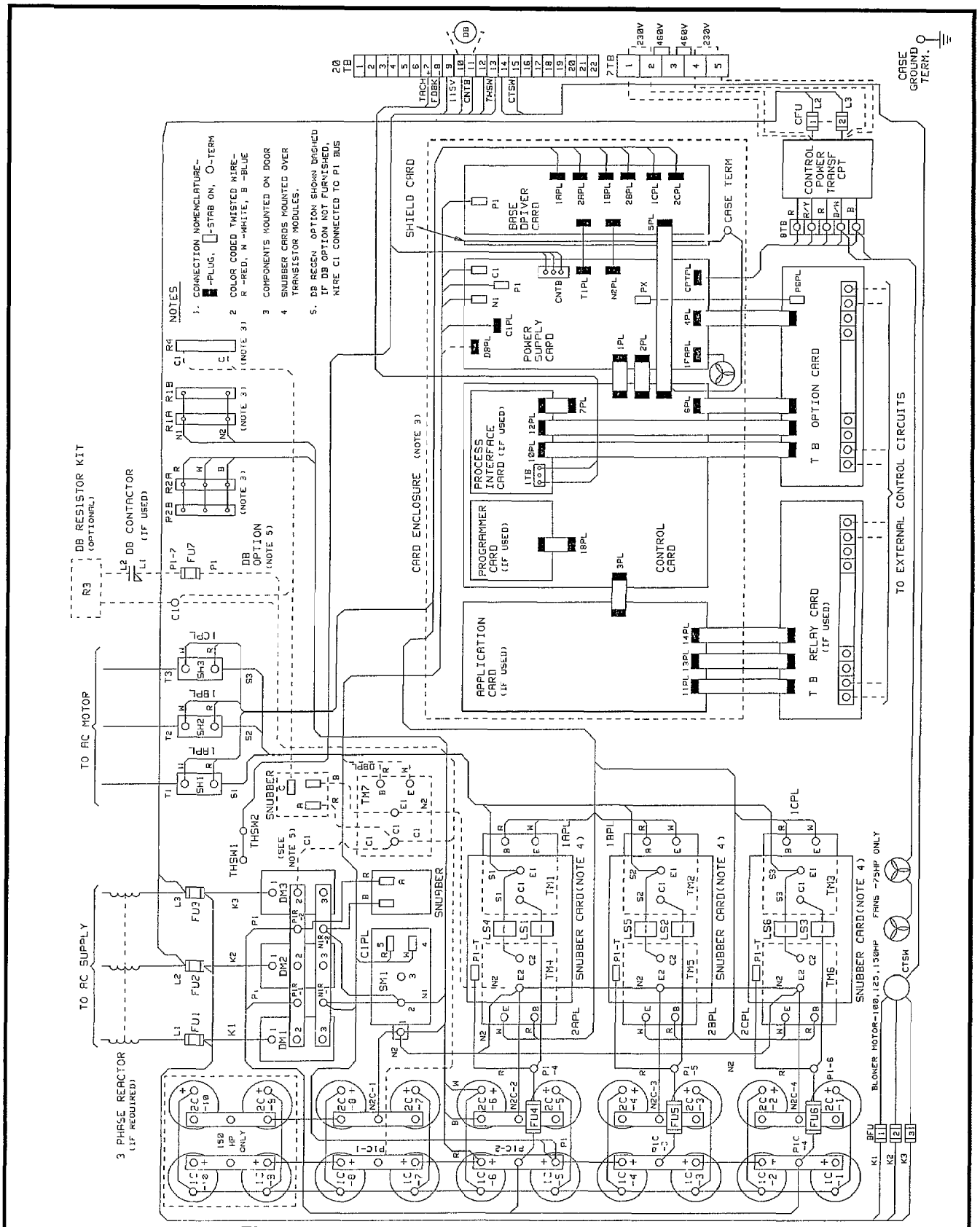


Figure 4-8. Connection Diagram (75 To 125 HP, 460V)

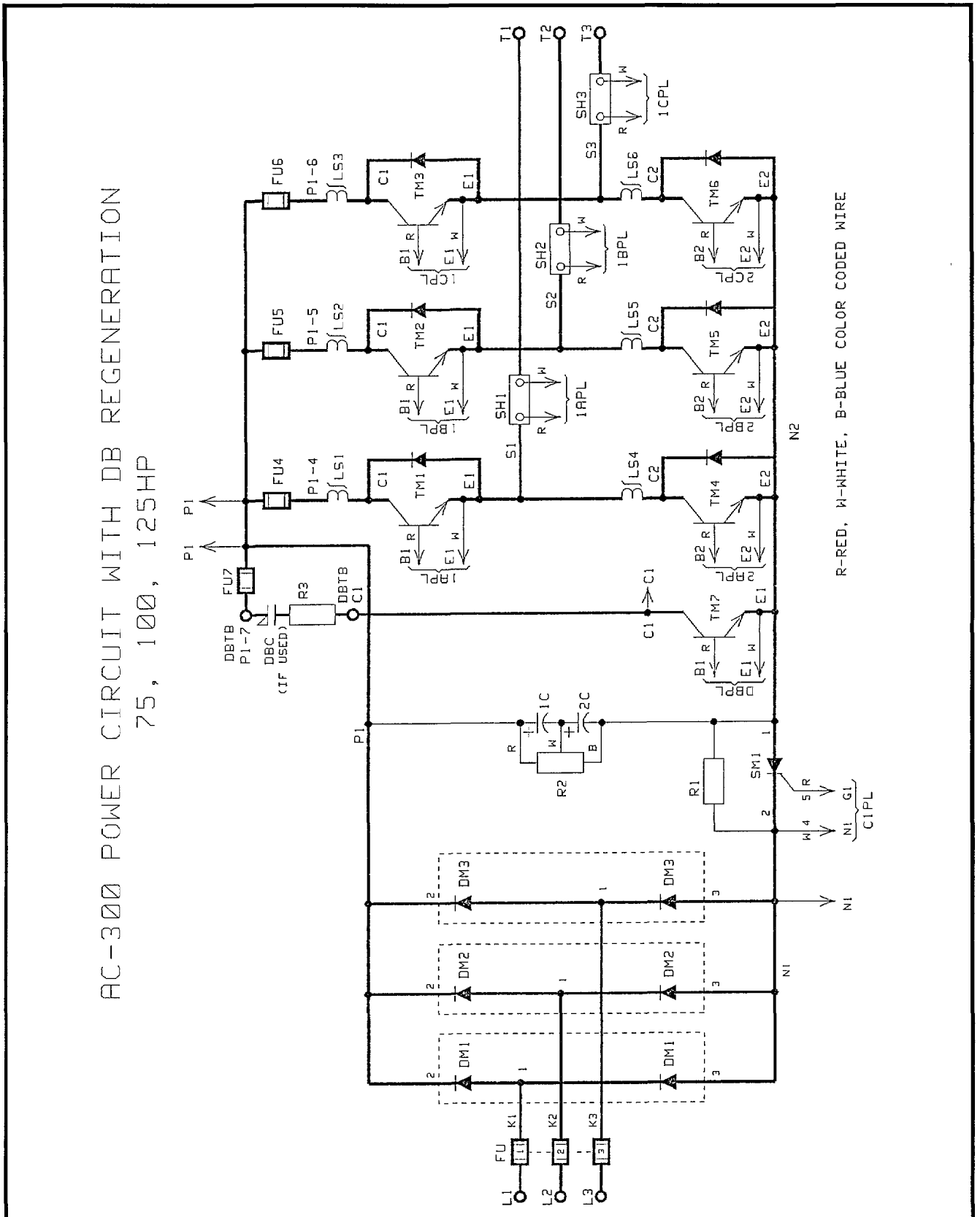


Figure 4-9. Power Circuit With DB Regeneration (75 To 125 HP, 460V)

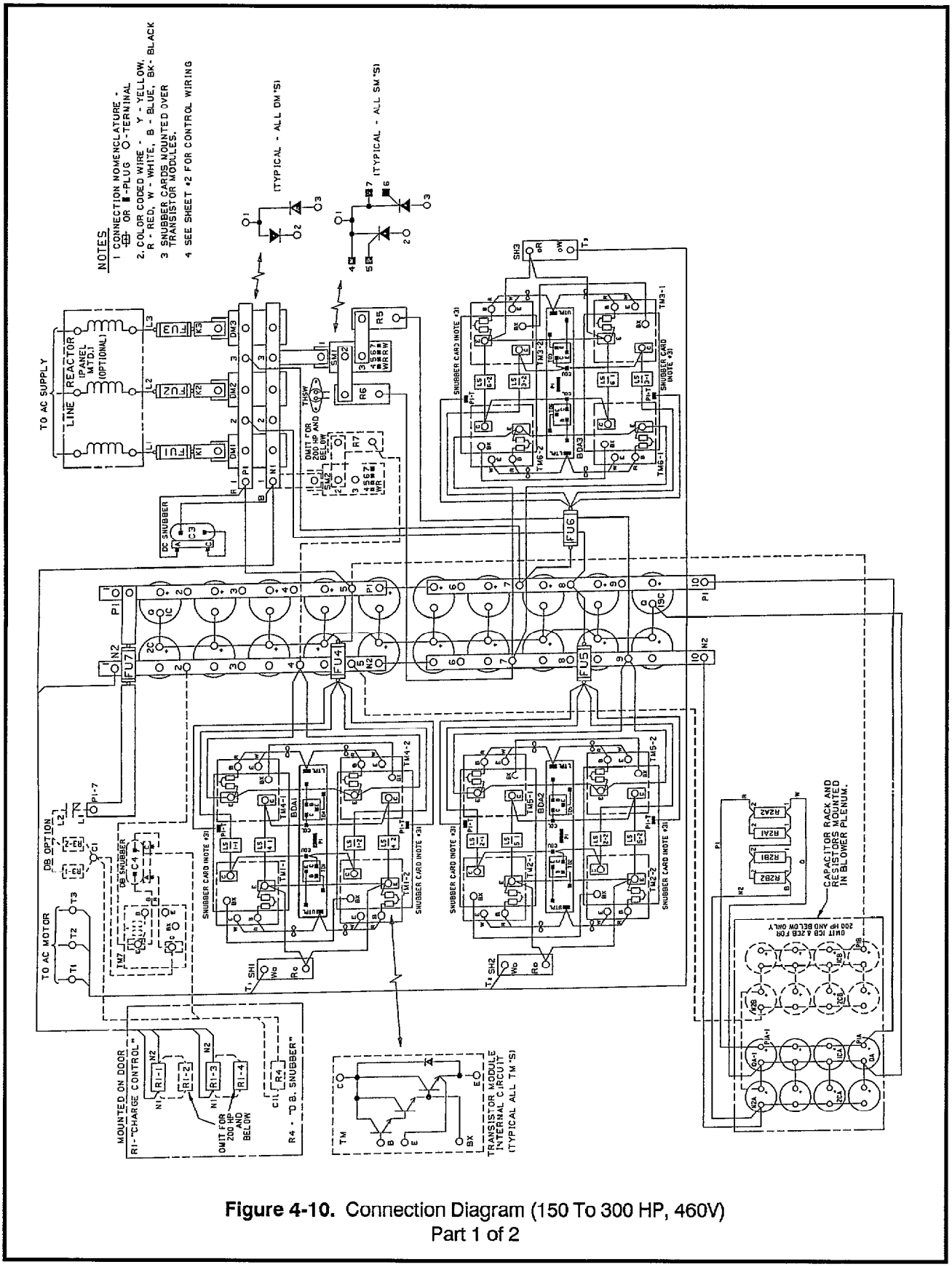


Figure 4-10. Connection Diagram (150 To 300 HP, 460V)
Part 1 of 2

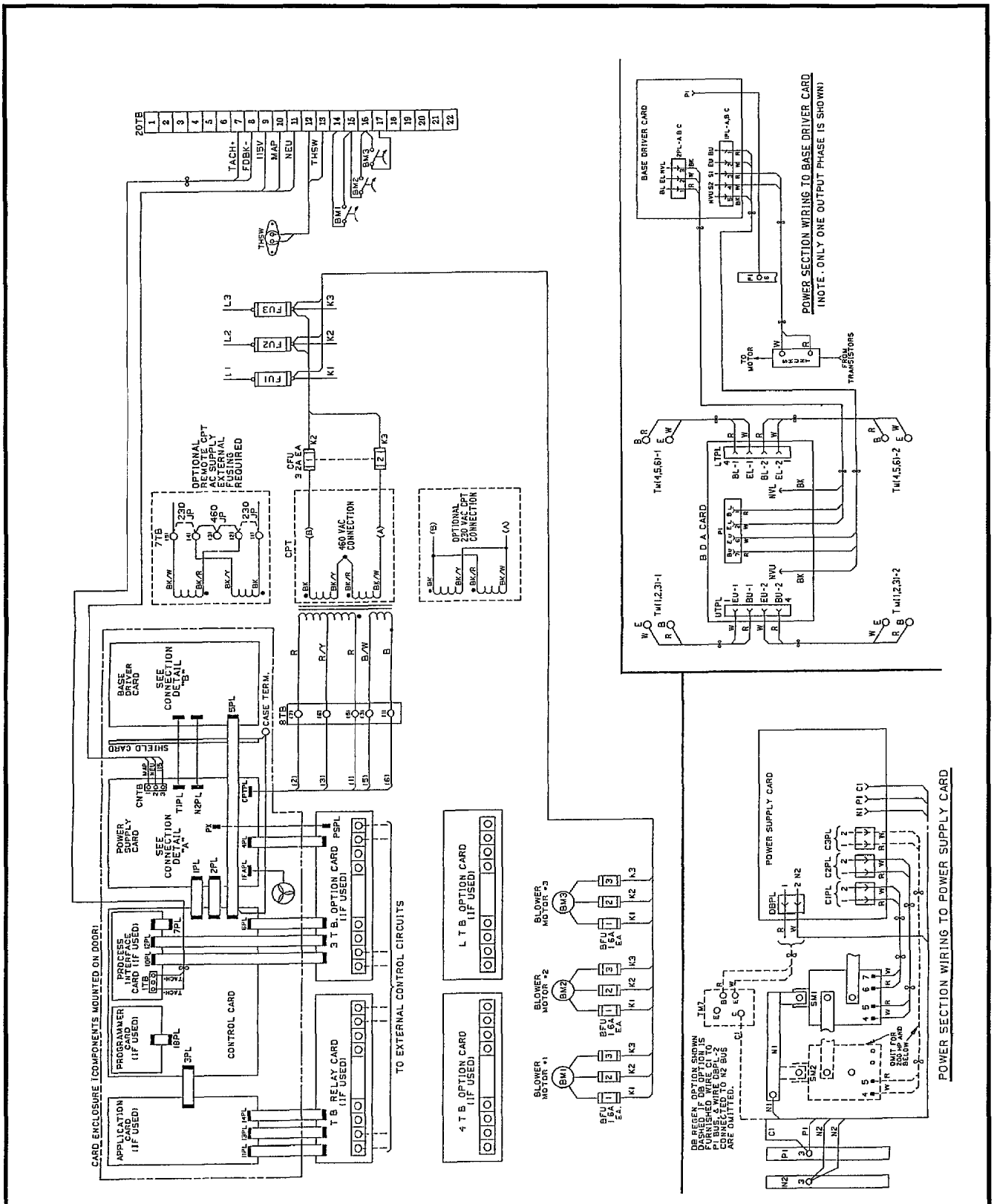


Figure 4-10. Connection Diagram (150 To 300 HP, 460V)
Part 2 of 2

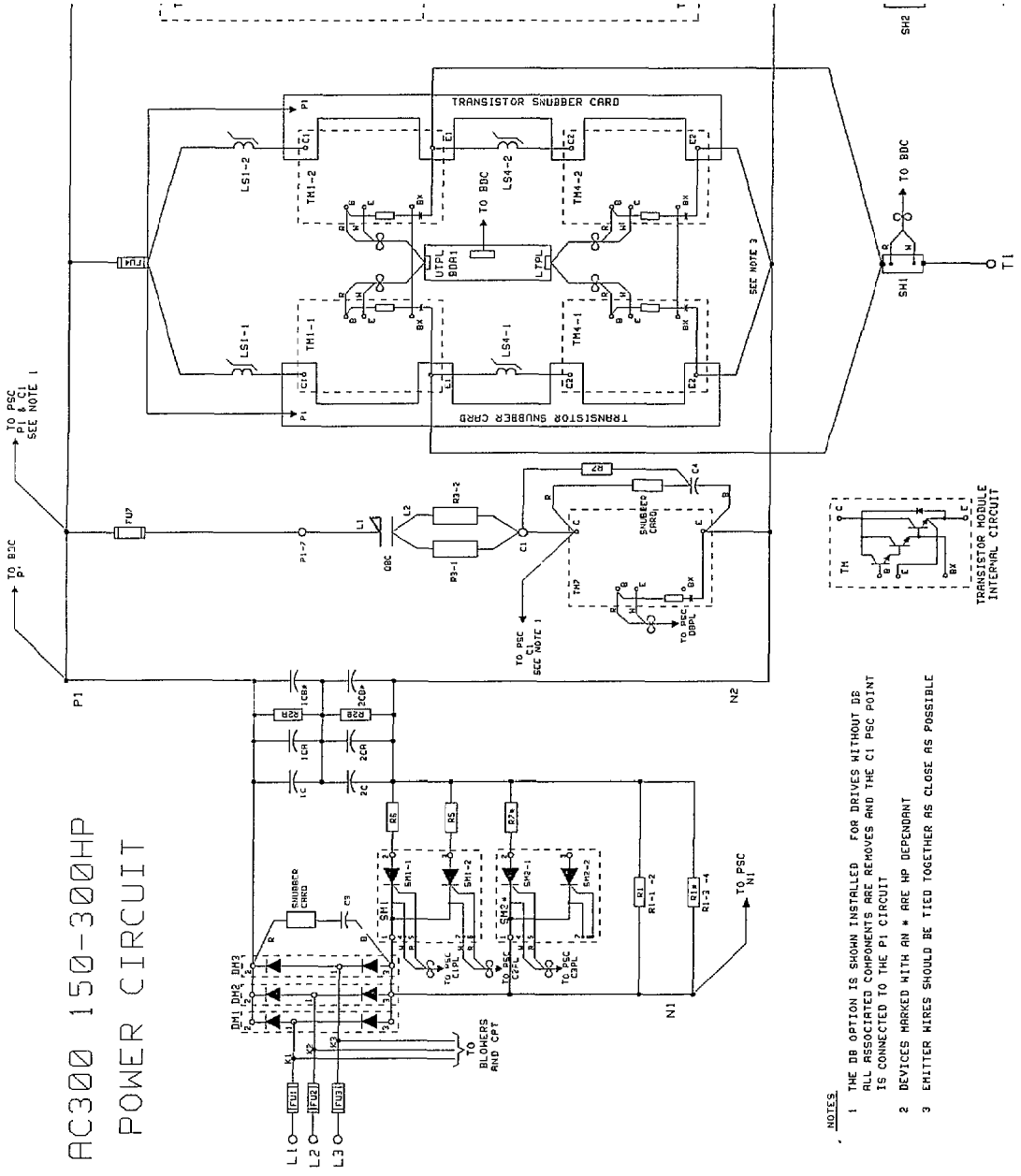


Figure 4-11. Power Circuit With DB Regeneration (150 To 300 HP, 460V)

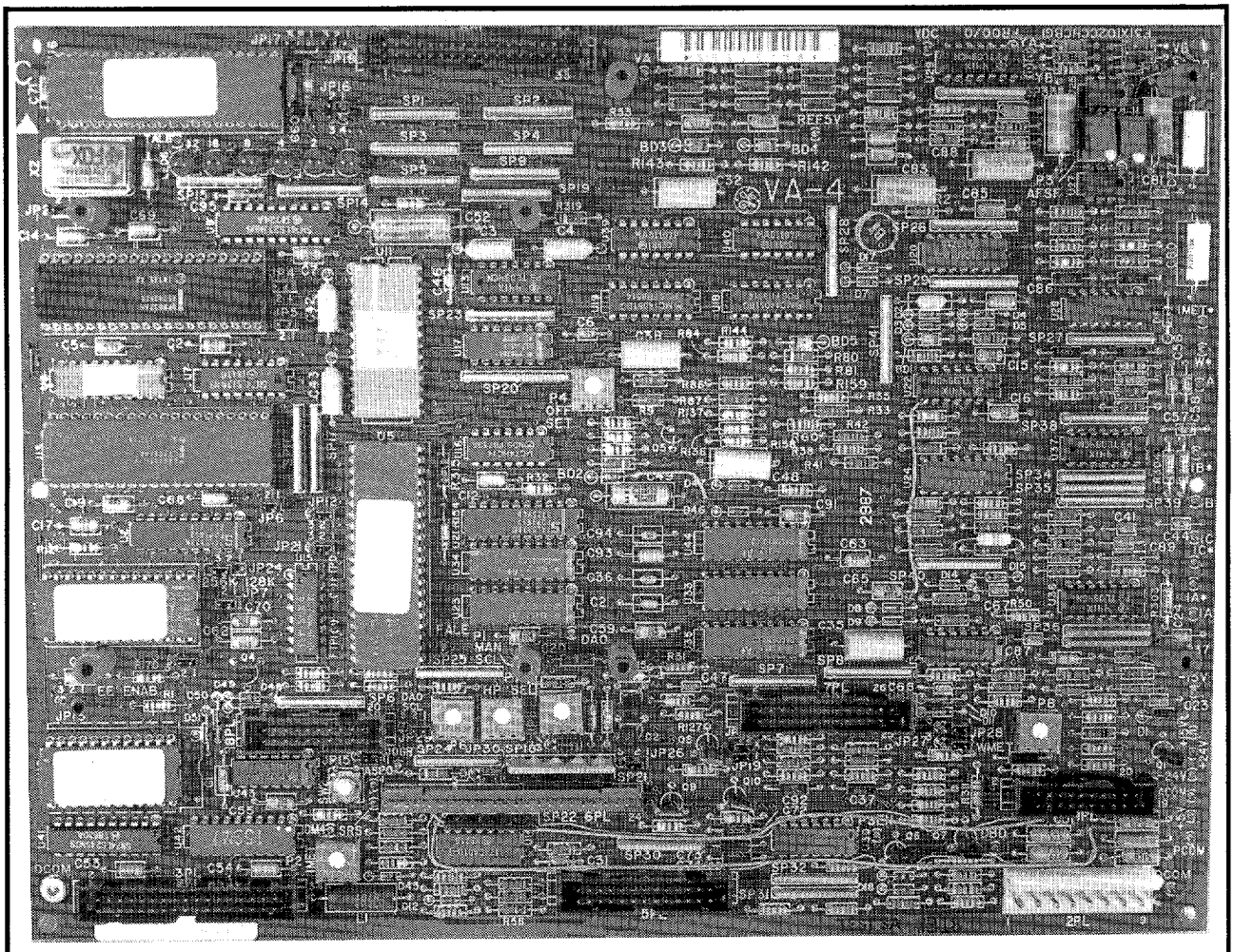


Figure 4-12. Control Card

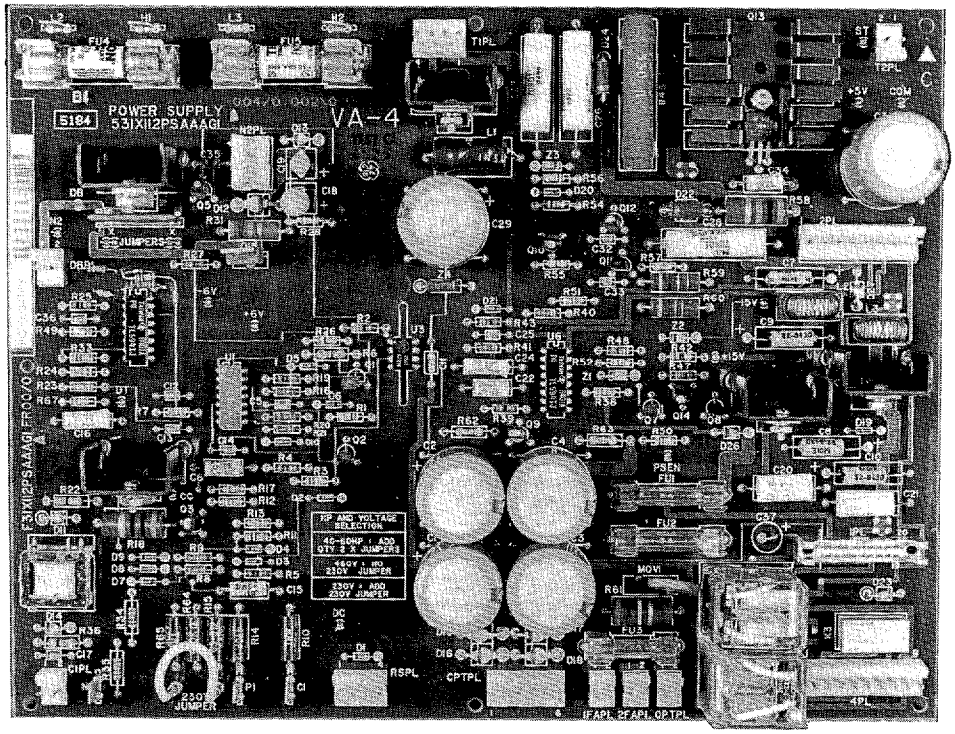


Figure 4-13. Power Supply Card

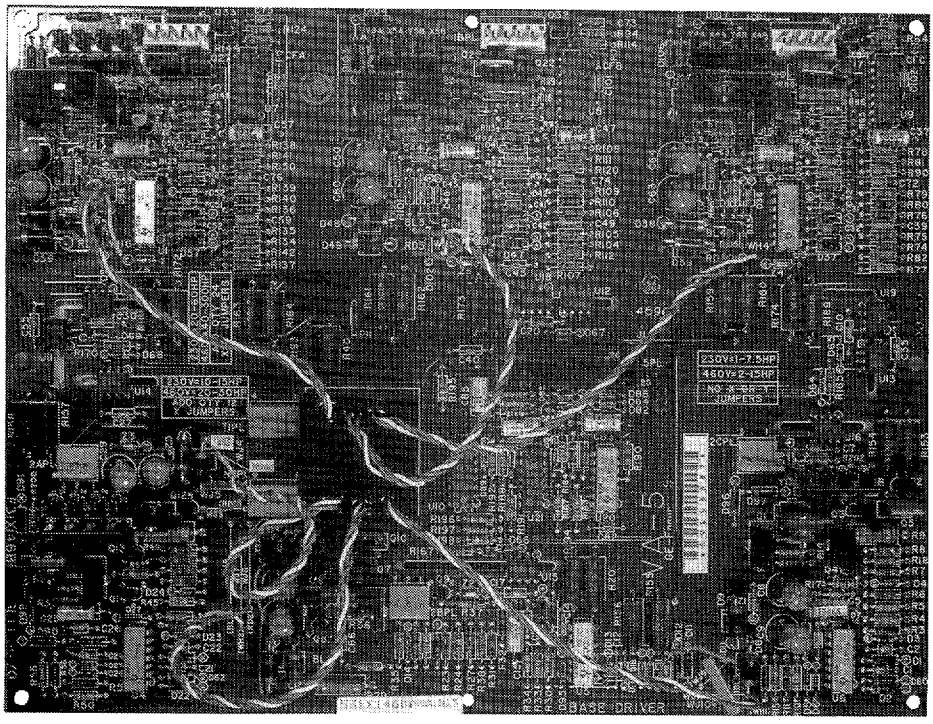


Figure 4-14. Base Drive Card

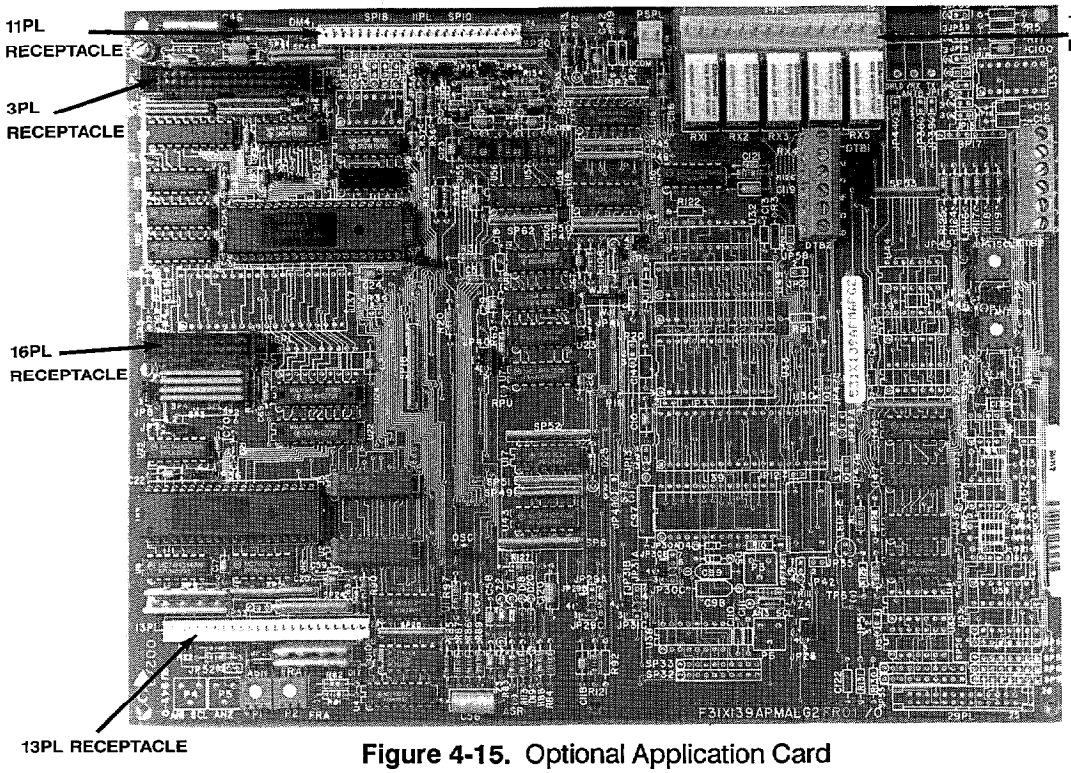


Figure 4-15. Optional Application Card

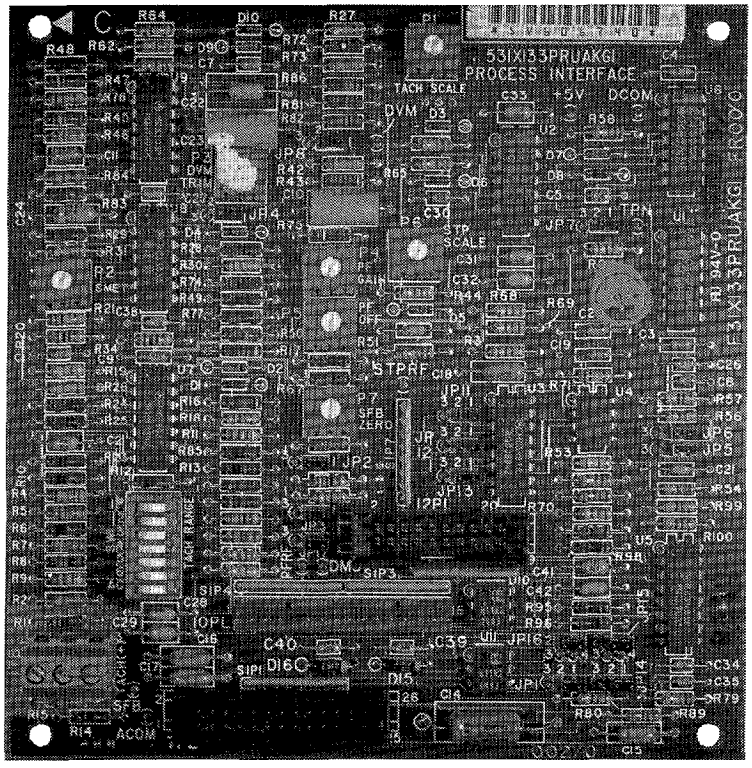


Figure 4-16. Optional Process Interface Card

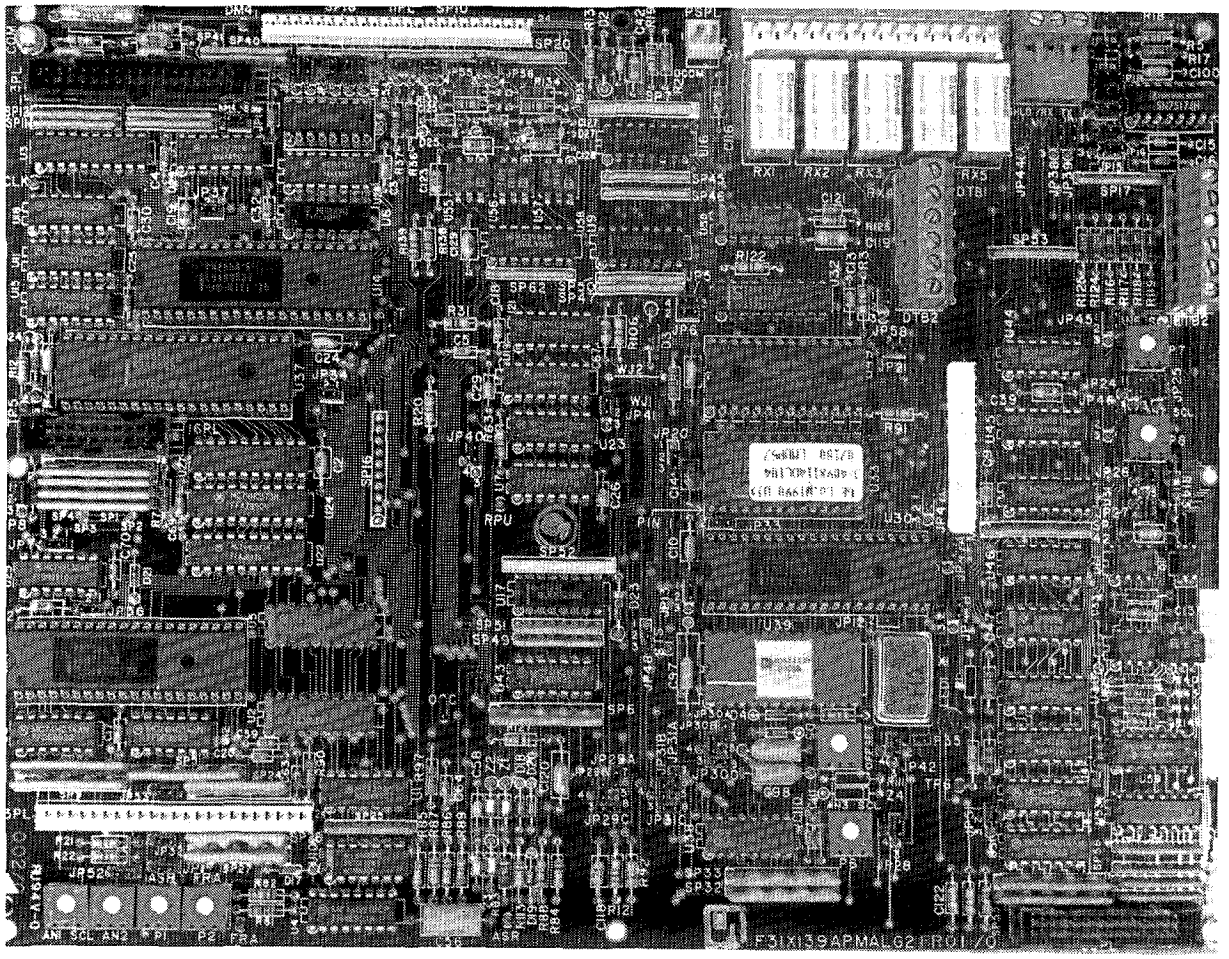


Figure 4-17. Optional Microapplication Card

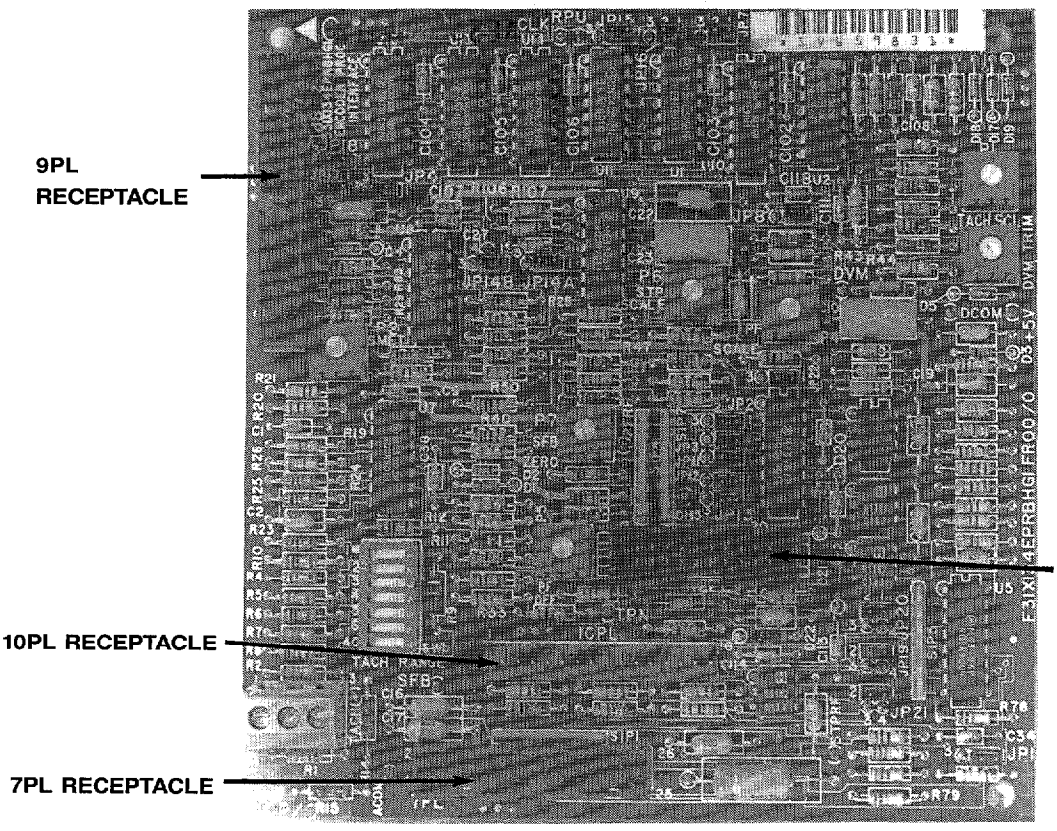


Figure 4-18. Optional Encoder Process Interface Card

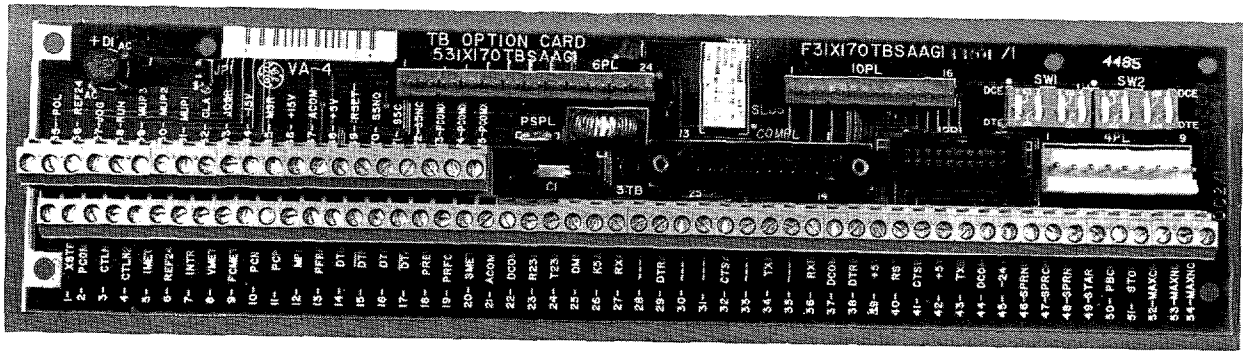


Figure 4-19. Optional 3TB Terminal Board Card

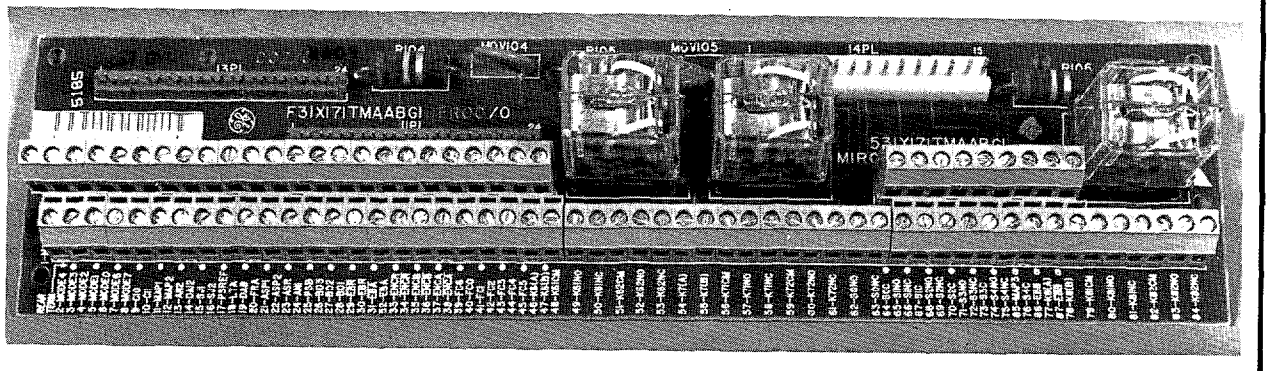


Figure 4-20. Optional 4TB Terminal Board Card

5. ELECTRICAL CONNECTOR PIN CHARTS

This section contains electrical connector pin charts for all connectors in the controller. The major connectors are listed in numerical order and any additional smaller connectors are listed at the end of the section under the heading **"ADDITIONAL CONNECTORS"**.

CONNECTOR 1PL -- BETWEEN CONTROL CARD AND POWER SUPPLY CARD (PSC)

Pin	Nomen-clature	Description
1	DCOM	Digital Common.
2	DBD	
3	DCOM	
4	PSEN	Power Supply Enable - indicates when all power supplies are in range.
5	DCOM	
6	--	NO CONNECTION
7	--	NO CONNECTION
8	--	NO CONNECTION
9	FRC	Fault relay control - Drives the 24 volt fault relay on PSC.
10	-24V	Unregulated -24V return path for FRC and RP.
11	RP	K2 (MAX) relay pickup.
12	RUN	Drive run signal from RUN (K1) relay contact on PSC.
13	REF24	Supplies +/-24V to RUN (K1) relay on PSC depending on setting of jumper JP19 and JP20 on Control card.
14	W5V	Watchdog Reference - Provides 5V reference from PSC to undervoltage detector on Control card.
15	--	NO CONNECTION
16	--	NO CONNECTION
17	--	NO CONNECTION
18	--	NO CONNECTION
19	DCOM	
20	--	NO CONNECTION

CONNECTOR 2PL -- BETWEEN CONTROL CARD AND POWER SUPPLY CARD (PSC)

Pin	Nomen-clature	Description
1	ACOM	Analog signal return for +/-15V.
2	-15V	-15V regulated analog power supply input.
3	+15V	+15V regulated power supply input.
4	+24V	+24V unregulated raw power supply input.
5	-24V	-24V unregulated raw power supply input.

CONNECTOR 2PL -- BETWEEN CONTROL CARD AND POWER SUPPLY CARD (PSC) (continued)

Pin	Nomen-clature	Description
6	+5V	+5V regulated digital power supply input.
7	DCOM	Digital signal return for +5V.
8	+5V	
9	PCOM	Raw power supply return for +/-24V.

CONNECTOR 3PL -- BETWEEN CONTROL CARD AND OPTIONAL APPLICATION CARD *or* MICROAPPLICATION CARD

Pin	Nomen-clature	Description*
1	D0	U4 data bus bit 0 (least significant bit).
2	D1	U4 data bus bit 1.
3	DCOM	Digital common.
4	D2	U4 data bus bit 2.
5	D3	U4 data bus bit 3.
6	D4	U4 data bus bit 4.
7	DCOM	
8	D5	U4 data bus bit 5.
9	D6	U4 data bus bit 6.
10	D7	U4 data bus bit 7.
11	DCOM	
12	A8	U4 address bus bit 8.
13	A9	U4 address bus bit 9.
14	A10	U4 address bus bit 10.
15	DCOM	
16	A11	U4 address bus bit 11.
17	SEL	Signal from U3. Used for Microapplication card synchronization.
18	ALE	Address latch enable from U4.
19	DCOM	
20	/RD	Read control line from U4.
21	/WR	Write control line from U4.
22	A15	U4 address bus bit 15, used for card I/O select.
23	RST	Card reset signal from Control card (active high).
24	+5V	+5V regulated digital supply to card.
25	+5V	
26	+5V	
27	PCOM	+/-24V power supply return.
28	+24V	+24V non-regulated raw power supply to card.
29	-24V	-24V non-regulated power supply to card.
30	DM4	Provides "dummy" connection from the card.
31	+15V	+15V regulated analog supply to card.
32	-15V	-15V regulated analog supply to card.
33	ACOM	Analog signal return for +/-15V.
34	OPTA	Input to A/D converter from card multiplexer.

Descriptions apply to both **Application and **Microapplication** cards.*

CONNECTOR 4PL -- BETWEEN POWER SUPPLY CARD (PSC) AND OPTIONAL 3TB TERMINAL BOARD CARD

Pin	3TB Term. #	Nomen-clature	Description
1	46	SPRNO	Fault Relay Normally Open Contact.
2	47	SPRCM	Fault Relay Common Connection.
3	48	SPRNC	Fault Relay Normally Closed Contact.
4	49	START	Connection Point for Start PB to "RUN" relay (K1).
5	50	PBCM	Common Point for Start and Stop Pushbuttons.
6	51	STOP	Connection for Stop Pushbutton.
7	52	MAXCM	MA Auxiliary Relay Common Connection.
8	53	MAXNO	MA Auxiliary Relay Normally Open Contact.
9	54	MAXNC	MA Auxiliary Relay Normally Closed Contact.
			NOTE: *Contact rating of MAX relay is 1.2 amps at 115 volts AC or 28 volts DC. Contact rating of fault relay is 1 amp at 115 volts AC or 28 volts DC.*

CONNECTOR 5PL -- BETWEEN CONTROL CARD AND BASE DRIVE CARD

Pin	Nomen-clature	Description
1	U/DA	Up/down counter Phase A.
2	+15V	
3	U/DB	Up/down counter Phase B.
4	DCOM	Shield wire.
5	U/DC	Up/down counter Phase C.
6	+15V	
7	ENA	Enable signal for U/DA counter.
8	DCOM	Shield wire.
9	ENB	Enable signal for U/DB counter.
10	--	NO CONNECTION
11	ENC	Enable signal for U/DC counter.
12	DCOM	Shield wire.
13	CFA	Current feedback Phase A.
14	+5V	
15	CFB	Current feedback Phase B.
16	DCOM	Shield wire.
17	CFC	Current feedback Phase C.
18	+5V	
19	VFA	Voltage feedback Phase A.
20	DCOM	Shield wire.
21	VFB	Voltage feedback Phase B.
22	--	NO CONNECTION
23	VFC	Voltage feedback Phase C.
24	DCOM	Shield wire.
25	DCN	Resistance isolated - DC Link Feedback.
26	DCP	Resistance isolated + DC Link Feedback.

CONNECTOR 6PL -- BETWEEN CONTROL CARD AND OPTIONAL 3TB TERMINAL BOARD CARD

Pin	3TB Term. #	Nomen-clature	Description
1	69	RSET	A hard reset of drive is initiated when RSET is connected to +15V. Must be open to run. RSET immediately turns off all inverter transistors, drops out the fault relay and holds all microprocessors in a reset state until RSET is opened. The drive will not automatically restart if reset while running.
2	68	+5V	Not for use external to drive.
3	67	ACOM	Not for use external to drive.
4	66	+15V	Signal voltage for System Reference pot, RSET circuit and others.
5	65	MSR	Manual system reference input to drive, ±15V max. Positive for forward motoring with POL at zero or open.
6	64	-15V	Signal voltage for System Reference pot.
7	63	JOGR	Jog reference input (can be selected to bypass linear time), ±14V = Top speed. Programmable to 7V = top speed for fast update rate. Refer to Custom Software Description.
8	62	CLA	Current limit adjust - Adds to software setting of current limit. +10V increases current limit by 125% of rated, -10V decreases it by 125%.
9	61	MUP1	Multiuse point 1 - connects to undedicated point MUP1.
10	60	MUP2	Multiuse point 2 - connects to undedicated point MUP2.
11	59	MUP3	Multiuse point 3 - connects to undedicated point MUP3.
12	58	RUN1	Alternate run command input point in place of "RUN" relay. Do not pull high if run relay is also being used. Drive runs when connected to 3TB56 (6PL-14), REF24. With JP19 and JP20 on the Control card in the 1-2 and 3-4 positions respectively, input is active (controller runs) when RUN is pulled down to -24V DC. Input is inactive (controller stops) when RUN is open or driven to +24 VDC. With JP19 and JP20 in the 1-3 and 2-4 positions respectively, input is active (controller runs) when RUN is driven to +24V DC. Input is inactive (controller stops) when RUN is open or pulled down to -24 VDC.
13	57	JOG	Jog command input point. Activated, deactivated same as RUN above.
14	56	REF24	Selectable + or -24V supply for RUN, JOG, POL, REV, or XTSP input.

CONNECTOR 6PL -- BETWEEN CONTROL CARD AND OPTIONAL 3TB (contd.)

Pin	3TB Term. #	Nomen-clature	Description
15	55	POL	Control input which reverses the direction of the drive for any applied reference; e.g. activating POL low is equivalent to reversing the polarity of the input at MSR. Activated/ Deactivated same as RUN above. Refer to Custom Software Description and elementaries if used.
16	1	XSTP	Must be activated to enable drive. Activated/Deactivated same as RUN described previously.
17	2	PCOM	Not for use on external drive.
18	3	CTLN1	Control On - Must be connected to CTLN2 to allow the drive to run. Opening this connection unconditionally preconditions the drive.
19	4	CTLN2	See CTLN1.
20	5	IMET	Analog load meter output proportional to motor torque pro ducing current IX. (See P2 on the Control card, TABLE 1 for details.) Load not to exceed 1000 ohms/volt.
21	6	REF24	Selectable $\pm$24V supply for RUN, JOG, REV, POL or XSTP input.
22	7	SLD5	NOT USED
23	8	WMET	Analog frequency meter output. (See P8 on the Control card, TABLE 1, for details.) Maximum load is 5 milliamps.
24	9	FCMET	NOT USED

CONNECTOR 7PL -- BETWEEN CONTROL CARD AND OPTIONAL PROCESS INTERFACE CARD *or* ENCODER PROCESS INTERFACE CARD

Pin	Nomen-clature	Description*
1	ACOM	Analog signal return.
2	DVM	Output from digital voltmeter.
3	SFB	Output from analog tach circuit.
4	ASFB	Amplified signal from analog tach circuit, used for improved resolution at low speed operation.
5	ACOM	
6	-15V	-15V supply to analog circuits on Process Interface (P.I.) card.
7	+15V	+15V supply to analog circuits on P.I. card.
8	+5V	+5V supply to logic circuits on P.I. card.
9	+5V	
10	TRS	RS422 driver tri-state command to the P.I. card.
11	DCOM	Digital signal return.
12	DIR	133 Card: Signal from digital tach interface signalling direction of motor rotation when a quadrature track tach is used. 134 Card: NOT USED
13	TPN	133 Card: Net pulse train signal from digital tach or AC analog tach interface used to determine tach speed. 134 Card: Jumpered to carry the encoder marker pulse channel for absolute position detection.
14	PRF	133 Card: Pulse train from digital reference interface used to determine desired reference when used. 134 Card: NOT USED
15	DCOM	
16	DCOM	
17	RXD	Input from either the RS422 or RS232C serial link receiver from the Process Interface card.
18	TXD	Output to the RS422 and RS232C serial link drivers from the Process Interface card.
19	/DTR	Data terminal ready - Output indicating the Process Interface card is ready to receive a byte on the serial link. Low when ready, high when not ready.
20	/CTS	Clear to send - Input from serial link signalling when the receiver on the other end of the link is ready to receive a byte. When low, drive may transmit. When high, drive must not send anything out on TXD.
21	RST	Reset line from Control card to Process Interface card, active high.
22	+5V	
23	+5V	
24	DCOM	
25	DCOM	
26	-24V	-24V raw power to Process Interface card.

Descriptions apply to both **Process Interface (133) and **Encoder Process Interface (134)** cards except where each is shown.*

CONNECTOR 9PL -- BETWEEN CONTROL CARD AND OPTIONAL ENCODER PROCESS INTERFACE CARD

Pin	Nomen-clature	Description
1	P10	U/D counter output to TUP.
2	P11	U/D counter output to TUP.
3	P12	U/D counter output to TUP.
4	P13	U/D counter output to TUP.
5	P14	U/D counter output to TUP.
6	P15	U/D counter output to TUP.
7	P16	U/D counter output to TUP.
8	P17	U/D counter output to TUP.
9	P20	U/D counter output to TUP.
10	P21	U/D counter output to TUP.
11	P22	U/D counter output to TUP.
12	P23	U/D counter output to TUP.
13	DM5	Multi-use point 5, connects to 10PL-26 (3TB-25).
14	MKAPLS	Marker pulse output to LUP.
15	--	NO CONNECTION
16	--	NO CONNECTION
17	--	NO CONNECTION
18	--	NO CONNECTION
19	--	NO CONNECTION
20	--	NO CONNECTION
21	--	NO CONNECTION
22	--	NO CONNECTION
23	DCOM	Digital common.
24	DCOM	Digital common.
25	--	NO CONNECTION
26	--	NO CONNECTION
27	--	NO CONNECTION
28	--	NO CONNECTION
29	--	NO CONNECTION
30	--	NO CONNECTION
31	--	NO CONNECTION
32	--	NO CONNECTION
33	--	NO CONNECTION
34	--	NO CONNECTION

CONNECTOR 10PL -- BETWEEN OPTIONAL 3TB TERMINAL BOARD CARD AND OPTIONAL PROCESS INTERFACE CARD *or* ENCODER PROCESS INTERFACE CARD

Pin	3TB Term. #	Nomen-clature	Description*	
1	10	PCN	Optional negative input for process follower current (1-5, 4-20, 10-50, or 5-65mA.) or voltage (0-10 VDC) reference.	
2	11	PCP	Optional positive input for process follower reference.	
3	12	MR1	Optional reference for manual system reference potentiometer to allow coordinated max speed scaling when PFRF is connected to Control card reference input via auto/manual selection.	
4	13	PFRF	Optional process follower interface output (0-10V) which connects to a speed reference or other analog process input.	
5	14	DTB	133 Card:	Optional square wave tach quadrature input (track B high). Operating Voltage: -0.5V to +0.8V is logic 0 and +2.4V to +15V is logic 1.
			134 Card:	Encoder channel B input.
6	15	/DTB	133 Card:	Optional inverted square wave differential tach quadrature input (track B low).
			134 Card:	Encoder channel B input.
7	16	DTA	133 Card:	Optional square wave tach quadrature input (track A high). Operating Voltage: -0.5V to +0.8V is logic 0 and +2.4V to +15V is logic 1.
			134 Card:	Encoder channel A input.
8	17	/DTA	133 Card:	Optional inverted square wave differential tach quadrature input (track A low).
			134 Card:	Encoder channel A input.
9	18	PREF	133 Card:	Optional pulse reference frequency positive input, 0 to 6KHz Max.
			134 Card:	NOT USED
10	19	PRFC	133 Card:	Optional high impedance common for pulse reference input.
			134 Card:	Encoder marker pulse channel inverting input.
11	20	SMET	Analog speed meter output, 10 VDC. See TABLE 1, P2 on Process Interface card or Encoder Process Interface card.	
12	21	ACOM	133 Card:	NOT USED
			134 Card:	Encoder marker pulse channel non-inverting input.
13	22	DCOM	133 Card:	Signal common. Not for use external drive.
			134 Card:	NO CONNECTION
14	23	R232	Optional RS232C receive line.	
15	24	T232	Optional RS232C transmit line.	
16	25	DM5	133 Card:	Dummy point 5 connects to undedicated point DM5.
			134 Card:	Spare connection that can be jumpered to 9PL.

NOTE: / = NOT; **Example:** /C = Not C

Descriptions apply to both **Process Interface (133) and **Encoder Process Interface (134)** cards except where each is shown.*

CONNECTOR 11PL -- BETWEEN OPTIONAL 4TB TERMINAL BOARD CARD AND OPTIONAL APPLICATION CARD *or* MICROAPPLICATION CARD

Pin	4TB Term. #	Nomen-clature	Description*	
1	30	EBR	Application:	NOT USED (No connection on 132 version card.)
			Microappl:	MUP 24-bit non-inverting differential or single-ended encoder** marker pulse input. This input can also be used to generate an external interrupt to MUP for both LAN and non-LAN applications. If JP22 is in 2-3 position, 11PL-1 is connected to LANTB-1 (TX) via JP39.
2	29	/EBR	Application:	NOT USED (No connection on 132 version card.)
			Microappl:	MUP 24-bit inverting differential encoder** marker pulse input. If JP23 is in 2-3 position, 11PL-2 is connected to LANTB-2 (/RX) via JP38.
3	85	MAP3	Multiuse point 3 - connects to connection point MAP3, which may be jumpered to another connection point (controlled by JP29). (No connection on 132 version Application card).	
4	25	PSB	Strobe for optional 16 bit parallel input on 16PL. Inputs must be set 50 microseconds before strobe, and strobe held for 100 microseconds. See TABLE 1, JP7 on Application card or Microapplication card.	
5	26	B03	Optional programmable 24V bit output #3 (5V TTL on 132 version Application card).	
6	27	B02	Optional programmable 24V bit output #2 (5V TTL on 132 version Application card).	
7	28	B01	Optional programmable 24V bit output #1 (5V TTL on 132 version Application card).	
8	32	EBA	Application:	NOT USED (+5V on 132 version card.)
			Microappl:	MUP 24-bit non-inverting differential or single-ended encoder** track A input.
9	31	/EBA	Application:	NOT USED (+5V on 132 version card.)
			Microappl:	MUP 24-bit inverting differential encoder** track A input.
10	86	EBB	Application:	NOT USED (DCOM on 132 version card.)
			Microappl:	MUP 24-bit non-inverting differential or single-ended encoder** track B input.

Descriptions apply to both **Application and **Microapplication (Microappl)** cards except where each is shown.*

***All encoder inputs are RS422 compatible or can be used with logic levels of up to +/-15 VDC. In the single-ended mode (JP9 in 2-3 position), the threshhold between logic 0 and logic 1 is 1.8 VDC.*

CONNECTOR 11PL -- BETWEEN OPTIONAL 4TB TERMINAL BOARD CARD AND OPTIONAL APPLICATION CARD *or* MICROAPPLICATION CARD (continued)

Pin	4TB Term. #	Nomen-clature	Description*
11	87	/EBB	Application: NOT USED (DCOM on 132 version card.) Microappl: MUP 24-bit inverting differential encoder** track B input.
12	33	ENCR	Optional RUP 12-bit non-inverting differential or single-ended encoder** marker pulse input.
13	34	/ENCR	Optional RUP 12-bit inverting differential encoder** marker pulse input.
14	35	ENCB	Optional RUP 12-bit non-inverting differential or single-ended encoder** Track B input.
15	36	/ENCB	Optional RUP 12-bit inverting differential encoder** Track B input.
16	37	ENCA	Optional RUP 12-bit non-inverting differential or single-ended encoder** Track A input.
17	38	/ENCA	Optional RUP 12-bit inverting differential encoder** Track A input.
18	39	FC6	Bit 6 of optional fault code 24V output (5V TTL on 132 version Application card).
19	40	FC0	Bit 0 of optional fault code 24V output (5V TTL on 132 version Application card).
20	41	FC1	Bit 1 of optional fault code 24V output (5V TTL on 132 version Application card).
21	42	FC2	Bit 2 of optional fault code 24V output (5V TTL on 132 version Application card).
22	43	FC3	Bit 3 of optional fault code 24V output (5V TTL on 132 version Application card).
23	44	FC4	Bit 4 of optional fault code 24V output (5V TTL on 132 version Application card).
24	45	FC5	Bit 5 of optional fault code 24V output (5V TTL on 132 version Application card).

Descriptions apply to both **Application and **Microapplication (Microappl)** cards except where each is shown.*

***All encoder inputs are RS422 compatible or can be used with logic levels of up to +/-15 VDC. In the single-ended mode (JP9 in 2-3 position), the threshhold between logic 0 and logic 1 is 1.8 VDC.*

CONNECTOR 12PL -- BETWEEN OPTIONAL 3TB TERMINAL BOARD CARD AND OPTIONAL PROCESS INTERFACE CARD *or* ENCODER PROCESS INTERFACE CARD

Pin	3TB Term. #	Nomen-clature	Description*
1	--	--	NO CONNECTION
2	--	--	NO CONNECTION
3	--	--	NO CONNECTION
4	--	--	NO CONNECTION
5	--	--	NO CONNECTION
6	--	--	NO CONNECTION
7	32	CTSA or /C	Optional inverted RS422 clear-to-send input from serial link signaling when the Control card may transmit a byte, active high. (EIA standard nomenclature is CTSA.)
8	29	DTRA or /D	Optional inverted RS422 data-terminal-ready output from Control card signaling when it is ready to receive a byte on the serial link, active high. (EIA standard nomenclature is DTRA.)
9	34	TXA or /T	Optional inverted RS422 transmitted data signal from Control card to serial link. (EIA standard nomenclature is TXA.)
10	27	RXA or /R	Optional inverted RS422 received data signal from serial link to Control card. (EIA standard nomenclature is RXA.)
11	36	RXB or R	Optional non-inverted RS422 received data signal from serial link to Control card. (EIA standard nomenclature is RXB.)
12	43	TXB or T	Optional non-inverted RS422 transmitted data signal from Control card to serial link. (EIA standard nomenclature is TXB.)
13	38	DTRB or D	Optional non-inverted RS422 data-terminal-ready output from Control card. (EIA standard nomenclature is DTRB.)
14	41	CTSB or C	Optional non-inverted RS422 clear-to-send input from serial link. (EIA standard nomenclature is CTSB.)
15	40	RST	Reset signal from Control card, active high.
16	39	+5V	+5V DC supply. Not for use external to drive.
17	42	+5V	+5V DC supply. Not for use external to drive.
18	37	DCOM	Signal common. Not for use external to drive.
19	44	DCOM	Signal common. Not for use external to drive.
20	45	-24V	-24V DC unregulated supply.

Descriptions apply to both **Process Interface (133) and **Encoder Process Interface (134)** cards.*

CONNECTOR 13PL -- BETWEEN OPTIONAL 4TB TERMINAL BOARD CARD AND OPTIONAL APPLICATION CARD *or* MICROAPPLICATION CARD

NOTE: *Refer to the Custom Instruction Book, software jumper description, for definitions of I/O through 13PL for particular application.*

Pin	4TB Term. #	Nomen-clature	Description*
1	6	MODE 0	Programmable mode select input. Refer to elementaries and Custom Software Description** for information.
2	5	MODE 1	Programmable mode select input. Refer to elementaries and Custom Software Description** for information.
3	4	MODE 2	Programmable mode select input. Refer to elementaries and Custom Software Description** for information.
4	3	MODE 3	Programmable mode select input. Refer to elementaries and Custom Software Description** for information.
5	2	MODE 4	Programmable mode select input. Refer to elementaries and Custom Software Description** for information.
6	1	MODE 5	Programmable mode select input. Refer to elementaries and Custom Software Description** for information.
7	7	MODE 6	Programmable mode select input. Refer to elementaries and Custom Software Description** for information.
8	8	MODE 7	Programmable mode select input. Refer to elementaries and Custom Software Description** for information.
9	9	CI0	Application: NO CONNECTION (-24V on 132 version card.) Microappl: Programmable control mode select input.
10	10	CI1	Application: NO CONNECTION (ACOM on 132 version card.) Microappl: Programmable control mode select input.
11	11	MAP1	Multiuse point 1 - connects to connection point MAP1, which may be jumpered to another connection point. (ACOM on 132 version Application card.)
12	12	MAP2	Multiuse point 2 - connects to connection point MAP2, which may be jumpered to another connection point. (-15V on 132 version Application card.)
13	13	AN2	Application: NO CONNECTION (+15V on 132 version card.) Microappl: General purpose MUP +/-10 VDC analog input. May be scaled to higher ranges of up to 48 VDC by pot P5.
14	14	DA12	Optional analog output from 12-bit D/A converter. Range = -10V to +10V, 1 MA, with 450 ohms output impedance. Update rate is 30 to 40 times/second.

Descriptions apply to both **Application and **Microapplication (Microappl)** cards except where each is shown.*

***Functions of MODE 0 - MODE 7 are determined by Addresses 000-015 (ADR000 - ADR015) in the Custom Software Description.*

CONNECTOR 13PL -- BETWEEN OPTIONAL 4TB TERMINAL BOARD CARD AND OPTIONAL APPLICATION CARD *or* MICROAPPLICATION CARD (continued)

Pin	4TB Term. #	Nomen-clature	Description*
15	15	SJ1	Optional analog input to summing junction #1. Operating range = ±10V; maximum input = ±15V.
16	16	SJ2	Optional analog input to summing junction #2. Operating range = ±10V; maximum input = ±15V.
17	17	PSREF	Optional analog position or process reference input (±10V).
18	18	LTA	Optional analog linear time adjust. Positive voltages increase timing, negative voltages decrease timing. ±10 volts maximum. 10 volts is equal to 66 second adjust.
19	19	DA8	Optional analog output from 8-bit D/A converter. See DA12 for specifications. Normal update rate of 180 times/sec. Reduced to 30 to 40 times/sec if fast update selected for DA0.
20	20	FRA	Optional analog field reference adjust. Positive voltages up to +10V increase field current. Negative voltages down to -10V decrease field current. See TABLE 1 Application card or Microapplication card JP32 and P2.
21	21	ASP1	Optional spare analog input #1 (8-bit resolution, programmable).
22	22	ASP2	Optional spare analog input #2 (12-bit resolution, programmable). **NOTE:** *Not available when recalibrate tach (RECAL TACH) mode is active.*
23	23	ASR	Optional auto speed reference input to drive, ±15V maximum, positive for forward cell firing while motoring. ASR is the active speed reference in the auto mode when analog auto reference is selected.
24	24	AN1	Application: NO CONNECTION (ACOM on 132 version card.) Microappl: General purpose MUP +/-10 VDC analog input. May be scaled to higher ranges of up to 48 VDC by pot P4.

Descriptions apply to both **Application and **Microapplication (Microappl)** cards except where each is shown.*

CONNECTOR 14PL -- BETWEEN OPTIONAL 4TB TERMINAL BOARD CARD AND OPTIONAL APPLICATION CARD *or* MICROAPPLICATION CARD

Pin	4TB Term. #	Nomen-clature	Description*
1	62	S0NO	Optional SLD #0 normally open contact.
2	63	S0NC	Optional SLD #0 normally closed contact.
3	64	S0C	Optional SLD #0 common contact.
4	65	S1NO	Optional SLD #1 normally open contact.
5	66	S1NC	Optional SLD #1 normally closed contact.
6	67	S1C	Optional SLD #1 common contact.
7	68	S2NO	Optional SLD #2 normally open contact.
8	69	S2NC	Optional SLD #2 normally closed contact.
9	70	S2C	Optional SLD #2 common contact.
10	71	S3NO	Optional SLD #3 normally open contact.
11	72	S3NC	Optional SLD #3 normally closed contact.
12	73	S3C	Optional SLD #3 common contact.
13	74	S4NO	Optional SLD #4 normally open contact.
14	75	S4NC	Optional SLD #4 normally closed contact.
15	76	S4C	Optional SLD #4 common contact.

Descriptions apply to both **Application and **Microapplication** cards.*

CONNECTOR 16PL -- LOCATED ON OPTIONAL APPLICATION CARD *or* MICROAPPLICATION CARD

Pin	Nomen-clature	Description*
1	DCOM	Digital common.
2	+5V	NOT USED
3	PI0	Optional parallel word input least significant binary bit input. For all inputs refer to TABLE 1, JP8 description for voltage level select. Input must be set for 50 microseconds prior to strobe.
4	PI1	Optional parallel binary bit input 1.
5	PI2	Optional parallel binary bit input 2.
6	PI3	Optional parallel binary bit input 3 (or PI0 - PI3 = BCD units digit).
7	PI4	Optional parallel binary bit input 4.
8	PI5	Optional parallel binary bit input 5.
9	PI6	Optional parallel binary bit input 6.
10	PI7	Optional parallel binary bit input 7 (or PI4 - PI7 = BCD tens digit).
11	PI8	Optional parallel binary bit input 8.
12	PI9	Optional parallel binary bit input 9.
13	PI10	Optional parallel binary bit input 10.
14	PI11	Optional parallel binary bit input 11 (or PI8 - PI11 = BCD hundreds digit).
15	PI12	Optional parallel binary bit input 12.
16	PI13	Optional parallel binary bit input 13.
17	PI14	Optional parallel binary bit input 14 (or PI12 - PI14 = BCD thousands digit).
18	PI15	Optional parallel sign bit input.
19	DCOM	Digital common.
20	DCOM	Digital common.

Descriptions apply to both **Application and **Microapplication** cards.*

CONNECTOR 18PL -- BETWEEN CONTROL CARD AND OPTIONAL PROGRAMMER CARD

Pin	Nomen-clature	Description
1	----	NO CONNECTION
2	----	NO CONNECTION
3	----	NO CONNECTION
4	----	NO CONNECTION
5	----	NO CONNECTION
6	----	NO CONNECTION
7	----	NO CONNECTION
8	----	NO CONNECTION
9	----	NO CONNECTION
10	----	NO CONNECTION
11	RXT	Input from the Programmer card serial link driver to Control card.
12	TXT	Output from the Control card to the Programmer card serial link receiver.
13	/DTRD	Data terminal ready - Output from Control card indicating that it is ready to receive a byte from the Programmer card. Signal goes low when ready, high when not ready.
14	/CTSD	Clear to send - Input from Programmer card signalling when it is ready to receive a byte. Signal low when ready, high when not ready.
15	RST	Reset line from Control card to the Programmer card, active high.
16	+5V	+5V regulated power supply to the Programmer card.
17	+5V	
18	DCOM	Signal and power return for the Programmer card.
19	DCOM	
20	-24V	-24V unregulated power supply to Programmer card.

TERMINAL BOARD 1TB LOCATED ON OPTIONAL PROCESS INTERFACE CARD *or* ENCODER PROCESS INTERFACE CARD

Pin	Nomen-clature	Description
1	TKN	Negative tachometer feedback signal. Maximum input is 310 VDC or 344 VAC.
2	--	NO CONNECTION - Provides electrical clearance.
3	TKP	Positive tachometer feedback signal.

TERMINAL BOARD LANTB LOCATED ON OPTIONAL MICROAPPLICATION CARD

Pin	Nomen-clature	Description
1	TX	Non-inverting transmitter output and receiver input in the half-duplex (LAN) mode. Transmitter output in the single-ended mode.
2	/RX	Inverting transmitter output and receiver input in the half-duplex (LAN) mode. For non-serial applications, may be used as a general purpose MUP input for application software. To use this mode, place jumpers JP16, JP18, JP19 in the 2-3 position and leave JP17 open.
3	--	NO CONNECTION - Used to ground the shield at one end of the link or to daisy chain the shield at intermediate drops.

ADDITIONAL CONNECTORS

CONNECTOR 1APL -- BETWEEN BASE DRIVE CARD AND PHASE A SHUNT AND TM1 (up to 125 HP)

Pin	Nomen-clature	Description
1	B1A	Red wire to TM1 base.
2	E1A	White wire to TM1 emitter.
3	S1A	Red wire to shunt 1.
4	S2A	White wire to shunt 1.

CONNECTOR 1APL -- BETWEEN BASE DRIVE CARD AND PHASE A SHUNT AND BDA CARD (150-300 HP)

Pin	Nomen-clature	Description
1	B1A	Red wire to Base Drive Amplifier (BDA) card (BDAPL).
2	E1A	White wire to Base Drive Amplifier card (BDAPL).
3	S1A	Red wire to shunt 1.
4	S2A	White wire to shunt 1.
5	VN1A	Black wire to Base Drive Amplifier card (NVU) (6V power supply).

CONNECTOR 1BPL -- BETWEEN BASE DRIVE CARD AND PHASE B SHUNT AND TM2 (up to 125 HP)

Pin	Nomen-clature	Description
1	B1B	Red wire to TM2 base.
2	E1B	White wire to TM2 emitter.
3	S1B	Red wire to shunt 2.
4	S2B	White wire to shunt 2.

CONNECTOR 1BPL -- BETWEEN BASE DRIVE CARD AND PHASE B SHUNT AND BDA CARD (150-300 HP)

Pin	Nomen-clature	Description
1	B1B	Red wire to Base Drive Amplifier (BDA) card (BDAPL).
2	E1B	White wire to Base Drive Amplifier card (BDAPL).
3	S1B	Red wire to shunt 2.
4	S2B	White wire to shunt 2.
5	VN1B	Black wire to Base Drive Amplifier card (NVU) (6V power supply).

CONNECTOR 1CPL -- BETWEEN BASE DRIVE CARD AND PHASE C SHUNT AND TM3 (up to 125 HP)

Pin	Nomen-clature	Description
1	B1C	Red wire to TM3 base.
2	E1C	White wire to TM3 emitter.
3	S1C	Red wire to shunt 3.
4	S2C	White wire to shunt 3.

CONNECTOR 1CPL -- BETWEEN BASE DRIVE CARD AND PHASE C SHUNT AND BDA CARD (150-300 HP)

Pin	Nomen-clature	Description
1	B1C	Red wire to Base Drive Amplifier (BDA) card (BDAPL).
2	E1C	White wire to TM3 Base Drive Amplifier card (BDAPL).
3	S1C	Red wire to shunt 3.
4	S2C	White wire to shunt 3.
5	VN1C	Black wire to Base Drive Amplifier card (NVU) (6V power supply).

CONNECTOR T1PL -- BETWEEN BASE DRIVE CARD AND POWER SUPPLY CARD

Pin	Nomen-clature	Description
1	TP1	+24V unregulated Power Supply card to Base Drive Card.
2	TP2	Switching transformer voltage from Power Supply card to Base Drive Card.
3	TS4	DCOM: Return for unregulated switching +5V from Base Drive Card to Power Supply card.
4	TS5	Switching +5V unregulated from Base Drive Card to Power Supply card.

CONNECTOR N2PL -- BETWEEN BASE DRIVE CARD AND POWER SUPPLY CARD

Pin	Nomen-clature	Description
1	TS9	Switching +6V from Power Supply card to Base Drive card.
2	TS8	Common for +/-6V.
3	TS7	Switching -6V from Power Supply card to Base Drive card.
4	--	NO CONNECTION

CONNECTOR 2APL -- BETWEEN BASE DRIVE CARD AND TM1 (up to 125 HP)

Pin	Nomen-clature	Description
1	B2A	Red wire to B1 of TM1.
2	E2A	White wire to E1 of TM1.

CONNECTOR 2APL -- BETWEEN BASE DRIVE CARD AND BDA CARD (150-300 HP)

Pin	Nomen-clature	Description
1	B2A	Red wire to Base Drive Amplifier (BDA) card (BDAPL).
2	E2A	White wire to Base Drive Amplifier card (BDAPL).
3	VN2A	Black wire to Base Drive Amplifier card (NVL) (6V power supply).

CONNECTOR 2BPL -- BETWEEN BASE DRIVE CARD AND TM2 (up to 125 HP)

Pin	Nomen-clature	Description
1	B2B	Red wire to B1 of TM2.
2	E2B	White wire to E1 of TM2.

CONNECTOR 2BPL -- BETWEEN BASE DRIVE CARD AND BDA CARD (150-300 HP)

Pin	Nomenclature	Description
1	B2B	Red wire to Base Drive Amplifier (BDA) card (BDAPL).
2	E2B	White wire to Base Drive Amplifier card (BDAPL).
3	VN2B	Black wire to Base Drive Amplifier card (NVL) (6V power supply).

CONNECTOR 2CPL -- BETWEEN BASE DRIVE CARD AND TM3 (up to 125 HP)

Pin	Nomenclature	Description
1	B2C	Red wire to B1 of TM3.
2	E2C	White wire to E1 of TM3.

CONNECTOR 2CPL -- BETWEEN BASE DRIVE CARD AND BDA CARD (150-300 HP)

Pin	Nomenclature	Description
1	B2C	Red wire to Base Drive Amplifier (BDA) card (BDAPL).
2	E2C	White wire to Base Drive Amplifier card (BDAPL).
3	VN2C	Black wire to Base Drive Amplifier card (NVL) (6V power supply).

CONNECTOR DBPL -- BETWEEN POWER SUPPLY CARD AND TM7 (OPTIONAL DB)

Pin	Nomenclature	Description
1	B1D	Red wire to base (B1) of TM7.
2	E1D	White wire to emitter (E1) of TM7.

NOTE: *For 150-300 HP controllers with no DB option, pin (2) to N2 buss is included.*

CONNECTOR C1PL -- BETWEEN POWER SUPPLY CARD AND SM1 (up to 125 HP)

Pin	Nomenclature	Description
1	G1	Red wire to gate of SM1 for regulation of DC Link.
2	N1	White wire to cathode of SM1 and negative DC Link.

CONNECTOR C1PL -- BETWEEN POWER SUPPLY CARD AND SM1 (150-300 HP)

Pin	Nomenclature	Description
1	G1	Red wire to gate of SM1 for charge control of DC Link.
2	N1	White wire to cathode of SM1 for charge control of DC Link.

CONNECTOR C2PL -- BETWEEN POWER SUPPLY CARD AND SM1 (150-300 HP)

Pin	Nomenclature	Description
1	G1	Red wire to gate of SM1 for charge control of DC Link.
2	N1	White wire to cathode of SM1 for charge control of DC Link.

CONNECTOR C3PL -- BETWEEN POWER SUPPLY CARD AND SM2 (250-300 HP)

Pin	Nomen-clature	Description
1	G1	Red wire to gate of SM2 for charge control of DC Link.
2	N1	White wire to cathode of SM2 for charge control of DC Link.

CONNECTOR CPTPL -- BETWEEN POWER SUPPLY CARD AND CONTROL POWER TRANSFORMER

Pin	Nomen-clature	Description
1	--	38 VAC, center tapped for controller power supplies.
2	--	38 VAC, center tapped for controller power supplies.
3	--	Center tap point for 38 VAC.
4	--	NO CONNECTION
5	--	115 VAC for cooling fan (motor side).
6	--	115 VAC for cooling fan (grounded side).

CONNECTOR 1FAPL -- BETWEEN POWER SUPPLY CARD AND CONTROLLER COOLING FAN

Pin	Nomen-clature	Description
1	X_1	115 VAC for controller cooling fan (motor side).
2	X_2	115 VAC for controller cooling fan (grounded side).

CONNECTOR 2FAPL -- BETWEEN POWER SUPPLY CARD AND CONTROLLER COOLING FAN

Pin	Nomen-clature	Description
1	X_1	115 VAC for controller cooling fan (motor side).
2	X_2	115 VAC for controller cooling fan (grounded side).

TERMINAL BLOCK CNTB -- BETWEEN POWER SUPPLY CARD AND 20TB TERMINAL STRIP (75-300HP, 460V or 40-150HP, 230V)

Pin	Nomen-clature	Description
1	MAP	N.O. Max (MA Auxilary) contact (hot fused at 1 amp maximum total).
2	X_2	115 VAC (grounded).
3	FX_1	115 VAC (hot fused at 1 amp maximum total).

NOTES:

6. STARTUP PROCEDURE

This startup procedure is meant as a general guide for the initial checkout and startup of an AC-300 drive. It references some general hardware and software checks relative to most basic drive applications. Refer to the sections on "Hardware Jumpers and Pot Adjustments" and "Key Parameter Adjustments" and the accompanying "Parameter Map" (Figure 8-1) for more complete information on adjustments available on the AC-300 drive. Refer to the Custom Software Description and Elementaries for information on wiring and adjustments specific to your drive application.

NOTE: *Prior to beginning actual drive startup, read this entire manual and be aware of all pertinent information. If an optional Programmer was supplied with the drive, review the "Use of Optional Programmer Module" section, "Diagnostic Mode for Drive Setup". Special diagnostic functions are available which will greatly aid drive startup.*

NOTE: *Review all* ***WARNING*** *and* ***CAUTION*** *notices at the beginning of this instruction book before proceeding.*

WARNING - BURN AND SHOCK HAZARD:

Always read the complete instructions prior to applying power or troubleshooting the equipment. Follow the start-up procedure step by step. Read and heed all WARNING, CAUTION, and NOTE labels posted on the equipment.

BEFORE APPLYING AC POWER

1. Verify that all incoming wiring is in agreement with the elementary drawings supplied with the drive, and conforms to approved wiring practices (wire size, separation of power from control, etc.). Verify that all electrical terminal connections are tight.

2. Verify that incoming 3-phase voltage is in accordance with nameplate requirements (+10%, -5%), unless otherwise specified.

3. Confirm that all jumper (JP) and switch positions (SW) are in accordance with the jumper selection chart supplied in the controller pocket and Custom Software Description.
   - Refer to the card layout diagrams (Figures 7-1 thru 7-9 for the location of jumpers and switches, and to TABLE 1 for descriptions/functions.

4. Verify that the motor has been properly installed and that the shaft is free to rotate.

5. Check the incoming AC power fuses, all DC Link fuses, and all control power fuses using an ohmmeter set to the "x 1" ohms scale.

6. If specific information on the motor and application were **not** submitted prior to shipment, complete the following **AC-300 Parameter Quick Calculation Chart**; otherwise, proceed to the Apply AC Power paragraph of this section.

AC-300 PARAMETER QUICK CALCULATION CHART

The following is a method of calculating the basic parameters to satisfactorily run an AC-300 controller for most simple applications. If detailed motor and application information was available prior to factory testing of the controller, then all parameters were precalculated and entered.

WARNING - BURN AND SHOCK HAZARD: If the drive system is to be operated above base frequency (60 Hz), the motor design must be reviewed with the motor manufacturer to verify that operation at these speeds is safe.

Record the following Motor Nameplate information:

Motor HP _____ Motor Amps _____
Motor Volts _____ Motor Hertz (at base frequency) _____
Motor RPM (at rated frequency) _____
Synchronous RPM (see NOTE) _____

NOTE: *For the synchronous RPM value, enter the next highest hundred RPM over Motor RPM. For example, use 1800 Synchronous RPM for a 1750 RPM Motor.*

Record the following necessary load information:

Maximum RPM required _____ Minimum RPM required _____
Acceleration Time (seconds) _____ Deceleration Time (seconds) _____
Controller Amps _____
(from controller nameplate output rating)

Calculate the following parameter values as shown from the information recorded above (round numbers to the nearest whole number):

Adr#	Mnemonic	Name/Formula	Value	
023	IEXSC =	900 ÷ (0.92 × Motor Hertz) =	_____	
033	HVBAS =	625 × Motor Hertz ÷ Motor Volts =	_____	(or 297 for 230 VAC)
034	LHVBA =	50 × Controller Amps ÷ Motor Amps =	_____	(or 25 for 230 VAC)
194	FLDPS =	32 × Motor Amps ÷ Controller Amps =	_____	
076	FLDMX =	50 × Motor Amps ÷ Controller Amps =	_____	
039	SLCMP =	31.5 × Controller Amps ÷ Motor Amps × (Synchronous RPM - Motor RPM) × (Motor Hertz ÷ Synchronous RPM) =	_____	
041	ILMMX =	Motor Current Limit 150 × Motor Amps ÷ Controller Amps =	_____	(Do not exceed 150 factor.)
040	ILGEN =	Regenerative Current Limit =	20 if DB Regeneration option not included.	
		138 × Motor Amps ÷ Controller Amps =	_____	If DB Regeneration option is included. (Do not exceed 138 factor.)
		17250 ÷ Controller HP =	_____	(for 150-300 HP)
079	CROSS =	0.92 × Motor Hertz =	_____	

Adr#	Mnemonic	Name/Formula	Value
081	FRQRG =	1,091,100 ÷ Motor Hertz ÷ Maximum RPM × Synchronous RPM =	_____
068	REFSF =	Maximum RPM =	_____
070	RPMSF =	Maximum RPM =	_____
071	REFLM =	1.01 × Maximum RPM =	_____
073	SPDLM =	1.1 × Maximum RPM =	_____
080	FRQLM =	Motor Hz × Maximum RPM ÷ Motor RPM =	_____
072	MINSP =	Minimum Speed 0.1 × Motor RPM =	_____ (normally)
057	ACELT =	Acceleration Time = *(0 to 9999 for 0 to 999.9 sec.; 10 = 1 sec.)*	_____
058	DCELT =	Deceleration Time = *(0 to 9999 for 0 to 999.9 sec.; 10 = 1 sec.)* **NOTE:** *If the DB option is not provided, the setting of DCELT should not be shorter than the motor coast time or a DC Link overvoltage trip will result.*	_____
196	WMETS =	Speed Meter Signal Scaling (3000 ÷ Motor Hz) × (Motor RPM ÷ Maximum RPM) =	_____

NOTE: *The parameters should be entered where called for in the "APPLY AC POWER" paragraph of this section.*

APPLY AC POWER

1. Verify that the Diagnostic LEDs are blinking one at a time from right to left or confirm that the programmer displays the word "OPERATE".

NOTE: *If at any time the LEDs remain locked, indicating a fault message, the Programmer displays a fault code, or either display is blank, refer to the "Troubleshooting" section for corrective procedures.*

2. Check for proper direction of rotation of all 3-phase blower motor fans (if equipped) in the power unit (75-300 HP) and on the AC motor(s).

3. Verify that the following power supply voltages appear at the appropriate Control Card test points by measuring from the supply test point to PCOM test point or by using the built-in DVM.

NOTE: *Using the DVM requires optional Programmer and Process Interface Card; refer to "Use of Optional Programmer Module" section.*

Test Point	Value
+24V	23.8 to 29.0 VDC
-24V	-23.8 to -29.0 VDC
+15V	14.25 to 15.75 VDC
-15V	-14.25 to -15.75 VDC
+5V	4.95 to 5.05 VDC

NOTE: *A Programmer is required for the following steps. For complete information on Programmer operation, refer to "Use of Optional Programmer Module", Section 11.*

4. Place the EEROM write protect device (JP13 on the Control card) in the "ENABLE" position (2-3) to allow programming. (The 1-2 position is the "PROTECT" position.)

CAUTION: Do not perform a "HARD RESET" or turn power "OFF" or "ON" with JP13 in the "ENABLE" (2-3) position.

NOTE: *In the following steps, the number or function encased in brackets [ ] indicates the Programmer key to be pressed.*

5. Access the Parameter Mode by pressing **[SET] [DRIVE#] [7] [7] [ENTER]** on the keypad.
   - -- The word "PARAMETER" should now be displayed and the black keypad functions are now usable.
   - -- To access a parameter address, press all three numbers on the keypad corresponding to the three-digit number of the desired address and the name of the address and its present value will be displayed.
   - -- To access a new parameter, press **[CLEAR/MODE]** and the three digits corresponding to the next parameter desired.

6. Verify that parameter address #003, DGNJP (ADR003-DGNJP) contains an odd number (1, 3, 9, etc.) by pressing **[0] [0] [3]** (must be an odd value to allow programming of new parameter values).
   - -- If this address contains an even number, add a "1" to the number and enter the new value.

     Example:
     ADR003 contains the number 16. Add 1 to obtain 17; press **[1] [7] [ENTER]**.
     - -- Press **[CLEAR/MODE]** and the three digits for the next desired parameter.

NOTE: *Confirm that the drive parameters agree with the "as shipped" parameter list inside the drive door or in the Custom Software Description prior to making any adjustments. Document any changes.*
*If parameters were not pre-calculated for the specific application, the **AC-300 Parameter Quick Calculation Chart** in the "Before Applying AC Power" paragraph should be completed, and those parameters entered into the controller before proceeding. Return JP13 to the 1-2 position after programming is finished.*

7. Enter the Diagnostic mode from the Parameter mode by pressing **[CLEAR/MODE]**.
   - -- "DIAGNOSTIC" chould appear.
   - -- The blue and black keypad functions will now control the drive.

NOTE: *The motor must be connected for the next step.*

WARNING - SHOCK AND POTENTIAL EQUIPMENT ROTATION HAZARD: The output starter (if supplied) will be energized and power applied to the motor in the next step; however, the motor should not rotate.

NOTE: *Refer to the "Use of Optional Programmer Module" section for complete information on the SCR test.*

8. Check the power section integrity by performing an SCR test (Diagnostic Test 12) by pressing **[TEST] [1] [2] [ENTER]** (display will show the same).
   - -- The output starter (if supplied) will be energized and power applied to the motor.
   - -- When test is completed (approx. 3 to 4 minutes), display will show either "PASSED" or a fault. If "PASSED" is displayed, proceed to the next step; if a fault is displayed, refer to the SCR Test instructions and the "Troubleshooting" section.

NOTE: *If a Programmer and Process Interface card are included in the drive, an on-board digital voltmeter (DVM) and electronic wire check are available to verify wiring and make the following adjustments. Refer to the "Use of Optional Programmer Module" section for instructions.*

9. Refer to the Custom Software Description of ADR000-REFJP and the controller elementaries to determinethe source of the reference(s) to the controller. Proceed per the following appropriate step(s) to adjust the reference(s).
   a. **Programmed Reference** - Verify that the settings of the reference parameters (if any) indicated by ADR000-REFJP are set correctly.
      - -- They may be changed by entering the Parameter mode, entering the parameter address, then entering the correct reference.

        Example: Run reference set by ADR061-RUNRF; ADR061 as shipped set at 3000 RPM and the desired reference is 2000 RPM.
        - -- Enter the Parameter mode and press **[0] [6] [1]** (Programmer now displays "RUNRF.3000")
        - -- Press **[2] [0] [0] [0] [ENTER]** (The reference is now set at 2000 RPM).

b. **Analog Reference** - There are three commonly used analog reference inputs to the drive: Manual System Reference (MSR) and Jog Reference (JOGR) through 3TB (and 6PL) to the Control card, and Auto System Reference (ASR) through 4TB (and 13PL) to the optional Application card.

NOTE: *Additionally, an analog current reference (process follower) may be brought into the optional Process Interface card. Adjustment of the process follower reference is described in TABLE 1 under Process Interface card.*

Adjustment of MSR (3TB65/6PL-5)

-- With the drive stopped, set the reference to MAXIMUM.
-- Adjust P1-REF SCALE on the Control card to obtain 4.0 VDC at test point SRS (Manual System Reference only; jumper JP1 on the Control card selects the manual system reference input voltage range).
-- Measure with a volt-ohmmeter (or the DVM diagnostic test) at the appropriate test point and PCOMX.
-- Reset the reference to 0% speed.

Adjustment of ASR (4TB23/13PL-23)

-- With the drive stopped, set the reference to MAXIMUM.
-- Adjust P1-ASR SCALE on the optional Application card to obtain 4.0 VDC at test point ASR also located on the optional Application card (Auto System Reference only; jumper JP1 on the optional Application card selects the reference input voltage range).
-- Measure with a volt-ohmmeter (or the DVM diagnostic test) at the appropriate test point and PCOMX.
-- Reset the reference to 0% speed.

Adjustment of JOGR (3TB63/6PL-7)

-- If an analog jog reference is supplied through JOGR, adjust the external source for the desired voltage between Control card test point JOGR and PCOMX on 3TB.

NOTE: *+/-14 VDC corresponds to full reference. This is programmable to 7 VDC = top speed for fast update rate (refer to Custom Software Description).*

-- Desired Jog Speed x 14 VDC/Top Speed = Volts at JOGR

Example: Top speed = 1800 RPM and desired JOG speed is 100 RPM.

-- 100 x 14 VDC/1800 = 0.78 VDC at JOGR
-- Adjust the external reference source to 0.78 VDC.

NOTE: *The following steps will use Diagnostic Run (Test 2-DGR) on the Programmer module to set-up the feedback(s) and load adjustment. Refer to the "Use of Optional Programmer Module", Section 11, Diagnostic Run before proceeding.*

NOTE: *If an output contactor or motor starter is used, a jumper or switch must be connected between 3TB52 (4PL-7) and 3TB53 (4PL-8) to energize the contactor when required.*

WARNING - SHOCK AND EQUIPMENT ROTATION HAZARD:

The next step will apply power to the motor and cause it to rotate. Verify that the shaft is free to rotate. Do not bypass any protective contacts from the motor starter circuit (if used).

10. Calibrate feedback adjustment (if used) and motor operation as follows:
    a. With zero reference applied, press **[DGR]** and slowly increase the Diagnostic Run reference until the motor starts to turn at a steady, smooth speed.
    b. Verify that all output currents are balanced to within +/-10% of the average of these currents with a clamp-on ammeter or by measuring the AC voltage at test points CFA, CFB, and CFC on the Power Supply/Interface card. (At no load, motors larger than 1 HP normally draw approximately 307 Full Load Amps; motors less than 1 HP may be higher.)
    c. Slowly increase reference to full speed and verify stable motor and load operation (refer to Troubleshooting section if problems are experienced).
    d. Reduce reference to zero and press **[STOP]** on the Programmer module.

11. Calibrate motor speed feedback for analog tach feedback setting using Diagnostic Run as follows (if no tach or other feedback device is used, proceed to step 12):
    a. Check that JP8 and SW1 on the Process Interface card are set properly for the type of tach and maximum tach volts at top speed (refer to TABLE 1 for proper setting).
    b. With the drive stopped, adjust P7-SFB ZERO on the Process Interface card for 0 VDC at test point SFB (also located on the Process Interface card).
    c. Start the drive by pressing **[DGR]** on the Programmer module and increase the reference to maximum.
        -- Check for 4 VDC at test point SFB and adjust, if necessary, with P1-TACH SCALE.
        -- Check actual motor speed with a hand-held tach and verify the proper tach output volts.
    d. After top speed has been calibrated, proceed to step 12 with the motor still running at top speed.

12. Scale the load current as follows (if necessary, stop the drive and connect the load if not already connected):
    a. Run the motor in Diagnostic Run at top speed and at as close to rated drive current as possible.
    b. Verify that the % Load readout on the Programmer module is approximately - *%Load = 100 x Actual RMS Current/Controller Nameplate Rated Current*
    c. Set all load meter outputs using IMET, P2 on Control card.
    d. Reduce the Diagnostic Run reference to zero and press **[STOP]**.

NOTE: *The speed reference referred to in the following steps is the **Normal Operating Reference**, not the Diagnostic Run Reference.*

13. Set the speed reference (speed pot, N/C reference, etc.) at its minimum setting and start the drive (press "START" on an operator station or call for a "RUN" from the control).

14. Ensure that all "STOP" circuits are functional.

NOTE: *"STOP" from an operator station (115 VAC START/STOP circuit - 3TB49, 50, 51) or removing a "RUN" input from 3TB8 will not function if the drive is started from the Programmer keypad.*

15. Restart the drive and slowly increase the speed reference until the motor starts to rotate.
    -- If the motor does not rotate, or rotates in the wrong direction, check the Diagnostic LEDs or the Programmer module display. If DC tach feedback is provided, a tach polarity fault may be indicated. Refer to the fault codes in the Troubleshooting section for corrective action.

WARNING - ELECTRICAL BURN AND SHOCK HAZARD: Exercise caution as high voltage may be present in the following steps.

NOTE: *If experiencing difficulty in making the adjustments in step 16, it may be necessary, due to the tolerance of the tach, to verify the actual tach output voltage/RPM. Check the tachometer's actual speed and output voltage with a hand-held tach and voltmeter to verify the proper SW1 setting on the Process Interface card. If no tach is used, SFB will always be 0 VDC in step 16.*

16. Start the drive and gradually increase speed reference to maximum.
    -- Check for 4 VDC at test points SRS and SFB on the Control and Process Interface cards. (If ASR is used, also check test point ASR on the Application card.)
    -- Adjust P1-TACH SCALE (requires optional Process Interface card) if necessary to reach the correct top speed (only on drives with an AC or DC tach).
    -- Confirm rated motor V/Hz.

17. Slowly decrease the speed and verify stability through the entire speed range.
    -- If the drive is not stable, refer to the Stability Adjustment procedure in the Troubleshooting section.

18. Confirm (after drive is stable throughout the entire speed range) that the motor current does not exceed the nameplate rating (with load).

Before continuing, it is recommended that the following be performed:

-- The list of "as shipped" parameter values provided with the controller be checked against the actual parameters in the controller.

NOTE: *Other parameters as sold and shipped with the drive can be accessed and changed as necessary (a complete list of pertinent parameters is contained in the Custom Software Description shipped with the drive). A partial listing of "Key" Parameters and their descriptions is located in the "Software Adjustments" section of this Instruction Book. Final responsibility for parameter settings resides with the customer. Document all changes to parameter settings in the space provided in the Custom Software Description.*

-- All incoming signals be checked for proper value and correct connection. Refer to the "Use of Programmer Module" section, Test 20 - Electronic Wire Check.

19. Change parameter address #003, DGNJP (ADR003-DGNJP) back to even number (2, 4, 8, etc.). (This must be an even value to prevent accidental programming of parameter values).
    -- Press **[0] [0] [3]**.
    -- If this address contains an odd number, subtract a "1" from the number and enter the new value.

    Example:
    ADR003 contains the number 17. Subtract 1 to obtain 16; press **[1] [6] [ENTER]**.
    -- Press **[CLEAR/MODE] [CLEAR/MODE] [CLEAR/MODE]** to exit the Parameter mode. ("OPERATE" should now appear on the display and the red keypad functions will now control the drive.)

20. Place the EEROM write protect device (JP13 on the Control card) back to the "PROTECT" position (1-2) to prevent accidental programming. (The 2-3 position is the "ENABLE" position.)

THE CONTROLLER IS NOW READY FOR OPERATION.

NOTES:

7. HARDWARE ADJUSTMENTS

The following section details adjustments available in hardware and software. The hardware portion describes the location and function of all jumpers and potentiometers on the AC-300 controller cards. The software portion describes key software "parameter" adjustments available and provides a "Parameter Map" (Figure 8-1) that relates the adjustments to electrical hardware equivalents.

HARDWARE JUMPER AND POT ADJUSTMENTS

Most of the jumper selections and pot adjustments have been factory set. When one of the cards has to be changed, make sure the replacement card switches, jumpers, and pots are set to be the same as on the replaced card. For a full description of the jumper and pot adjustments, see TABLE 1.

NOTE: *If the Control card is replaced, the EEROM (U12) from the original card should be put into the replacement card. If the same failure symptoms still exist, the new EEROM, shipped with the replacement card, can be reinstalled but must be reprogrammed per the customer software jumper and adjustment values supplied in the Custom Software Description or parameter list in the drive door pocket.*

Hardware jumpers and/or pots are located on the Control card, Power Supply card, Base Driver card, optional Process Interface card, optional Encoder Process Interface card, optional Application card, optional Microapplication card, and optional Programmer card. Figures 7-1 thru 7-9 show the locations of pots, jumpers, test points and other key components of these cards.

Most of the pots and jumpers are used in the interface circuits of the microprocessor software control circuit. Some of the jumpers are only for manufacturer's use. The test data sheets supplied with each controller (in the drive door pocket) indicate the position of these adjustments as they were set in the factory. Adjustments not set in the factory can easily be made by the startup procedure in this book (or in the Startup Guide, GEJ-7152) and by use of TABLE 1.

NOTE: *Only high impedance digital voltmeters or the optional "on-board" DVM (Test 3) should be used to make indicated measurements.*

NOTE: *Before changing any jumper positions, compare actual jumper positions with "as shipped" jumper positions through use of the Custom Software Description.*

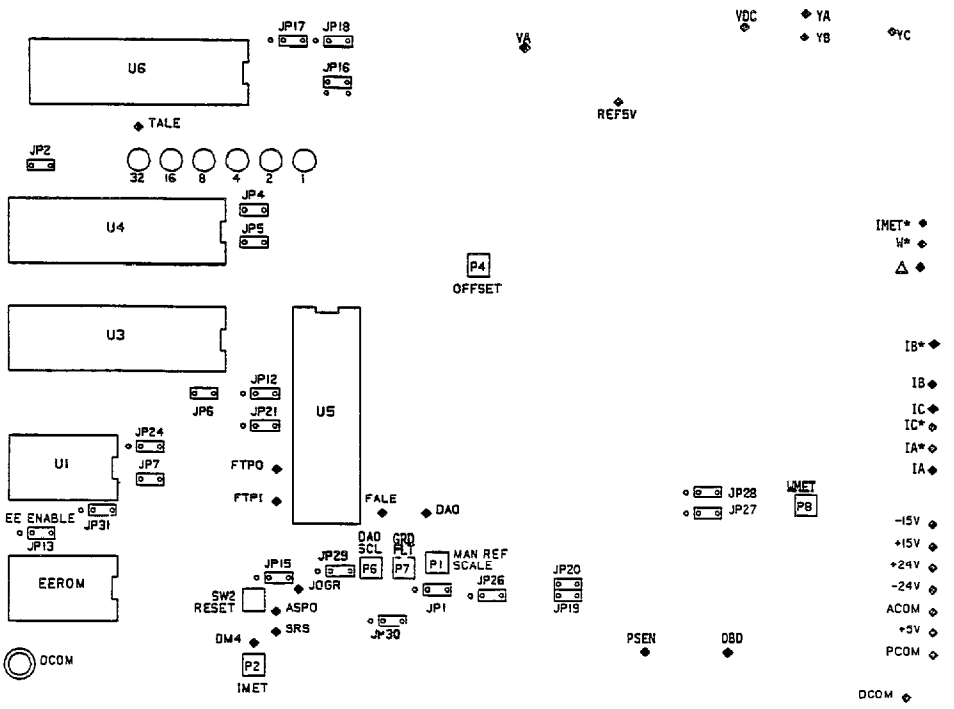


Figure 7- 1. Control Card Layout Diagram

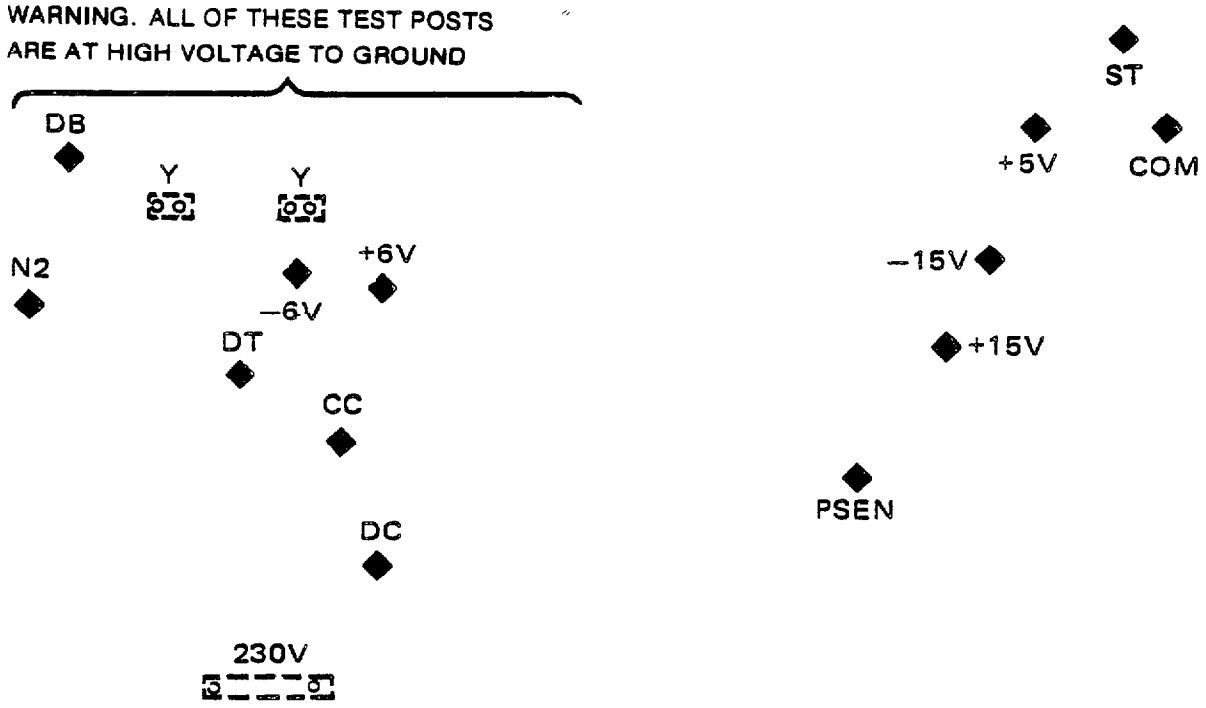


Figure 7- 2. Power Supply Card Layout Diagram

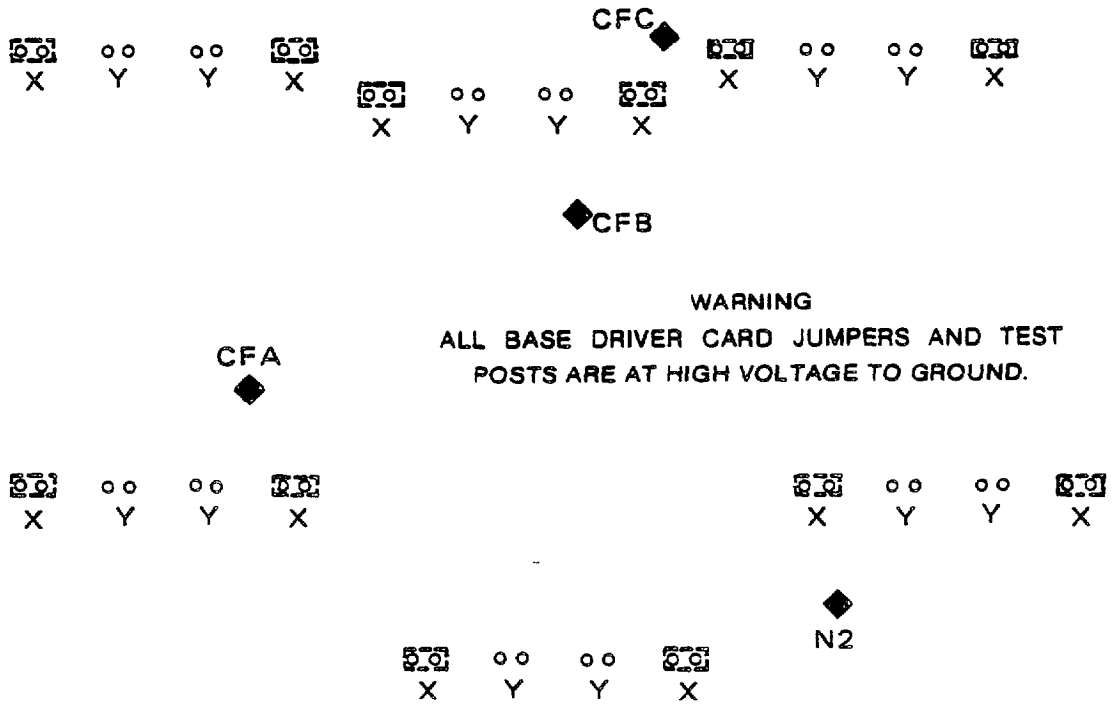


Figure 7-3. Base Drive Card Layout Diagram

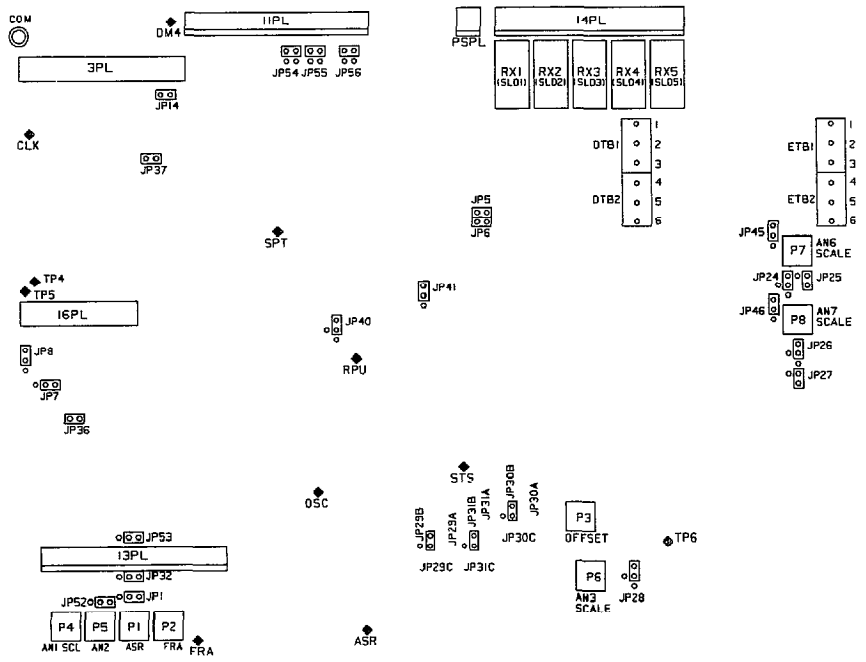


Figure 7-4. Optional Application Card Layout Diagram

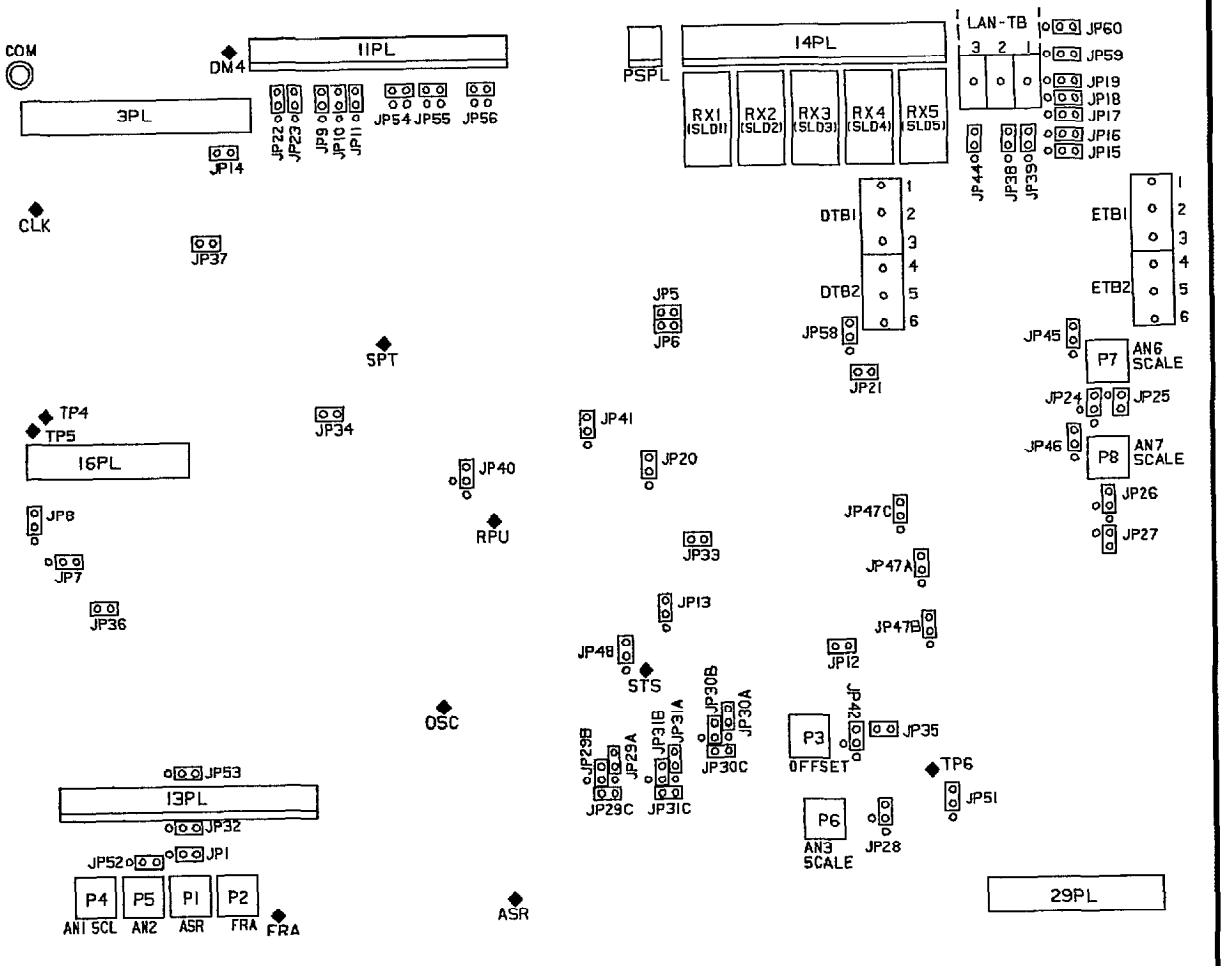


Figure 7-5. Optional Microapplication Card Layout Diagram

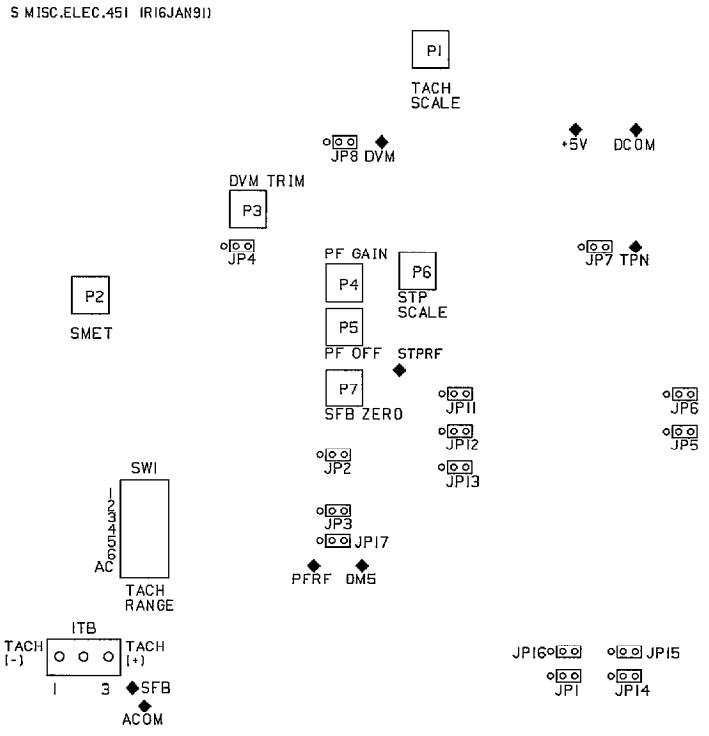


Figure 7-6. Optional Process Interface Card Layout Diagram

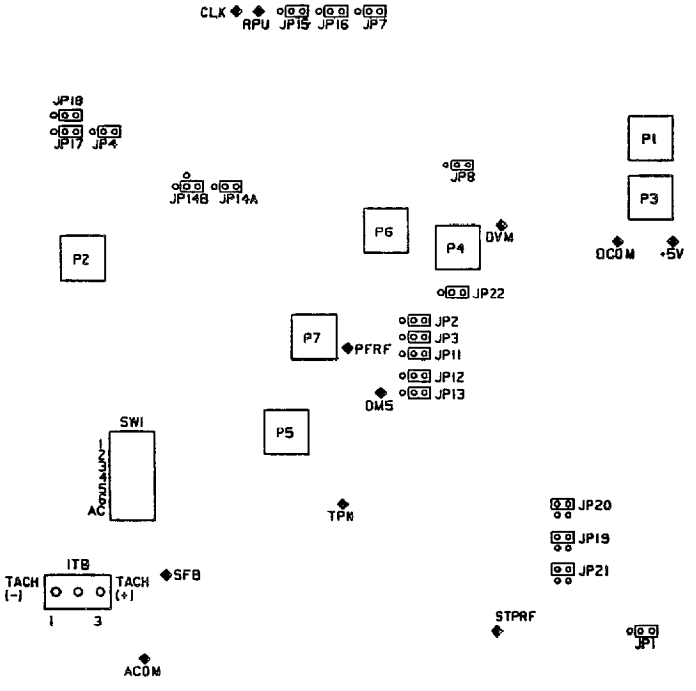


Figure 7-7. Optional Encoder Process Interface Card Layout Diagram

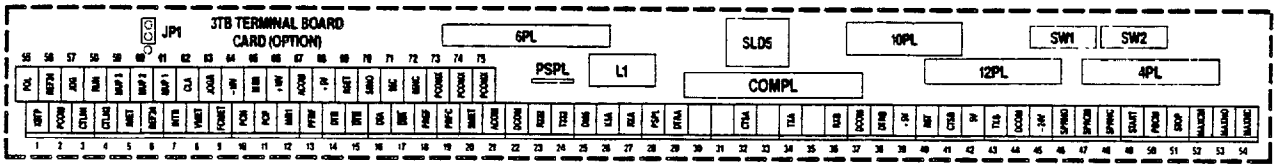


Figure 7-8. Optional 3TB Terminal Board Card Layout Diagram

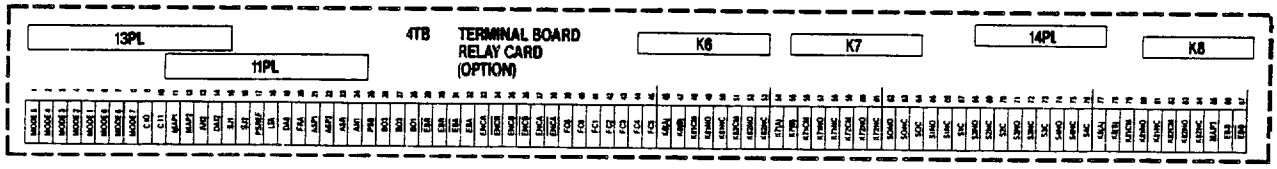


Figure 7-9. Optional 4TB Terminal Board Card Layout Diagram

TABLE 1: JUMPER AND POT ADJUSTMENTS

Location	Jumper or Pot	Number	Nomenclature	Description
Control Card	Pot	P1	MAN REF	To adjust test point SRS to four (4) volts when top speed reference is at MSR, 3TB65 (6PL-5; connector 6PL, pin number 5). Used in conjunction with jumper JP1.
		P2	IMET	This pot is used to adjust the output of the analog load meter circuit. Adjust the output voltage to be from 1 to 8 volts at rated load. Maximum current load on this signal is 1mA. (Also see JP15.)
		P4	OFFSET	Adjusted by the manufacturer, no need to be adjusted in the field unless directed to do so by fault codes. Refer to adjustment procedure in "Use of the Optional Programmer Module" section if a Programmer is available.
		P6	DA0 SCL	This pot is used to adjust the gain of the 12-bit digital-to-analog converter (D/A) output from the Control card. Full CCW sets the gain at 0.25, full CW sets the gain at 1.25. The output saturates at 12 VDC.
		P7	GRD FLT	This pot is used to adjust the level of ground fault current for FL26-ITRIP shutdown. Full CCW provides trip at minimum current levels. Full CW disables the ground fault protection.
		P8	WMET	This pot is used to adjust the output of the analog frequency meter circuit. Output is adjustable from 5 to 10 volts at rated voltage. Maximum current load on this signal is 1mA.
	Jumper	JP1	MAN REF	This jumper selects the maximum voltage range for the manual speed reference input. **1-2:** 8 to 15 volts **2-3:** 5 to 8 volts
		JP2 JP4 JP5 JP6 JP7		JP2 - JP7 should be set as follows: **1-2:** Set in this position for proper operation.

Location	Jumper or Pot	Number	Nomenclature	Description
Control Card (continued)	Jumper	JP12		Allows Transistor test without Programmer **1-2:** Disables test (normal operation). **2-3:** Place in this position for test before power-up. A test will be executed the first time a RUN is commanded. Subsequent RUN commands will execute normally. (Not used with elevators.)
		JP13		This jumper provides a hardware write protection for the software jumpers and adjustments. **1-2:** Protected (should be in this position for maximum security and reliability of the drive). **2-3:** Enables reprogramming. Do not perform a "hard" reset or turn power on or off in this position.
		JP15		This jumper selects whether the (IMET) load meter driver output is to be bipolar or absoluted. (Also see P2.) **1-2:** Absoluted (always positive) **2-3:** Bipolar (zero center)
		JP16 JP17 JP18		JP16, JP17, JP18 should be set as follows: **Set by factory** - refer to jumper listing in Custom Instruction Book or drive door pocket listing for proper setting.
		JP19 JP20		JP19, JP20 set the polarity of RUN, JOG, POL, and XSTOP. **1-2 & 3-4:** Inputs are **active** when pulled down between -24 to 0.4 VDC. Inputs are **inactive** when open or tied to +3.5 to +24 VDC. **1-3 & 2-4:** Inputs are **active** when tied between +3.5 to +24 VDC. Inputs are **inactive** when open or held at -24 to 1.5 VDC.
		JP21		Allows interactive card test. **1-2:** Disables card test (normal operation). **2-3:** Initiates card test if drive is not running. To exit card test, move JP21 back to the 1-2 position and hard reset the drive (push SW2).
		JP24 JP26 JP27 JP28 JP30 JP31		JP24, JP26, JP27, JP28, JP30, JP31 set as follows: **Set by factory** - refer to jumper listing in Custom Instruction Book or drive door pocket listing for proper setting.

Location	Jumper or Pot	Number	Nomenclature	Description
Control Card (continued)	Jumper	JP29		This jumper selects whether the D/A output has a fixed gain or adjustable gain. **1-2:** Fixed gain (see TABLE 4, RAM Variable Scaling). **2-3:** Gain adjustable by P6-DA0 SCL.
		JP32		Set in 1-2 position for proper operation.
	Switch	SW2	Hardware Reset	This switch causes a hardware reset ("hard" reset) of the controller when pressed. Used to clear some drive faults (refer to TABLE 6, Specific Fault Troubleshooting). The power supplies are toggled "OFF" and "ON" and the drive will return to the initial power up condition.
Power Supply Card	Jumper	JP1		Undervoltage trip detection **1-2:** For 460 or 230 VAC supply **2-3:** For 380 or 208 VAC supply
		JP2		Installed for all HP ratings
		JP3 JP4		JP3 and JP4 installed for: 40 to 300 HP on 460V controllers 20 to 150 HP on 230V controllers
		JP	230V	The JP 230V is required for operating the controller from 230V AC line voltage. Do not use 230V jumper for 460V controllers.
3TB Terminal Board	Jumper	JP1		JP1 selects whether optional SLD5 mounted on 3TB is controlled by the Control card or by an external contact closure. **1-2:** Control card control of SLD5, programmed by ADR085, 086 and 087. **2-3:** External control of SLD5 by connecting 3TB26, K5A to 3TB56, REF24.
	Switch	SW1 SW2		SW1 and SW2 select the mode of serial link communication through COMPL on 3TB (depress switch as shown by dot to set for up or down position). For use with Data Terminal Equipment (DTE): **Switch** **Set Position** SW1-1,2, 3: Up SW1-4: Down SW2-1, 3, 4: Down SW2-2: Up For use with Data Communication Equipment (DCE), set all switches in **down** position.

Location	Jumper or Pot	Number	Nomenclature	Description
Base Drive Card	Jumper	JPX (Qty 12) JPY (Qty 12)		**No JPX or JPY jumpers connected:** Programs proper base drive to power transistors for 2 - 15 HP, 460 VAC (or 1 - 7.5 HP, 230 VAC) controllers. **All JPX jumpers connected and No JPY jumpers connected:** Programs proper base drive for 20 - 30 HP, 460 VAC (or 10 - 15 HP, 230 VAC) controllers. **All JPX and JPY jumpers connected:** Programs proper base drive for 40 - 300 HP, 460 VAC (or 20 - 150 HP, 230 VAC) controllers.
Process Interface Card (optional)	Switch	SW1	Tach Range Select	These switches provide coarse scaling of the tach circuit as shown in the following chart. DIP switch positions are selected for the legend of the side depressed. To select position 3 ON, depress the switch at position 3 next to the ON legend.
	Pot	P1	TACH SCALE	Used to scale the tach circuit to obtain 4.0 volts at test point SFB at maximum speed.

SW1 Position No. "Closed" or "ON"	Tach Volt At Maximum Speed DC Tach	AC Tach
All "open" or "off"	10 - 12.5V	11.1 - 13.9V
1	12.5 - 15.5V	13.9 - 17.2V
2	15.5 - 19V	17.2 - 21.1V
1, 2	19 - 22V	21.1 - 24.4V
3	22 - 25V	24.4 - 27.8V
1, 3	25 - 30V	27.8 - 33.3V
1, 2, 3	30 - 37.5V	33.3 - 41.7V
4	37.5 - 46V	41.7 - 51.1V
1, 2, 4	46 - 56V	51.1 - 62.2V
1, 2, 3, 4	56 - 70V	62.2 - 77.8V
5	70 - 87V	77.8 - 96.7V
2, 3, 5	87 - 105V	96.7 - 117V
1, 2, 4, 5	105 - 120V	117 - 133V
1, 2, 3, 4, 5	120 - 144V	133 - 160V
6	144 - 180V	160 - 200V
1, 2, 4, 6	180 - 205V	200 - 228V
5, 6	205 - 250V	228 - 278V
1, 2, 3, 4, 5, 6	250 - 310V	278 - 344V

NOTE: *SW1-7 must be OFF (or open) for DC tachs or ON (or closed) for AC tachs. Maximum voltages less than 10 volts are not to be used.*

Location	Jumper or Pot	Num-ber	Nomen-clature	Description
Process Interface Card (optional) (continued)	Pot	P2	SMET	Used to adjust the analog output signal at 10PL-9 of the speed metering circuit from 4 to 10 volts at maximum motor speed. Maximum current load on this signal is 2mA. (Also see JP4.)
		P3	DVM	With DVM test point connected to +5V and optional Programmer set to DVM operation, adjust P3 until the Programmer displays 5.00.
		P4 P5	PF GAIN PF OFF	These pots allow trimming the gain and offset of the circuit that provides up to 10 volts from 1 to 5, 4 to 20, 5 to 50, or 5 to 65mA transducer inputs. To adjust, start with PF gain at the full CCW position and proceed as follows: -- Apply the current reference corresponding to minimum speed. -- Adjust PF OFF for 0 volts at test point PFRF. -- Adjust the current reference corresponding to maximum speed. -- Adjust PF GAIN for 10.0 volts at PFRF. Repeat as necessary **NOTE:** *To prevent reversing when the current reference is less than the minimum current, the drive should be set up in a non-reversing mode.*
		P6	STP SCALE	Set Point Scale - Used to adjust scaling of definable input at test point STPRF. Refer to elementaries and engineering notes if used.
		P7	SFB ZERO	This pot adjusts test point SFB to be zero with zero tach feedback. Set speed reference to zero (jumper MSR or ASR to PCOMX if used) and adjust P7 for zero motor speed when a run is commanded.
	Jumper	JP1		JP1 selects circuitry for the serial link communication mode. **1-2:** Enables RS422 Reception. **2-3:** Enables RS232C Reception.
		JP2 JP3		JP2 and JP3 select the process follower mode per the following table: **JP2** **JP3** **Value** **1-2:** **1-2:** 5-65 mA current loop follower. **1-2:** **2-3:** 10-50 mA current loop follower. **2-3:** **1-2:** 4-20 mA current loop follower. **2-3:** **2-3:** 1-5 mA current loop follower or 10V voltage follower.

Location	Jumper or Pot	Number	Nomenclature	Description
Process Interface Card (optional) (continued)	Jumper	JP4		JP4 selects the speed meter (SMET) circuit mode. (Also see P2.) **1-2:** Unipolar speed meter driver mode. **2-3:** Bipolar meter drive mode (DC tach only).
		JP5		JP5 selects digital tach quadrature mode **1-2:** Quadruples frequency of digital tach input (only when JP6 is in doubler mode). **2-3:** Disables quadrature tach mode.
		JP6		JP6 selects digital tach doubler mode **1-2:** Doubles frequency of digital tach input at 3TB16. **2-3:** Disables doubler tach mode.
		JP7		JP7 selects type of digital tach. **1-2:** AN tach linearization mode or reluctance digital tach. **2-3:** Square wave digital tach.
		JP8		JP8 selects analog tach filter mode. **1-2:** Enables 4MS filter in analog tach feedback (required for AC tach operation). **2-3:** Disables 4MS filter - Required for speed regulator performance above 20 radians (DC or Digital Tach only).
		JP9 JP10		JP9, JP10 are the digital tach or encoder mode selects **1-2:** Differential RS422 compatible with terminating resistor. **2-3:** Single-ended input at 10PL-5 and 10PL-7. **Open:** Differential operation with no terminating resistor. **NOTE:** *Jumpers JP9 and JP10 have been replaced by jumpers JP15 and JP16 on later version cards.*
		JP11		TXD Mode Select **1-2:** 12PL-9, 12PL-12 RS422 transmitted data mode. **2-3:** 12PL-9 RS232 transmitted data mode.
		JP12		DTR Mode Select **1-2:** 12PL-8, 12PL-13 RS422 DTR handshake mode. **2-3:** 12PL-8 RS232 handshake mode.
		JP13		DTR RS232 Logic Polarity **1-2:** 12PL-8 high = drive ready to receive **2-3:** 12PL-8 low = drive ready to receive

Location	Jumper or Pot	Number	Nomenclature	Description
Process Interface Card (optional) (continued)	Jumper	JP14		CTS Mode Select **1-2:** For RS422 handshaking **2-3:** For RS232 handshaking **NOTE:** *For Series Six interfaces using the CCM card, set JP1 and JP11 in the 1-2 position and JP12, JP13, and JP14 in the 2-3 position. 12PL-7 should be tied to +5V. (Not applicable to AAG1, ABG1, or ACG1 version cards.)*
		JP15		DTA - /DTA maximum input voltage select (3TB16, 17) (See **NOTE** for JP16.) **3-4:** 1.5 VDC **2-3:** 5.0 VDC **1-2:** 10 VDC **1-4:** 15 VDC
		JP16		DTB - /DTB maximum input voltage select (3TB14, 15) **3-4:** 1.5 VDC **2-3:** 5.0 VDC **1-2:** 10 VDC **1-4:** 15 VDC **NOTE:** *When replacing **"AJ"** or earlier version card with **"AK"** or later version card, there is **not** a complete circuit for single-ended tachometers. With inputs at DTA (3TB16) and DTB (3TB14) only, /DTA (3TB17) and /DTB (3TB15) **must** be connected to PSPL (3TB28) to complete the circuit through the optical isolator LED. Also, common and +15V of tachometer should be twisted pair wiring routed through balun core with common tied to PSPL.*
		JP17		Voltage Follower Enable **1-2:** Voltage Follower Disabled **2-3:** Voltage Follower Enabled

Location	Jumper or Pot	Num-ber	Nomen-clature	Description
Application Card (optional)	Pot	P1	ASR SCALE	P1 is used in conjunction with JP1 to trim test point ASR to four (4) volts for top speed reference applied to 13PL-23 (4TB23).
		P2	FRA SCALE	P2 is used to scale the field reference adjust input at 13PL-20 (4TB20). One volt amplitude at test point FRA increases/decreases the FLDMX parameter setting by 204 counts.
		P7	AN6 SCALE	P7 is used to provide a variable input load for general purpose amplifier #1. Refer to elementaries and operating notes for further information, if used (version APM card, Figure 7-4).
		P8	AN7 SCALE	P8 is used to provide a variable input load for general purpose amplifier #2. Refer to elementaries and operating notes for further information, if used (version APM card, Figure 7-4).
	Jumper	JP1		JP1 selects the voltage range for the auto speed reference input (ASR) at 4TB23 (13PL-23). **1-2:** 8 to 15 volts. **2-3:** 5 to 8 volts.
		JP2		JP2 selects whether the encoder Track A input at 11PL-16 (4TB37) is single-ended or differential. **1-2:** RS422 Differential **2-3:** TTL compatible single-ended **NOTE:** *Jumpers JP2, JP3, and JP4 have been replaced by jumpers JP54, JP55, and JP56 on later version cards.*
		JP3		JP3 selects whether the encoder Track B input at 11PL-14 (4TB35) is single-ended or differential. **1-2:** RS422 Differential. **2-3:** TTL compatible single-ended. **NOTE:** *Jumpers JP2, JP3, and JP4 have been replaced by jumpers JP54, JP55, and JP56 on later version cards.*
		JP4		JP2 selects whether the encoder marker track input at 11PL-12 (4TB33) is single-ended or differential. **1-2:** RS422 Differential. **2-3:** TTL compatible single-ended. **NOTE:** *Jumpers JP2, JP3, and JP4 have been replaced by jumpers JP54, JP55, and JP56 on later version cards.*

Location	Jumper or Pot	Number	Nomenclature	Description
Application Card (optional) (continued)	Jumper	JP5 JP6		JP5 and JP6 select polarity of the eight logic inputs, MODE 0-MODE 7, at 13PL-1 thru 8 (4TB1 thru 8). **1-2 & 3-4:** The control is active low. Inputs are active when pulled below 0.8 volts. **1-3 & 2-4:** The control is active high. Inputs are active when pulled above 3.5 volts or driven by TTL high output.
		JP7		JP7 selects whether the sixteen-bit parallel input is latched on the high or low level of the strobe. **1-2:** Data latched on low to high transition of PSB. **2-3:** Data latched on low level of PSB.
		JP8		JP8 selects whether the sixteen-bit parallel input and strobe signals are driven by an active high or low driver. **1-2:** Lines internally pulled up to +5V. Must be driven by active pull-down to common driver. **2-3:** Lines internally pulled down to DCOM. Must be driven by an active pull-up (above 3.5 volt) drive.
		JP14		JP14 set as follows: **1-2:** Should be in this position for proper operation. **Open:** Only for manufacturer's use.
		JP24, JP25 JP26, JP27 JP28, JP29 JP30, JP31 JP29A, 29B, 29C JP30A, 30B, 30C JP31A, 31B, 31C		JP24 thru JP31 define functions for the on-board general purpose amplifiers (GP AMP #1 and GP AMP #2). Refer to drive elementaries, operating notes and Custom Software Description for more information (if used). **NOTE:** *Jumpers JP29, JP30, JP31 have been replaced by jumpers JP29A, JP29B, JP29C, and JP30A, JP30B, JP30C, and JP31A, JP31B, JP31C on later version cards.*
		JP32		JP32 selects the scaling range of P2-FRA SCALE. **1-2:** Normal range. **2-3:** Extended range.
		JP36 JP37		JP36 & JP37 set as follows: **1-2:** Should be in this position for proper operation. **Open:** Only for manufacturer's use.
		JP40 JP41		JP40 & JP41 select whether the encoder input is used as a feedback or a pulse reference. **JP40 JP41** **1-2: 1-2:** Normal encoder operation. **2-3: 1-2:** Enables encoder track A as a pulse reference frequency input at 3TB37 (11PL-16, ENCA), 40 kHz maximum. **2-4: --** Not used.

Location	Jumper or Pot	Number	Nomenclature	Description
Application Card (optional) (continued)	Jumper	JP45		JP45 selects whether P7-AN6 SCALE is in or out of the GP AMP #1 circuit: **1-2:** Connects P7 into amplifier circuit. **2-3:** Disconnects P7 from amplifier circuit.
		JP46		JP46 selects whether P8-AN7 SCALE is in or out of the GP AMP #2 circuit: **1-2:** Connects P8 into amplifier circuit. **2-3:** Disconnects P8 from amplifier circuit.
		JP54		Jumper sets max input voltage to optical isolators from ENCR (11PL-12, 4TB33-34). **1-2:** 10 volts **2-3:** 5 volts **3-4:** 1.5 volts **Open:** 15 volts
		JP55		Jumper sets max input voltage to optical isolators from ENCB (11PL-14, 4TB35-36). **1-2:** 10 volts **2-3:** 5 volts **3-4:** 1.5 volts **Open:** 15 volts
		JP56		Jumper sets max voltage input to optional isolators from ENCA (11PL-16, 4TB37-38). **1-2:** 10 volts **2-3:** 5 volts **3-4:** 1.5 volts **Open:** 15 volts
Programmer Card (optional)	Jumper	JP1 JP3 JP4 JP5		The necessary jumper positions are primarily dictated by the connector used: **CONNECTOR ON CARD JP1 JP3 JP4 JP5** 18PL Control **1-2: 1-2: 1-2: 2-3:** 12PL Proc. Inter. **1-2: 2-3: 2-3: 2-3:** **NOTE:** *Older version programmers may have JP2 instead of JP5. Set JP2 in 1-2 position when provided.*
Micro-Application Card (optional)	Pot	P1	ASR SCALE	P1 is used in conjunction with JP1 to trim test point ASR to four (4) volts for top speed reference applied to 13PL-23 (4TB23).
		P2	FRA SCALE	P2 is used to scale the field reference adjust input at 13PL-20 (4TB20). One volt amplitude at test point FRA increases/decreases the FLDMX parameter setting by 204 counts.

Location	Jumper or Pot	Number	Nomenclature	Description
Micro-Application Card (optional) (continued)	Pot	P3	OFFSET	P3 is used to trim the zero offset of the MUP bus A-D converter.
		P4	AN1 SCALE	P4 is used to scale the analog input AN1 at 13PL-24. Gain of this input is 0 bits/volt in the full CCW position and 204.8 bits/volt in the full CW position. Full scale 12-bit conversion is +2047/-2048 bits.
		P5	AN2 SCALE	P5 is used to scale the analog input AN2 at 13PL-13. Gain of this input is 0 bits/volt in the full CCW position and 204.8 bits/volt in the full CW position. Full scale 12-bit conversion is +2047/-2048 bits.
		P6	AN3 SCALE	P6 is used to scale the analog input AN3 that is available through MAP3 (11PL-3). Gain of this input is 0 bits/volt in the full CCW position and 204.8 bits/volt in the full CW position. Full scale 12-bit conversion is +2047/-2048 bits. (See JP28 and JP29 also.)
		P7	AN6 SCALE	P7 is used to provide a variable input load for general purpose amplifier #1. Refer to elementaries and operating notes for further information.
		P8	AN7 SCALE	P8 is used to provide a variable input load for general purpose amplifier #2. Refer to elementaries and operating notes for further information.
	Jumper	JP1		Selects the voltage range for the auto speed reference input. **1-2:** (Def) 8 to 18 VDC. **2-3:** 5 to 8 VDC.
		JP2		Selects whether the shared bus encoder track A input at 11PL-16 is single-ended or differential. **NOTE:** *This jumper has been replaced by jumper JP54 on later version cards.* **1-2:** RS422 differential. **2-3:** (Def) TTL compatible single-ended.
		JP3		Selects whether the shared bus encoder track B input at 11PL-14 is single-ended or differential. **NOTE:** *This jumper has been replaced by jumper JP55 on later version cards.* **1-2:** RS422 differential. **2-3:** (Def) TTL compatible single-ended.

Location	Jumper or Pot	Number	Nomenclature	Description
Micro-Application Card (optional) (continued)	Jumper	JP4		Selects whether the shared bus encoder marker track input at 11PL-12 is single-ended or differential. **NOTE:** *This jumper has been replaced by jumper JP56 on later version cards.* **1-2:** RS422 differential. **2-3:** (Def) TTL compatible single-ended.
		JP5 JP6		These select whether the 13 control inputs at MODE0 - MODE7 and CI0 - CI4 are active high or active low. **JP5** **JP6** **Level** **3-4:** **1-2:** Control is active low; activated when the input is pulled below 0.8 VDC. **2-4:** **1-3:** Control is active high; activated when the input is pulled above 3.5 VDC or driven by a TTL high output.
		JP7		Selects whether the 16-bit parallel input is latched on the rising or falling edge of the strobe. **1-2:** Data input latched on falling edge of PSB. **2-3:** Data input latched on low to high transition of PSB.
		JP8		Selects whether the 16-bit parallel input and strobe signals are driven by an active high or low driver. **1-2:** Lines internally pulled up to +5 VDC. Must be driven by an active pull-down to common driver. **2-3:** Lines internally pulled down to DCOM. Must be driven by an active pull-up above 3.5 VDC driver.
		JP9		EBR Input Jumper - Selects whether the MUP encoder marker track input (or external interrupt) at 11PL-1 is single-ended or differential. **1-2:** RS422 compatible differential with 330 ohm terminating resistor. **2-3:** TTL compatible single-ended. **Open:** Differential with no terminating resistor.
		JP10		EBA Input Jumper - Selects whether the MUP encoder track A input at 11PL-8 is single-ended or differential. **1-2:** RS422 compatible differential with 330 ohm terminating resistor. **2-3:** TTL compatible single-ended. **Open:** Differential with no terminating resistor.

Location	Jumper or Pot	Number	Nomenclature	Description
Micro-Application Card (optional) (continued)	Jumper	JP11		EBB Input Jumper - Selects whether the MUP encoder track B input at 11PL-10 is single-ended or differential. **1-2:** RS422 compatible differential with 330 ohm terminating resistor. **2-3:** TTL compatible single-ended. **Open:** Differential with no terminating resistor.
		JP12		MUP Oscillator Enable **1-2:** (Def) Always (normal operation) **Open:** Never (manufacturing test only)
		JP13		Allows selection of source of interrupt INTO for MUP. **1-2:** LAN serial communication **2-3:** (Def) MUP encoder marker pulse detection
		JP14		U28 and U29 Enable **1-2:** (Def) Always (normal operation) **Open:** Never (manufacturing test only)
		JP15		Controls mode of serial driver. **1-2:** Passive pull-up for multidrop LAN communications. **2-3:** Not Used.
		JP16		Serial Mode Select #1 - Partially controls serial link modes. **1-2:** Connects TX to RX for half-duplex differential operation at LANTB; used for LAN communications. **2-3:** Connects RX to JP19-3 for full-duplex single-ended operation at LANTB or for inhibiting serial communication for for use as a non-serial input.
		JP17		Serial Mode Select #2 - Partially controls serial link modes. **1-2:** Connects /TX to /RX for half-duplex differential operation at LANTB; used for LAN communications. **2-3:** Connects /RX to common to inhibit serial communication. **Open:** Allows full-duplex single-ended or non-serial operation at LANTB.
		JP18		Serial Mode Select #3 - Partially controls serial link termination. **1-2:** Connects terminating resistor (for first and last drops of differential LAN system). **2-3:** Disconnects terminating resistor.

Location	Jumper or Pot	Number	Nomenclature	Description
Micro-Application Card (optional) (continued)	Jumper	JP19		Serial Mode Select #4 - Partially controls serial link termination. **1-2:** Connects terminating resistor (for first and last drops of differential LAN system). **2-3:** Disconnects terminating resistor. Connects JP16-3 to 1.8 VDC bias for single-ended operation.
		JP20		Selects type of EPROM for MUP. **1-2:** (Def) 128K EPROM on Microapplication card. **2-3:** 256K or 512K EPROM on Microapplication card. **NOTE:** *128K EPROM: U33=409X106AB106 through AD106 and 409X111AA106 through AC106. 256K EPROM: U33=409X113_ _ _ _ _ or 409X114_ _ _ _ _.*
		JP21		MUP RAM Enable **1-2:** Always (normal operation). **Open:** Never (manufacturing test only).
		JP22		11PL-1 Select - Allows multiple use of 11PL-1 input. **1-2:** Encoder marker or external interrupt non-inverting input. **2-3:** Bring in LAN-TX line via 11PL (alternate LANTB-1). **NOTE:** *JP38 must also be in 2-3 position.*
		JP23		11PL-2 Select - Allows multiple use of 11PL-2 input. **1-2:** Encoder marker inverting input. **2-3:** Bring in LAN/RX line via 11PL (alternate LANTB-2). **NOTE:** *JP39 must also be in 2-3 position.*
		JP24		Controls signal to **inverting input** of General Purpose Amplifier #1. **1-2:** (Def) GP AMP #1 inverting input tied to ACOM. **2-3:** GP AMP #1 inverting input tied to pin 5 of JP29 (if JP29 in 2-5 position, then MAP3 tied to GP AMP #1 inverting input). **2-4:** GP AMP #1 inverting input tied to pin 4 of JP30 (if JP30 in 2-4 position, then MAP2 tied to GP AMP #1 inverting input).

Location	Jumper or Pot	Number	Nomenclature	Description
Micro-Application Card (optional) (continued)	Jumper	JP25		Controls signal to **non-inverting input** of General Purpose Amplifier #1. **1-2:** (Def) GP AMP #1 non-inverting input tied to ACOM. **1-3:** GP AMP #1 non-inverting input tied to pin 5 of JP29 (if JP29 in 2-5 position, then MAP3 tied to GP AMP #1 non-inverting input). **1-4:** GP AMP #1 non - inverting input tied to pin 4 of JP31 (if JP31 in 2-4 position, then MAP1 tied to GP AMP #1 non-inverting input).
		JP26		Controls signal to **inverting input** of General Purpose Amplifier #2. **1-2:** (Def) GP AMP #2 inverting input tied to ACOM. **2-3:** GP AMP #2 inverting input tied to DM4. **2-4:** GP AMP #2 inverting input tied to pin 4 of JP31 (if JP31 in 2-4 position, then MAP1 tied to GP AMP #2 inverting input).
		JP27		Controls signal to **non-inverting input** of General Purpose Amplifier #2. **1-2:** (Def) GP AMP #2 non-inverting input tied to ACOM. **1-3:** GP AMP #2 non-inverting input tied DM4. **1-4:** GP AMP #2 non-inverting input tied to pin 4 of JP30 (if JP30 in 2-4 position, then MAP2 tied to GP AMP #2 inverting input).
		JP28		MAP3, Select #2 - Controls which signal is connected to JP29-4. **1-2:** (Def) AN6 output from GP AMP #1. **2-3:** AN7 output from GP AMP #2. **2-4:** AN3 input from MAP3.
		JP29		MAP3, Select #1 - Controls function of MAP3. **1-2:** (Def) MAP3 is digital control input CI4. **2-3:** MAP3 is tied to digital output bit B06. **2-3:** MAP3 is tied to pin 2 of the JP28 (JP28 position then determines function of MAP3). **2-5:** MAP3 is tied to pin 3 of JP24/JP25 (if JP24 in 2-3 position, then MAP3 is the inverting input to GP AMP #1; if JP25 in 1-3 position, MAP3 is the non-inverting input to GP AMP #1). **NOTE:** *This jumper has been replaced by jumpers JP29A, JP29B, JP29C on later version cards.*

Location	Jumper or Pot	Number	Nomenclature	Description
Micro-Application Card (optional) (continued)	Jumper	JP29A		**6-7:** Park position.
		JP29B		**1-6:** Input LI-5 is digital control input CI4. **1-2:** MAP3 tied to digital control input CI4. **2-3:** MAP3 tied to GP amp outputs (JP28). **2-5:** MAP3 tied to pin 3 of JP24/JP25. **2-3:** MAP3 tied to digital output B06.
		JP29C		**3-8:** K17 (29PL-23) to B06 output.
		JP30		MAP2 Select - Controls function of MAP2. **1-2:** (Def) MAP2 is digital control input CI3. **2-3:** MAP2 is tied to digital output bit B05. **2-3:** MAP2 is tied to pin 4 of JP24 and pin 4 of JP27 (if JP24 in 2-4 position, then MAP2 is the inverting input to GP AMP #1; if JP27 in 1-4 position, then MAP2 is the non - inverting input to GP AMP #2). **2-5:** MAP2 is analog input AN5. **NOTE:** *This jumper has been replaced by jumpers JP30A, JP30B, JP30C on later version cards.*
		JP30A		**6-7:** Park position.
		JP30B		**1-6:** Input LI-4 is digital control input CI3. **1-2:** MAP2 tied to digital control input CI3. **2-4:** MAP2 tied to GP AMPs (JP27, JP24). **2-5:** MAP2 tied to AN5. **2-3:** MAP2 tied to digital output B05.
		JP30C		**3-8:** K16 (29PL22) to B05 output.
		JP31		MAP1 Select - Controls function of MAP1. **1-2:** (Def) MAP1 is digital control input CI2. **2-3:** MAP1 is tied to digital output bit B04. **2-4:** MAP1 is tied to pin 4 of JP25 and pin 4 of JP26 (if JP25 in 1-4 position, then MAP1 is the non - inverting input to GP AMP #1; if JP26 in 2-4 position, then MAP1 is the inverting input to GP AMP #2). **2-5:** MAP1 is analog input AN4. **NOTE:** *This jumper has been replaced by jumpers JP31A, JP31B, JP31C on later version cards.*
		JP31A		**6-7:** Park position.
		JP31B		**1-6:** Input LI-3 is digital control CI2. **1-2:** MAP1 tied to digital control input CI2. **2-4:** MAP1 tied to GP AMPs (JP25, JP26). **2-5:** MAP1 tied to AN4. **2-3:** MAP1 tied to digital output B04.
		JP31C		**3-8:** K15 (29PL21) to B04 output.

Location	Jumper or Pot	Number	Nomenclature	Description
Micro-Application Card (optional) (continued)	Jumper	JP32		Selects range of FRA pot P2. **1-2:** (Def) Normal FRA pot range. **2-3:** Extended FRA pot range.
		JP33		MUP Enable. **1-2:** (Def) Always (normal operation). **Open:** Never (manufacturing test only).
		JP34		U37 Enable. **1-2:** (Def) Always (normal operation). **Open:** Never (manufacturing test only).
		JP35		U39 Enable. **1-2:** (Def) Always (normal operation). **Open:** Never (manufacturing test only).
		JP36		U13 Enable. **1-2:** (Def) Always (normal operation). **Open:** Never (manufacturing test only).
		JP37		U14 Enable. **1-2:** (Def) Always (normal operation). **Open:** Never (manufacturing test only).
		JP38		LAN Source Select - Allows LAN interface through 11PL instead of through LANTB. **1-2:** (Def) Normal operation through LANTB. **2-3:** Bring in LAN/RX line via 11PL-1. **Note:** *JP22 must also be in position 2-3.*
		JP39		LAN Source Select - Allows LAN interface through 11PL instead of through LANTB. **1-2:** (Def) Normal operation through LANTB. **2-3:** Bring in LAN TX line via 11PL-2. **Note:** *JP23 must also be in position 2-3.*
		JP40		Controls the RUP Encoder up/down counter direction. **1-2:** (Def) Normal control by the encoder direction of rotation. **2-3:** Counts up only. For use when 11PL-16 (ENCA) is used as a pulse reference frequency input. **2-4:** Direction controlled by input level at 11PL-14 (ENCB).
		JP41		Pulse Reference Mode Select - Used in conjunction with JP40 to allow the RUP encoder circuit to be used as a pulse reference input. **1-2:** (Def) Encoder operation. **2-3:** Pulse reference operation.

Location	Jumper or Pot	Number	Nomenclature	Description
Micro-Application Card (optional) (continued)	Jumper	JP42		MUP Timer 1/External Interrupt Mode Select - Allows various application options with regard to this MUP resource. **1-2:** (Def) Normal operation without LAN - No special MUP Timer 1 nor external interrupt selected. **2-3:** External interrupt from EBR with a LAN and without MUP encoder. **2-4:** External pulse counting capability from EBA with or without LAN when MUP encoder is not used.
		JP43		U30 Port 1.3 Select - Provides compatibility needed for 8032 or 80C252 microprocessors. **1-2:** (Def) 8032 use; port 1.3 is a spare I/O port. **2-3:** 80C252 use; port 1.3 allows for simultaneous LAN and MUP encoder operation.
		JP44		U30 Port 1.4 Select - Provides optional functions of this microprocessor I/O pin. **1-2:** (Def) Spare I/O port or non-LAN control of the RS422 driver. **2-3:** Enables direct access to control input C10.
		JP45		Selects whether P7 is in or out of circuit. **1-2:** (Def) Enables pot P7 in GP amp #1 circuitry. **2-3:** Disables P7 in GP amp #1 circuitry.
		JP46		Selects whether P8 is in or out of circuit. **1-2:** (Def) Enables pot P8 in the GP amp #2 circuitry. **2-3:** Disables pot P8 in GP amp #2 circuitry.
		JP47		Used in conjunction with JP43 for using 80C252 microprocessor. **3-4:** (Def) Microprocessor U30 is type 8032. **4-5:** Microprocessor U30 is type 80C252 (allows simultaneous LAN & MUP encoder operation). **NOTE:** *This jumper has been replaced by jumpers JP47A, JP47B; JP47C on later version cards.*
		JP47A		**1-2:** (Def) Normal position. **2-3:** For use with 80C252 microprocessor.
		JP47B		**1-2:** No Connection (for future use). **2-3:** Enable LI-14 as logic input.

Location	Jumper or Pot	Number	Nomenclature	Description
Micro-Application Card (optional) (continued)	Jumper	JP47C		**1-2:** No Connection (future use). **2-3:** No Connection (future use).
		JP48		Selects type of EPROM for MUP. **1-2:** For 27256 (JP20 must be 2-3). **2-3:** For 27512 (JP20 must be 2-3). **NOTE:** *For U33 part numbers 49X113AA107 through 409X113AB107, 409X114AA107 and 409X114BA107 set JP48 in 1-2 position. Otherwise set in 2-3 position.*
		JP51		**1-2:** (Def) LI-13 non-inverting as MUP interrupt. **2-3:** LI-13 inverting MUP interrupt.
		JP52		**1-2:** Enables 13PL-9 as C10 input. **2-3:** Enables 29PL-1 as LI-1 input.
		JP53		**1-2:** Enables 13PL-11 as C11 input. **2-3:** Enables 29PL-2 as LI-2 input.
		JP54		Jumper sets max input voltage to optical isolators from ENCR (11PL-12). **1-2:** 10 volts **2-3:** 5 volts **3-4:** 1.5 volts **Open:** 15 volts
		JP55		Jumper sets max input voltage to optical isolators from ENCB (11PL-14). **1-2:** 10 volts **2-3:** 5 volts **3-4:** 1.5 volts **Open:** 15 volts
		JP56		Jumper sets max voltage input to optional isolators from ENCA (11PL-16). **1-2:** 10 volts **2-3:** 5 volts **3-4:** 1.5 volts **Open:** 15 volts
		JP57		Not Used - Set 1-2
		JP58 JP59 JP60		These select standard or optically isolated LAN communications mode. **1-2:** (Def) Standard LAN selected. **2-3:** Optically isolated LAN selected. **NOTE:** *JP58, JP59, JP60 must all be set either 1-2 or 2-3.*

Location	Jumper or Pot	Number	Nomenclature	Description
Encoder Process Interface Card (optional)	Switch	SW1	Tach Range Select	These switches provide coarse scaling of the tach circuit as shown in the following chart. DIP switch positions are selected for the legend of the side depressed. To select position 3 ON, depress the switch at position 3 next to the ON legend.

SW1 Position No. "Closed" or "ON"	Tach Volt At Maximum Speed DC Tach	AC Tach
All "open" or "off"	10 - 12.5V	11.1 - 13.9V
1	12.5 - 15.5V	13.9 - 17.2V
2	15.5 - 19V	17.2 - 21.1V
1, 2	19 - 22V	21.1 - 24.4V
3	22 - 25V	24.4 - 27.8V
1, 3	25 - 30V	27.8 - 33.3V
1, 2, 3	30 - 37.5V	33.3 - 41.7V
4	37.5 - 46V	41.7 - 51.1V
1, 2, 4	46 - 56V	51.1 - 62.2V
1, 2, 3, 4	56 - 70V	62.2 - 77.8V
5	70 - 87V	77.8 - 96.7V
2, 3, 5	87 - 105V	96.7 - 117V
1, 2, 4, 5	105 - 120V	117 - 133V
1, 2, 3, 4, 5	120 - 144V	133 - 160V
6	144 - 180V	160 - 200V
1, 2, 4, 6	180 - 205V	200 - 228V
5, 6	205 - 250V	228 - 278V
1, 2, 3, 4, 5, 6	250 - 310V	278 - 344V

NOTE: *SW1-7 must be OFF (or open) for DC tachs or ON (or closed) for AC tachs. Maximum voltages less than 10 volts are not to be used.*

	Pot	P1	TACH SCALE	Used to scale the tach circuit to obtain 4.0 volts at test point SFB at maximum speed.
		P2	SMET	Used to adjust the analog output signal at 10PL-9 of the speed metering circuit from 4 to 10 volts at maximum motor speed. Maximum current load on this signal is 2mA. (Also see JP4.)
		P3	DVM	With DVM test point connected to +5V and optional Programmer set to DVM operation, adjust P3 until the Programmer displays 5.00.

Location	Jumper or Pot	Number	Nomenclature	Description
Encoder Process Interface Card (optional) (continued)	Pot	P4 P5	PF GAIN PF OFF	These pots allow trimming the gain and offset of the circuit that provides up to 10 volts from 1 to 5, 4 to 20, 5 to 50, or 5 to 65mA transducer inputs. To adjust, start with PF gain at the full CCW position and procede as follows: -- Apply the current reference corresponding to minimum speed. -- Adjust PF OFF for 0 volts at test point PFRF. -- Adjust the current reference corresponding to maximum speed. -- Adjust PF GAIN for 10.0 volts at PFRF. Repeat as necessary **NOTE:** *To prevent reversing when the current reference is less than the minimum current, the drive should be set up in a non-reversing mode.*
		P6	STP SCALE	Set Point Scale - Used to adjust scaling of definable input at test point STPRF. Refer to elementaries and engineering notes if used.
		P7	SFB ZERO	This pot adjusts test point SFB to be zero with zero tach feedback. Set speed reference to zero (jumper MSR or ASR to PCOMX if used) and adjust P7 for zero motor speed when a run is commanded.
	Jumper	JP1		RXD Mode Select **1-2:** Enables RS422 reception on serial link data line. **2-3:** Enables RS232C reception.
		JP2 JP3		JP2 and JP3 select the process follower mode per the following table: **JP2 JP3 Range** **1-2: 1-2:** 5-65ma current loop follower. **1-2: 1-2:** 5-50ma current loop follower. **2-3: 1-2:** 4-20ma current loop follower. **2-3: 2-3:** 1-5ma current loop follower or 10V voltage follower.
		JP4		SMET Type Select. **1-2:** Unipolar speed meter driver mode. **2-3:** Bipolar meter driver mode (DC tach only).
		JP5		Marker Pulse Mode Select. **1-2:** Differential RS422 compatible, with 0.33 ohm termination. **2-3:** Single ended input at PCP. **NOTE:** *Jumper JP5 has been replaced on later version cards with jumper JP21.*

Location	Jumper or Pot	Number	Nomenclature	Description
Encoder Process Interface Card (optional) (continued)	Jumper	JP7		Marker Pulse Enable. **1-2:** Not Used. **2-3:** Enables marker pulse to TUP.
		JP8		Analog Tach Filter Select. **1-2:** Enables 4ms filter in analog tach feedback; required for AN tach operation. **2-3:** Disables 4ms filter; required for speed regulator performance above 20 radians.
		JP9 & JP10		Digital Tach or Encoder Mode Select. **1-2:** Differential RS422 compatable with termination resistor. **2-3:** Single-ended input at 10PL-5 and 10PL-7. **Open:** Differential operation with no terminating resistor. **NOTE:** *Jumpers JP9 and JP10 have been replaced on later version cards with jumpers JP19 and JP20.*
		JP11		TXD Mode Select. **1-2:** 12PL-9, 12PL-12 RS422 transmitted data mode. **2-3:** 12PL-9 RS232 transmitted data mode.
		JP12		DTR Mode Select. **1-2:** 12PL-8, 12PL-13 RS422 DTR handshake mode. **2-3:** 12PL-8 RS232 DTR handshake mode.
		JP13		DTR RS232 Logic Polarity. **1-2:** 12PL-8 high=drive ready to receive. **2-3:** 12PL-8 low=drive ready to receive. **NOTE:** *For Series 6 interfaces using the CCM card, set JP1 and JP11 to 1-2 and JP12-JP13 to 2-3. 12PL-7 should be tied to +5V.*
		JP14		CTS Mode Select. **1-2:** For RS422 CTS handshaking. **2-3:** For RS232 CTS handshaking. **NOTE:** *Jumper JP14 has been replaced on later version cards with jumpers JP14A and JP14B.*
		JP14A JP14B		CTS Handshake Mode Select. **JP14A** / **JP14B** / **Description** **1-2:** / **1-2:** / Enable Differential CTS Handshaking. **1-2:** / **2-3:** / Enable Single-Ended CTS Handshaking. **2-3:** / **2-3:** / Disable CTS Handshaking.

Location	Jumper or Pot	Number	Nomenclature	Description
Encoder Process Interface Card (optional) (continued)	Jumper	JP15		Encoder Counter Direction Control. **1-2:** Normal control by encoder direction of rotation. **2-3:** Up only.
		JP16		Encoder A Quad B Enable. **1-2:** Normal A quad B encoder operation. **2-3:** Count on leading edge of track A only.
		JP17		LUP Marker Pulse Select. **1-2:** Marker pulse not connected to 9PL-14 (normal operation). **2-3:** Marker pulse circuit output passed to 9PL-14 for use by LUP.
		JP18		LUP DM5 Select. **1-2:** DM5 not connected to 9PL-13 (normal operation). **2-3:** DM5 connected to 9PL-13 for use by LUP.
		JP19		Digital Tach Input Signal DTA Scaling. **1-2:** 10 Volt Maximum Input. **2-3:** 5.0 Volt Maximum Input. **3-4:** 1.5 Volt Maximum Input. **1-4:** 15 Volt Maximum Input.
		JP20		Digital Tach Input Signal DTB Scaling. **1-2:** 10 Volt Maximum Input. **2-3:** 5.0 Volt Maximum Input. **3-4:** 1.5 Volt Maximum Input. **1-4:** 15 Volt Maximum Input.
		JP21		Digital Tach Input Signal Marker Scaling. **1-2:** 10 Volt Maximum Input. **2-3:** 5.0 Volt Maximum Input. **3-4:** 1.5 Volt Maximum Input. **1-4:** 15 Volt Maximum Input.
		JP22		Voltage Follower Enable. **1-2:** Voltage Follower Disabled. **2-3:** Voltage Follower Enabled.

TABLE 2: TEST POINTS

As a further aid to troubleshooting the AC-300 drive controller, a series of test points have been provided to allow for specific signal observation and/or measurements. Test points not defined in the following listing are for manufacturer's use only and should not have instrumentation connected to them.

CONTROL CARD TEST POINTS

Test Point Name	Description	Nominal Value
PCOM	System common. Use for all signal measurements referenced to common.	
ACOM	Common for analog circuit power supplies. (Not for use external to drive.)	
DCOM	Common for digital circuit power supply. (Not for use external to drive.)	
DA0	Scaled output from 12-bit digital-to-analog converter (DAC).	
JOGR	Jog reference input at 6PL-7 (3TB-63).	
-15V	-15 VDC power supply.	-14.25 to -15.75 VDC
+15V	+15 VDC power supply.	+14.25 to +15.75 VDC
+5V	+5 VDC power supply.	+4.95 to +5.05 VDC
DBD	Operating condition signal from Power Supply card: -- Low in normal operation. -- High (+5V) indicated DC Link under or over voltage or phase loss on AC supply. -- Oscillating (pulse train) when Dynamic Braking is operatiing.	
VA, VB	2-Phase Representation of Motor Voltage Feedback (AC signal).	
YA, YB	2-Phase Representation of Volts/Hertz, or flux (AC signal).	
SRS	Scaled speed reference signal.	4 volts at top speed reference
DM4	Multi-use point 4, connects to DM4 point on Application card option.	

CONTROL CARD TEST POINTS (continued)

Test Point Name	Description	Nominal Value
PSEN	Power Suply Enable-indicates when all power supplies in range.	+12 to +13.5V
VDC	DC Line Voltage.	4V @ 650 VDC (Link)
IC IC*	Motor Phase-C Current Feedback. Motor Phase-C Current Reference.	
IB IB*	Motor Phase-B Current Feedback. Motor Phase-B Current Reference.	
IA IA*	Motor Phase-A Current Feedback. Motor Phase-A Current Reference.	
IMET*	Motor IMET Torque Current.	2V at rated load
W*	Motor Frequency (Speed).	4.3 VDC at maximum frequency (speed) Programmable ADR196-WMETS

BASE DRIVE CARD TEST POINTS

Test Point Name	Description	Nominal Value
CFA	Phase A Current Feedback Signal (see following **WARNING**).	Pulse train
CFB	Phase B Current Feedback Signal (see following **WARNING**).	Pulse train
CFC	Phase C Current Feedback Signal (see following **WARNING**).	Pulse train
N2	N2 Bus Potential (see following **WARNING**).	

WARNING-ELECTRIC SHOCK HAZARD: Certain test posts and jumper posts may be at high (i.e. 600V) voltage. Hazard of electrical shock exists and extreme caution must be exercised.

NOTE: *Base Drive Card test posts are at various high potentials to ground. They are for factory use only and NOT INTENDED FOR CUSTOMER USE.*

POWER SUPPLY CARD TEST POINTS

Test Point Name	Description	Nominal Value
DB	DB Transistor Base Drive (see following **WARNING**).	Pulse train if DB operating
N2	Negative DC Link potential (see following **WARNING**).	
DC	DB/CC DB duty/Charge Control signal (see following **WARNING**).	Pulse train if DB opening
CC	Charge Control (see following **WARNING**).	Pulse train
DT	DB Test (see following **WARNING**).	Pulse train if DB operating
PSEN	Power Supply Enable.	+12 to +13.5V
-15V	-15 VDC power supply.	-14.25 to -15.75 VDC
+15V	+15 VDC power supply.	+14.25 to +15.75 VDC
+5V	+5 VDC power supply.	+4.95 to +5.05 VDC
ST	Switching Transformer Voltage.	
COM	System Common (Not for use external to drive).	

WARNING-ELECTRIC SHOCK HAZARD: Certain test posts and jumper posts may be at high(i.e. 600V) voltage. Hazard of electrical shock exists and extreme caution must be exercised.

NOTE: *Power Supply Card test posts DB, N2, DC, CC, and DT are at high potential to ground. Use caution in taking measurements at these posts. All measurements should be made relative to the N2 test post. An oscilloscope or meter should be used.*

OPTIONAL PROCESS INTERFACE CARD TEST POINTS

Test Point Name	Description	Nominal Value
ACOM	Common of analog circuit power supplies.	
DCOM	Common of digital circuit power supply.	
5V	+5V DC power supply.	+4.95 to +5.05VDC
DVM	Input probe connection for digital voltmeter. Requires optional Programmer.	
SFB	Analog speed feedback signal. No value when digital tach or no tach being used in the drive system.	4 VDC at top speed
PFRF	Output of the reference interface circuitry.	10 volts at top speed
DM5	Dummy point 5, connects to 10PL-16 (3TB-25).	
STPRF	Access point to process follower amplifier. Used to bring in a setpoint controller reference by jumpering STPRF to DM5 and using 10PL-16 as the input.	

NOTE: *All analog and digital voltage signals should be referenced to 3TB-PCOMX only (unless shown differently on system elementary diagrams).*

OPTIONAL APPLICATION CARD TEST POINTS

Test Point Name	Description	Nominal Value
ASR	Auto System Reference signal.	10 volts at top speed reference
DA8	Only on version APG cards. Output from the 8-bit D/A converter.	±10 volts = maximum signal
DA12	Only on version APG cards. Output from the 12-bit D/A converter.	±10 volts = maximum signal
FRA	Field reference adjust signal.	±10 volts = full adjust
RPU	Encoder interface marker track, one pulse per revolution.	
OSC	500 KHz TTL oscillator used by the shared bus encoder interface and by the real time clock on MUP.	

OPTIONAL APPLICATION CARD TEST POINTS (continued)

Test Point Name	Description	Nominal Value
DM4	Dummy point 4, connects to 3PL-30. May be used to jumper signals from other test points to 3PL for custom applications, or to bring signals from the Control card to the general purpose amplifier circuits.	
CLK	NOT USED	
SPT	NOTUSED	

OPTIONAL ENCODER PROCESS INTERFACE CARD TEST POINTS

Test Point Name	Description	Nominal Value
ACOM	Common of analog circuit power supplies.	
DCOM	Common of digital circuit power supply.	
+5V	+5V DC power supply.	+4.95 to +5.05VDC
DVM	Input probe connection for digital voltmeter. Requires optional Programmer.	
SFB	Analog speed feedback signal. No value when digital tach or no tach being used in the drive system.	4 VDC at top speed
PFRF	Output of the reference interface circuitry.	10 VDC at top speed
DM5	Dummy point 5, connects to 10PL-16 (3TB-25).	
TPN	Net digital or AN tach pulse output.	
STPRF	Access point to process follower amplifier. Used to bring in a setpoint controllr reference by jumpering STPRF to DM5 and using 10PL-16 as the input.	
CLK	Net pulse input into the encoder up/down counter.	
RPU	Marker pulse input into the up/down counter reset circuitry.	

NOTE: *All analog and digital voltage signals should be referred to 3TB-PCOMX only (unless shown differently on system elementary diagrams).*

OPTIONAL MICROAPPLICATION CARD TEST POINTS

Test Point Name	Description	Nominal Value
ASR	Auto System Reference signal used to calibrate auto speed reference. Apply analog reference corresponding to top speed to 13PL-23 and adjust P1 (and JP1 if necessary) to obtain 4 volts.	4 volts at top speed reference
CLK	Output from shared bus encoder interface quadrature track combinatorial logic that generates one pulse at CLK for each transition at the encoder main track inputs ENCA and ENCB. For an "n" pulse per revolution encoder, the output at CLK will be 4n pulses per revolution. Used as input to an up/down counter that is read on the shared bus.	
FRA	Field reference adjust signal. Used to calibrate the analog input FRA (field forcing adjustment). One volt at this test pont increases/decreases the FLDMX parameter setting by 204 counts.	$\pm$10 volts = full adjust
OSC	460 KHz (G1) or 500 KHz (G2) TTL oscillator used by the shared bus encoder interface and by the real time clock on MUP.	
RPU	Output from shared bus encoder interface marker track logic. Generates one pulse per revolution and is used as a reset pulse for the up/down counter.	
SPT	Spare TTL output from the shared bus. NOT USED	
STS	TTL output from MUP's A-D converter that goes high while conversions are in progress.	
TP4	Spare TTL input or output point on the shared bus.	
TP5	Spare TTL input or output point on the shared bus.	
TP6	Spare TTL output point from MUP.	

NOTES:

8. SOFTWARE ADJUSTMENTS

TABLE 3 on the following pages is a listing of "standard" AC-300 software adjustments (parameters) common in most AC-300 drive applications. Included is the "AC-300 Parameter Map" (equivalent analog representation of parameters), the "Key Parameter Adjustments", and the "RAM Variable Scaling" table, that defines the scaling of RAM addresses available for interrogation of the drive control "circuit". The parameters can be examined and adjusted with the optional Programmer (refer to "Use of Optional Programmer Module") or via the serial link.

Additionally, a Custom Software Description is shipped with every drive. The Custom Software Description describes the setting and/or function of various programmable jumpers and adjustments (hardware and software) within the drives that are identified with the specific model number of that drive. Included in that is a listing of specific programmer options, their description and a set of software adjustment definitions pertinent to the specific drive.

PARAMETER MAP (Figure 8-1)

The Parameter Map is a diagnostic aid designed to reference specific software adjustments and correlate them to their specific regulating functions. For example, ADR015-BSLPG is represented as a potentiometer feedback across the base speed regulating amplifier, giving an effective linear gain adjustment of input to output.

Software adjustments are shown next to their symbol with the mnemonic and the parameter address in a box beside it. Hardware adjustments are represented by a symbol, the adjustment name, and the card it is located on (e.g.: Tach scale is a potentiometer, P1, on the Process Interface card). RAM addresses that can be used to examine outputs of certain stages of regulation through TEST 11, TEST 21, TEST 22, or D/A converters, are shown as a three-digit number in a circle connected to the regulator section it is associated with (e.g.: Speed loop output can be examined by using RAM address 053 in Test 11, 21, 22, or the D/A converters).

The optional Programmer display scaling adjustments are shown in the lower left section of the map as well as their relation to feedback.

KEY PARAMETER ADJUSTMENTS

The following parameter adjustments, listing Programmer mnemonic, address number, function name, and description, will prove helpful in setup and troubleshooting of the AC-300 drive. This is a partial list and should be used in conjunction with the "Parameter Map". For a more complete list of parameters contained in a specific drive, refer to the Custom Software Description furnished with that drive.

The following is a description of column headers:

NAME	Mnemonic displayed by Programmer when interrogating parameter.
ADR	Parameter address.
UNITS	Unit designation of displayed number (example - RPM).
MIN	Minimum allowable value.
MAX	Maximum allowable value.
STEP	Amount by which value can be incremented or decremented.
DEFAULT	Standard value if no special setting is required.
DESCRIPTION	Definition of parameter, including equation where required.

NOTE: *Parameters listed in TABLE 3 are grouped functionally per the parameter map (Figure 8-1).*

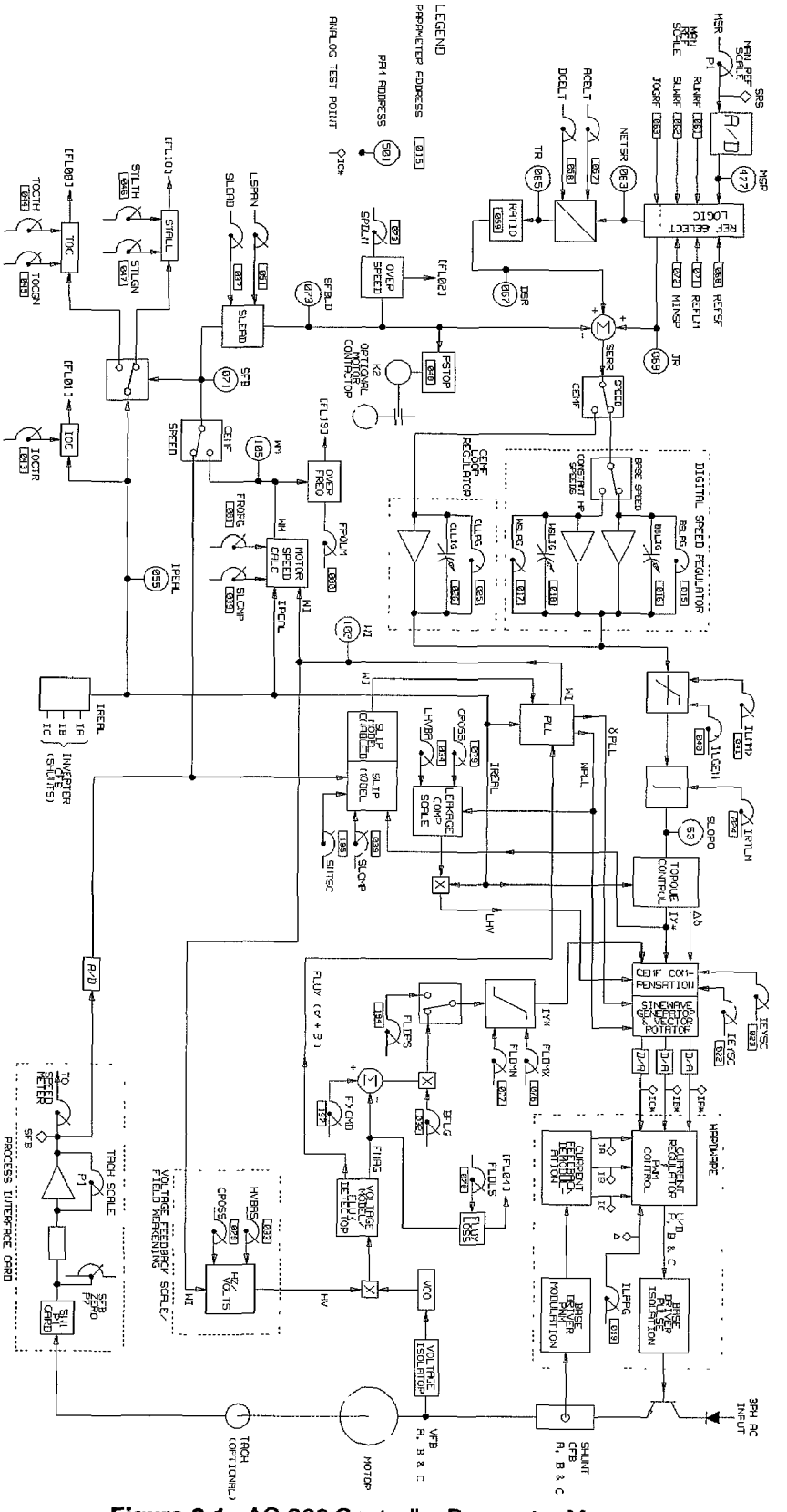


Figure 8-1. AC-300 Controller Parameter Map

TABLE 3: KEY PARAMETER ADJUSTMENTS

FLUX REGULATOR

Name	Adr.	Units	Min.	Max.	Step	Default
FLDMX	076	%Rated	0	100	1	50

Description: **Full Field Current** -- limits the maximum level of the magnetizing current component that the controller will use to regulate flux.
One unit = 1% rated controller current.

FLDMN	077	%Rated	0	100	1	0

Description: **Minimum Field Current** -- limits the minimum level of the magnetizing current component that the controller wil use to regulate flux.
One unit = 1% rated controller current.

FLDLS	078	%NOM	0	100	1	33

Description: **Field Loss Trip Setpoint** -- sets the % of flux reference level at which loss of flux feedback (V/Hz) will cause a controller trip.
One unit = 1% of nominal flux reference level.

CROSS	079	Hz	0	255	1	55

Description: **Field Crossover** -- sets the frequency at which the output voltage is regulated to a constant value and the volts per hertz ratio is reduced to attain motor speeds above base speed. One unit = 1 Hz.

BFLG	032	None	0	255	1	162

Description: **Base Field Regulator Gain** -- sets the gain of the flux regulator.

HVBAS	033	V/Hz	0	255	1	80

Description: **Base Voltage Scaling** -- sets the scale factor between the motor volts/hz and the internal representation of flux. For **230 volt** controllers, HVBAS = 297/(V/Hz); for **460 volt** controllers, HVBAS = 625/(V/Hz)

LHVBA	034	None	1	255	1	50

Description: **Leakage Inductance Compensation** -- sets the scale factor for the phase angle compensaton due to load current through leakage reactance at base speed.

DBTOT	049	%Duty/2	0	200	1	10

Description: **DB Timed Overvoltage Threshhold** -- sets the level of filtered duty cycle that causes a trip due to timed overvoltage of the DC link. An annunciated fault is displayed at five-sevenths (5/7) of this trip value. During regeneration, the dynamic braking resistor in the DC link is switched on and off with a constant frequency and variable duty cycle to attempt to limit DC link voltage.
DBTOT = Two x Duty Cycle (in percent)

LINEAR TIME

Name	Adr.	Units	Min.	Max.	Step	Default
ACELT	057	Decisec	0	9999	1	20

Description: **Acceleration Time** -- sets the rate at which motor operation will increase in response to commanded reference changes. For linear time, enter time in tenths of a second for acceleration from 0 to top speed. ACELT = Time (in sec.) x 10.

DCELT	058	Decisec	0	9999	1	20

Description: **Deceleration Time** -- sets the rate at which motor speed will decrease in response to command reference changes. For linear time, enter time in tenths of a second for deceleration from top speed to 0. DCELT = Time (in sec.) x 10.

REFERENCES AND SCALING

REFSF	068	None	0	6553	1	6553

Description: **Reference Scale Factor** -- sets the factor by which references entered from the serial link and Programmer card will be multiplied and by which speeds displayed on the Programmer card and sent to the serial link will be divided. Enter the value of speed reference which corresponds to top speed (4 volts at SFB). **Important** - affects settings of RUNRF, SLWRF, JOGRF, R1RF, R2RF, SLKRF, parallel input scaling, and speed reference display. **With Tach,** REFSF = Top Speed RPM; **for CEMF regulator,** REFSF = Base Speed (RPM) at rated armature volts.

RUNRF	061	RPM	0	6553	1	0

Description: **"Run" Key Reference** -- sets the preset reference for RUN commands from the serial link. Also may be used as the preset reference for manual start commands. Also serves as the top speed reference for motor operated rheostat (MRH) mode. May be a speed, torque or position depending on jumper setting in ADR006-RUNJP. Normally, RUNRF = Top Speed (RPM).

SLWRF	062	RPM	0	6553	1	0

Description: **"Slow" Key Reference** -- sets the preset reference for SLOW commands from the serial link. Normally, SLWRF = Slow Speed (RPM).

JOGRF	063	RPM	0	6553	1	400

Description: **"Jog" Key Reference** -- sets the preset reference for Jog commands from the serial link. Also may be used as the preset reference for manual jog commands. JOGRF = Jog Speed (RPM).

RPMSF	070	RPM	0	6553	1	6553

Description: **RPM Scale Factor** -- sets the speed in RPM which corresponds to 4 volts at SFB.
NOTE: *Change this parameter only while the drive is not running, and toggle the reset switch afterwards to make the change fully effective.*
IMPORTANT: *Affects setting of REFLM, MINSP, SPDLM, tapered current limit speed settings, and digital tach scaling. Same as setting for REFSF.*

REFERENCES AND SCALING (continued)

Name	Adr.	Units	Min.	Max.	Step	Default
REFLM	071	RPM	0	6656	1	6560

Description: **Maximum Speed** -- sets the level at which speed reference will be clamped if the input reference magnitude exceeds the level. Maximum value should not exceed 1.015 x RPMSF. Typically, REFLM = RUNRF + 100.

MINSP	072	RPM	0	6553	1	400

Description: **Minimum Speed** -- sets the level at which speed reference will be clamped if the input reference magnitude is less than the level. MINSP = desired Minimum Speed (RPM). Maximum value should not exceed RPMSF.

SPDLM	073	RPM	0	7936	1	7208

Description: **Speed Limit Trip Setpoint** -- sets the speed at which the drive will trip. SPDLM = RPM at drive overspeed trip. Should not exceed 1.2 x RPMSF.

SPEED REGULATOR

BSLPG	015	None	0	255	1	7

Description: **Base Speed Loop Proportional Gain** -- sets the proportional gain of the outer speed regulator below base speed when tachometer feedback is used.

BSLIG	016	None	0	255	1	8

Description: **Base Speed Loop Integral Gain** -- sets the integral gain of the outer speed regulator below base speed when tachometer feedback is used.

FTOFF	082	None	0	65536	1	0

Description: **Fine Tach Offset** -- sets the offset added to speed feedback to form feedback for the slip model (low speed tach operation) if auto-offset is disabled.

WSLPG	017	None	0	255	1	7

Description: **Weak Field Speed Loop Proportional Gain** -- sets the proportional gain of the outer speed regulator above crossover voltage.

WSLIG	018	None	0	255	1	8

Description: **Weak Field Speed Loop Integral Gain** -- sets the integral gain of the outer speed regulator above crossover voltage.

SLEAD	037	None	0	255	1	10

Description: **Speed Feedback Lead** -- sets the lead time constant on speed or CEMF feedback.

LSPAN	051	None	0	15	1	0

Description: **Speed Feedback Lead/Lag Span** -- sets the span of the tachometer lead/lag network. A setting of 0 is treated as a two-to-one span.

SPEED REGULATOR (continued)

Name	Adr.	Units	Min.	Max.	Step	Default
RSTOP	048	None	0	255	1	100

Description: **Regenerative Stop Speed** -- sets the speed at which the contactor opens during a regenerative stop.
One Unit = 0.03% speed (for Speed Reg.).
One Unit = 0.12% volts (for CEMF Reg.).

FRQLM	080	Hz	0	255	1	60

Description: **Maximum Frequency Limit** -- sets the maximum frequency at which the drive is expected to operate. An over-frequency trip is generated at 1.15 X FRQLM.
One unit = 1Hz.

CEMF REGULATOR

CLPPG	025	None	0	255	1	7

Description: **(Tachless) CEMF Loop Proportional Gain** -- sets the proportional gain of the outer frequency regulator when frequency is used as a feedback instead of a tachometer.

CLPIG	026	None	0	255	1	8

Description: **(Tachless) CEMF Loop Integral Gain** -- sets the integral gain of the outer frequency regulator when frequency is used as a feedback instead of a tachometer.

CLMPG	027	None	0	255	1	8

Description: **CEMF Limit Proportional Gain** -- sets the proportional gain of the spillover CEMF limit regulator.

CLMIG	028	None	0	255	1	10

Description: **CEMF Limit Integral Gain** -- sets the integral gain of the spillover CEMF limit regulator.

SLCMP	039	None	0	255	1	32

Description: **Slip Compensation** -- sets the scale factor for compensation of the open loop frequency to account for slip. Used in feedback path when tachless mode selected; used in slip model when active. Defining equation:
SLCMP = 31.5 x [I (inverter)/I (rated)] x Rated Slip (Where slip is in Hertz.)
Negative SLCMP can be obtained by programming in -1 to -128 in two's comp.: (65536 to 65408). Negative SLCMP is interpreted as zero SLCMP by slip model.

FRQRG	081	None	0	65536	1	18185

Description: **Frequency Range** -- scales the internal representation of open loop frequency compensated for slip to a number suitable for use as speed feedback to the speed regulator. FRQRG = 1,091,101/Applied Frequency (in electrical Hz) that produces mechanical rotation at "top speed" in no-load conditions.

CURRENT REGULATOR

Name	Adr.	Units	Min.	Max.	Step	Default
TQLIM	038	1.4°	0	127	1	64(90°)

Description: **Torque Control Limit** -- sets the maximum advance angle from phase lock achievable by the torque controller.

ILMMX	041	% Rated	0	255	2	150

Description: **Maximum Current Limit** -- sets the maximum current limit. One unit = 1% current.

ILGEN	040	% Rated	0	255	2	20

Description: **Generating Current Limit** -- limits the torque producing current when the motor is asked to regenerate. Overridden by ILMMX and by non-brake bit. One unit = 1% current.

IOCTR	043	% Rated	0	255	2	220

Description: **Instantaneous Overcurrent Trip Setpoint** -- sets the current level at which the drive will trip. One unit = 1% current.

TOCTH	044	% Rated	10	510	2	120

Description: **Timed Overcurrent Sense Threshold** -- sets the current level at which the TOC integrator begins to time-up. One unit = 1% current.

TOCGN	045	None	0	255	1	36

Description: **Timed Overcurrent Response** -- sets the time constant of the TOC integrator for the torque producing current. Refer to the Custom Software Description if multi-motor function is used.
TOCGN = 66,666/(Trip Time)(X-TOCTH)
NOTE: *X = current in %.*

TQCPG	020	None	0	255	1	32

Description: **Torque Regulator Proportional Gain** -- sets the gain of the phase advance compensation for use in weak field.

TQCTC	021	None	0	255	1	32

Description: **Torque Regulator Filter Time Constant** -- sets the filter time constant of the torque controller.

IRTLM	024	PU/S	1	255	1	50

Description: **Current Rate Limit** -- controls the maximum rate of change of current in response to commanded changes.
Relation is Current Rate = 0.9 (IRTLM) per unit per second.

FLDPS	194	% Rated	0	100	1	32

Description: **Field Preset** -- sets the fixed magnetizing current command to the current regulators during preflux and low frequency operation.

CURRENT REGULATOR (continued)

Name	Adr.	Units	Min.	Max.	Step	Default
IEXSC	023	None	0	255	1	15

Description: **I Extra Scale Factor** -- sets the scale factor for the extra in-phase torque producing current required vs. frequency for back EMF compensation.

OUTPUT SIGNALS

DAC0	090	None	0	65535	1	0

Description: **D/A #0 Output Control** -- controls what internal variable is sent to the 12-bit D/A converter (DAC0) on the Control card. Enter the desired RAM variable number into ADR090 to program DAC0.

RAM Adr.	Variable Name	RAM Adr.	Variable Name
053	Loop Output	071	SFB - Speed Feedback
055	IREAL	103	WI Inverter Frequency
061	Current Loop Output	105	WM Rotor Frequency
065	Timed Speed Reference	463	V/HZ
069	Jog Reference	501	TOC Timer

NOTE: *Refer to the "RAM Variable Scaling", TABLE 4 at the end of this listing for relative value of RAM addresses through various outputs.*

DAC8	096	None	0	65535	1	0

Description: **D/A #1 Output Control** -- controls what internal variable is sent to the 8-bit D/A converter (DAC8) on the Application card. See DAC0 for programming information.

DAC12	097	None	0	65535	1	0

Description: **D/A #2 Output Control** -- controls what internal variable is sent to the 12-bit D/A converter (DAC12) on the Application card. See DAC0 for programming information.

BIT #1	098	None	0	65535	1	0

Description: **Bit#1 Output Control** -- controls what internal bit is sent to bit output #1 (B01) on the Application card. Possible outputs include:

RAM Adr.	Variable Name	RAM Adr.	Variable Name
137	Fault	148	Reference Enable
140	Current Limit Flag	152	Precondition
141	Freq. Limit Flag		

NOTE: *In addition, the sign of all of the variables given in DAC0 output control may be used as outputs.*

BIT #2	099	None	0	65535	1	0

Description: **Bit#2 Output Control** -- controls what internal bit is sent to bit output #2 (B02) on the Application card. See Bit #1 for programming information.

BIT #3	100	None	0	65535	1	0

Description: **Bit#3 Output Control** -- controls what internal bit is sent to bit output #3 (B03) on the Application card. See Bit #1 for programming information.

OUTPUT SIGNALS (continued)

Name	Adr.	Units	Min.	Max.	Step	Default
WMETS	196	None	0	255	1	76

Description: **W-Meter Scale Factor** -- sets the scale factor used in the presentation of excitation frequency to the "WMET" (frequency meter) test point via the 8-bit D/A when the FUP diagnostic D/A mode has not been selected.
WMET = 5292/Max. Excitation Frequency

DIAGNOSTICS

DIAG1	122	None	0	65535	1	0

Description: **Diagnostic Variable #1** -- selects 1 of 2 variables which can be displayed on the serial link or Programmer card using "display diagnostics". See DAC0 output control for programming information.

DIAG2	123	None	0	65535	1	0

Description: **Diagnostic Variable #2** -- selects second variable which can be displayed on the serial link or Programmer card using "display diagnostics". See DAC0 output control for programming information.

SCHRL	127	None	1	26	1	18

Description: **Left Scaled Unit Character** -- determines which character is displayed as the left character of the scaled speed status message. 1 represents "A"; 2, "B"; 26, "Z".

SCHRR	128	None	1	26	1	13

Description: **Right Scaled Unit Character** -- see ADR127-SCHRL for programming information.

IRATD	129	None	0	3276	1	100

Description: **Displayed Arm Current Scaling** -- controls the factor by which per unit current is multiplied for display on the Programmer card.

VRATD	130	None	0	2621	1	100

Description: **Displayed Arm Volts Scaling** -- controls the factor by which per unit frequency voltage is multiplied for display on the Programmer card (goes to WMET).

FRATD	131	None	0	43690	1	75

Description: **Displayed Field Current Scaling** -- controls the factor by which per unit magnetizing current component is multiplied for display on the Programmer card.

SRATD	132	None	0	6200	1	100

Description: **Displayed Speed Scaling** -- controls the factor by which per unit speed is multiplied for display on the Programmer card.

TRATD	133	None	0	3276	1	100

Description: **Displayed Torque Scaling** -- controls the factor by which per unit torque is multiplied for display on the Programmer card.

HRATD	134	None	0	3276	1	100

Description: **Displayed HP Scaling** -- controls the factor by which per unit horsepower is multiplied for display on the Programmer card.

TABLE 4: RAM VARIABLE SCALING FOR PROGRAMMER AND DAC OUTPUTS

RAM Adr.	Variable Name	Value At 100% Signal Dec.	Value At 100% Signal Hex[1]	Volts On DAC[2] At 100% Signal
030	**IALPHA** Feedback	50	0032	
031	**IBETA** Feedback	50	0032	
053	**SLOPO** - Speed Loop Output	100	0064	2.0
055	**IREAL** Feedback	100	0064	2.0
063	**NETSR** - Net Speed Reference	6553	1999	8.0
065	**TR** - Timed Reference	6553	1999	8.0
067	**DSR** - Draw Speed Reference	6553	1999	8.0
069	**JR** - Jog Reference	6553	1999	8.0
071	**SFB** - Speed Feedback	6553	1999	8.0
073	**SFBLD** - Speed Feedback Lead	6553	1999	8.0
103	**WI** - Inverter Frequency	1638	0666	8.0
105	**WM** - Rotor Frequency	6553	1999	8.0
445	**Filtered** WI Feedback	1638	0666	8.0
448	**Filtered** Current Feedback	1638	0666	8.0
463	**Field** Current (Flux I)	100	0064	2.0
477	**MSR** - Manual Speed Reference	6553	1999	8.0
486	**ASR** - Automatic Speed Reference	6553	1999	8.0
501	**TOCT** - Timed Overcurrent Timer (trips at 1024)	1024	0400	0.3
492	**HP** - Horsepower	10238	27FE	3.1
056	**FUP, IX_COMM** - IX Command, FUP RAM Variable[3]			
060	**FUP, IY_COMM** - IY Command, FUP RAM Variable[3]			

NOTES: *(1) Negative numbers are represented in Two's complement.*

(2) D/A converters DAC8 and DAC12 have an op-amp with a gain of two.

(3) Use Parameter 125 to load the desired FUP Variable into, then program Parameter 122 or 123 with "733". Test 11 can then be used to monitor RUP RAM.

9. TROUBLESHOOTING

This section contains information on detecting and correcting problems which may be encountered when working on an AC-300 drive controller. The AC-300 controller contains a built-in fault detection system which indicates when a fault occurs and the nature of the fault. Tables 5, 6, and 7 are intended to aid in the correction of problems both indicated by the drive fault system and those not indicated by the fault display. Following Table 7 is a "Stability" Troubleshooting paragraph and Table 8, Stability Troubleshooting.

TABLE 5, "General Troubleshooting Chart" is for aid in troubleshooting problems which have no associated fault or error indication or cannot be remedied through the standard fault system. Table 5 includes a "Quick Reference Guide", an outline of the possible problem descriptions included in the chart.

TABLE 6 and TABLE 7, "Specific Fault Troubleshooting" and **"Specific Error Troubleshooting"** define drive fault and error codes. Fault codes (TABLE 6) are displayed on the Diagnostic LEDs and optional Programmer card, and broadcast on the optional serial link. Error codes (TABLE 7) are only related to Programmer or serial link errors, and therefore, only displayed by the Programmer card or broadcast on the serial link.

FAULT/ERROR CODES

The AC-300 drive controller displays faults with a series of Diagnostic LEDs (interpreted as a binary number) or with a fault message (fault number and name) on the optional Programmer card.

The LEDs on the Control card (in the normal condition) will flash from right to left:

One at a time in the standby mode.

Two at a time in the RUN or JOG modes.

These LEDs are also used to indicate the fault number. To determine the fault number from the Diagnostic LEDs, add the number value of the LEDs printed on the card above or below the LEDs (also add to this total the value of 64 if the LEDs are blinking). See Example below.

TYPES OF FAULTS

The faults listed in Table 6 may be divided into four categories:

Latched Fault (L)
The fault is latched. Once it occurs, it will not clear itself even if the cause of the fault no longer exists. The controller must be reset to clear it.

Non-Latched Fault (N)
The fault is Non-Latched. This type fault will clear itself if the cause of the fault no longer exists.

Trip Fault (T)
The fault trips the drive. This type fault causes the controller to remove power from the motor. The controller must be reset. All trip faults are latched.

Annunciated Fault (A)
The fault is annunciated only. If the drive is running it will not cause the drive to shut down. An annunciated fault will prevent the drive from starting if it is not running. Annunciated faults may or may not be latched.

There are several ways to reset the controller. A **software reset** may be executed from the Programmer card, or by serial command from a device connected to the serial link. A **hardware reset** may be executed by toggling the reset switch on the Drive Control card, by momentarily connecting 6PL-1 to 6PL-4 (3TB69 to 3TB66), or by momentarily interrupting incoming power to the drive. Most latched faults may be adequately cleared by a software reset, however, some faults should be cleared by a hardware reset. Fault types are indicated in Table 6. If a software reset does not seem to restore the drive to normal operation, a hardware reset should be attempted.

Example: FAULT CODE INDENTIFICATION

LED Indication	Binary Number						Decimal Number	Fault Number
	32	16	8	4	2	1		
Steady Display	0	1	1	1	0	0	16+8+4=28	FL28
Blinking Display	0	1	0	1	0	0	16+4+64=84	FL84

NOTE: *0=LED ON, 1=LED OFF, 1=BLINKING LED*

AC-300 GENERAL TROUBLESHOOTING CHART QUICK REFERENCE GUIDE

PROBLEM DESCRIPTION:

DRIVE FAULTS, LED INDICATION

1. All diagnostic LEDs off.
2. All diagnostic LEDs on.
3. Fault messages displayed (repetitive).
4. Fault messages displayed (non-repetitive, random).

PROGRAMMER PROBLEMS

5. Programmer display off.
6. Error messages displayed.
7. Programmer ineffective - no errors displayed.
8. Parameter changes ineffective.
9. Diagnostics do not perform.

FUSE BLOWING, MOV FAILURES, CARD FAILURES

10. Fuse blowing, AC or power fuses.
11. Power supply fuses blowing.
12. Repetitive card failures.
13. Line voltage problems.

OPERATIONAL PROBLEMS

14. 24 Volt logic not operating properly.
15. Drive/Motor does not run.
16. Drive runs in wrong direction.
17. Drive does not reverse.
18. Drive speed is incorrect.
19. Serial link does not function properly.
20. Erratic operation.
21. Stability problems.
22. Signal level detector chatter.
23. Too much coast time on deceleration (dynamic braking).

MOTOR PROBLEMS

24. Motor runs hot.
25. Motor vibration.

NOISE, WIRING, POWER SUPPLIES, ETC.

26. EEROM memory rewrites.
27. Noise or random nuisance faults.
28. Wiring procedures.
29. Checking power supplies.

TABLE 5: AC-300 GENERAL TROUBLESHOOTING CHART

PROBLEM	CHECK/ADJUST/REPLACE
1. All Diagnostic LEDs OFF	1. Check Programmer display for drive fault (if used). 2. Check for 3-phase AC input. 3. Check for open AC line fuses (FU1-3) or breaker. 4. Check for open fuses on the Power Supply card (FU1-5). 5. Ribbon cable(s) disconnected, open, loose, or intermittent. 6. Power supplies loaded down or failed - refer to "Checking Power Supplies", this section. 7. Attempt to enter the PARAMETER MODE on the Programmer card. Refer to Error Code List if any detected. A hard reset should clear this lock-up condition. 8. Replace Programmer card. 9. Replace Control card software.
2. All Diagnostic LEDs ON	1. Drive faulted - check fault code list. 2. RSET-3TB69 connected to +15V. 3. Replace Programmer card. 4. Replace Control card software.
3. Fault Messages Displayed (Repetitive)	1. See Specific Fault Troubleshooting, Table 6, for troubleshooting procedures related to specific drive fault number.
4. Fault Messages Displayed (Non-Repetitive, Random)	1. See problem description on "Noise or Random Nuisance Faults". 2. See "Wiring Procedure".
5. Programmer Display OFF	1. Check for proper insertion and connection of programmer plugs. 2. Check power supplies - refer to "Checking Power Supplies", this section. 3. Replace Programmer card.
6. Error Messages	1. See Error Message Listing in Table 7.
7. Programmer Ineffective - No Errors Displayed	1. Check for proper insertion and connection of programmer plugs. 2. Check jumpers on the Programmer card for proper position. 4. Is the drive in Auto mode - this mode inhibits certain mode references. 5. Replace Programmer card.
8. Parameter Changes Ineffective	1. Check JP13, write protect jumper on the Control card - must be in 2-3 or "ENABLE" position for programming. 2. Check setting in ADR003, DGNJP - must be an odd Number to allow reprogramming. 3. If the drive uses multi-motor operation for different gear ranges, spindle orient, etc., verify that the proper motor parameters are being adjusted. Check by setting the IOC trip setpoint of the motor parameters being adjusted to 10. Starting the drive will cause IOC fault if the proper motor parameters have been adjusted. 4. Check "Programmer Ineffective - No Errors Displayed" section. 5. Replace Programmer card. 6. Replace Control card software.

PROBLEM	CHECK/ADJUST/REPLACE
9. Diagnostics Do Not Perform	1. Verify that the particular function is available - refer to the Custom Software Description, software jumper descriptions. 2. Check JP13 on the Control card, must be in the 2-3 position for some Diagnostics tests to work. Refer to "Use of Optional Programmer" section. 3. Check the drive mode, Auto or Manual. The drive must be in Manual mode to run Diagnostic tests. 4. Check POL on 6PL-15 (3TB55) for reverse signal. 5. Refer to "Programmer Ineffective - No Errors Displayed", this section.
10. Fuse Blowing (AC Power Fuses)	1. Compare the fuse size with the Renewal Parts List. 2. Check for failed capacitors. 3. Check for out of tolerance, surging, or sagging AC line voltage - refer to "Line Voltage Problems" and "Noise or Random Nuisance Faults", this section. 4. Using a megger, check for a ground in drive motor circuit or blower circuit. (Review WARNINGS and CAUTIONS in Section 1.)
11. Power Supply Fuses Blowing	1. Verify that the fuses are correct size. 2. Check for any shorted or over-loaded power supplies (Refer to "Checking Power Supplies", this section). 3. Replace the Power Supply card.
12. Repetitive Card Failures	1. Check all connections at all terminal boards. 2. Verify that all ribbon cables and plugs are connected properly (i.e., not offset by one row or pin). 3. Check the stab-on connection points on the Power Supply and Base Drive cards. 4. Verify that all voltages (incoming AC and internal DC) are correct. 5. Check for moisture. 6. Check for damage due to airborne chemical contamination. 7. Check for intermittent external shorts to ground.
13. Line Voltage Problems	1. Verify that the line voltage is within drive specifications, -5% to +10%. 2. Verify that the AC line voltage is balanced line-to-line and line-to-ground. 3. Check for AC line notches or spikes. 4. Check for an ungrounded secondary on the power system transformer directly in front of the drive (if used). (The transformers are generally a delta/wye type with the neutral of the "wye" grounded through a high resistance or directly to ground.)
14. 24 Volt Logic Not Operating Properly	1. Check the $\pm$24 volt power supplies. 2. Check logic polarity control Jumpers per the Custom Software Description. 3. Check control logic connections at 3TB. 4. Check for proper polarity and common circuit if 24 volt supply is from a source other than drive. 5. Check that the drive is in the proper motor mode (0, 1, 2, or 3) for the particular motor or function being operated.

PROBLEM	CHECK/ADJUST/REPLACE
15. Drive/Motor Does Not Run	1. Fuses open (or optional circuit breaker open). 2. Fuses open on Power Supply card. 3. Check all hardware and software jumpers per the Custom Software Description. 4. Check control logic voltage and state at 3TB as well as power supply voltage levels. 5. Check that the drive is in the proper operate mode (AUTO or MANUAL). 6. Drive may be stalled in current limit.
16. Drive Runs in the Wrong Direction	1. Check for logic signal on 3TB55 (POL). 2. Motor connections reversed - correct the motor and tach leads. 3. Check all software jumpers per Custom Software Description. 4. Check the reference polarity. 5. Check logic polarity control jumpers per the Custom Software Description. 6. Check the Custom Software Description to verify that the reverse function is enabled when operating from the serial link.
17. Drive Does Not Reverse	1. Verify that the drive is a reversing drive. 2. Check the reference polarity. 3. Check control logic to POL, 3TB55 if used for reversing. 4. Check logic polarity control jumpers per the Custom Instructions or PC Software. 5. Check the Custom Software Description to verify that the function is enabled when operating from the serial link.
18. Drive Speed is Incorrect	1. Verify that the motor design is suitable for the application. 2. Check that the proper reference is enabled per the Custom Software Description. 3. Check for proper scaling of MSR input. 4. Check for proper ASR scaling. 5. Check for proper gain setting. 6. Check for noisy tach signal. Refer to "Wiring Procedures", this section. 7. Check that the drive is not in current limit. 8. Check that the drive is not in voltage limit. 9. Check that the Tach Scaling parameters are set correctly.
19. Serial Link Does Not Function Properly	1. Verify that all required jumpers to enable Serial Link are set correctly. 2. Check the baud rate of the drive compared to serial link device (hard reset must be performed afterward if baud rate is changed). 3. Check serial link connections at 3TB. 4. If RS422 is used, verify that the handshaking lines are properly connected. Verify that sending device is responding to the drive's DTR signal (low level) when the device is allowed to transmit to the drive. 5. Replace the Process Interface card.

PROBLEM	CHECK/ADJUST/REPLACE
20. Erratic Operation	1. Check power supplies for magnitude and ripple (refer to "Checking Power Supplies"). 2. Check connections on all ribbon cables. 3. Tighten all connections on terminal boards and all power connections. 4. Confirm correct hardware jumper setting per the Custom Software Description. 5. Check all parameters per the software test data sheet in the drive door or Custom Software Description. 6. Check for noisy ground connection - refer to "Wiring Procedures". 7. Check external control and reference connections.
21. Stability Problems	1. Check settings of all parameters per the software test data sheet inside the drive door pocket. 2. Verify that all contacts on the external contactor are closing, if used. 3. Refer to the "Troubleshooting Stability Problems" paragraph and Table 8 in this section.
22. Signal Level Detector Chatter	1. Verify SLD is not looking at a "noisy" signal (e.g., tach feedback, etc.). 2. Verify that enough hysteresis is programmed into the appropriate parameter address.
23. Too Much Coast Time on Deceleration	1. Check settings in Custom Software Description and confirm regeneration is enabled if dynamic braking is provided.
24. Motor Runs Hot	1. Motor may be overloaded, check for mechanical problems. 2. Check the motor winding temperature. If the motor has Class F insulation and the winding temperature is less than 130 C, this is normal. 3. Check the rotation of the motor blowers (if applicable). 4. If motor runs hot at very low speeds, check motor application guide for full-torque speed range. 5. Check Volts/Hz setting (ADR033, HVBAS).
25. Motor Vibration	1. Coast Stop Drive - if the vibration is electrically induced the vibration will disappear. If it does not disappear, check for mechanical causes. 2. Refer to "Stability" procedures. 3. Motor not mounted level. 4. Motor base vibrating.
26. EEROM Memory Rewrites	1. JP13 on the Drive Control card must be in the 1-2 position during "Power UP" or hard reset.
27. Noise or Random Nuisance Faults	1. Verify that the 5-volt DC supply is not used externally. 2. Check for suppression on all relays, contactors and starters. 3. Check all power supplies - refer to "Checking Power Supplies", this section. 4. Refer to "Wiring Procedures", this section.

PROBLEM	CHECK/ADJUST/REPLACE
28. Wiring Procedures	1. Check the wire sizes on the AC power connections. 2. Verify that the control wires (24-volt DC, 115-volt AC) do not run in the same conduit or run parallel, closer than four inches, to the power wires, causing noise pick-up. 3. Verify that all of the signal wires (i.e., tach, encoder, reference, load cell or RS-232C/422) are twisted pairs, shielded, and shields grounded at the drive. 4. Verify that the signal wires are not grounded at more than one end or other than the drive end only. 5. Verify that the signal wires do not run in the same conduit or parallel closer than four inches to the power wires. 6. Verify that there is only one system ground. 7. Verify that the ground bus is large enough to prevent voltage drops or noise pick up.
29. Checking Power Supplies	1. Check the +15V test point for +15VDC, $\pm$5%. Verify that there is less than or equal to 100 millivolts of ripple with an oscilloscope. 2. Check the -15V test point for -15VDC, $\pm$5%. Verify that there is less than or equal to 100 millivolts of ripple with an oscilloscope. 3. Check the +5V test point for +5VDC, $\pm$1%. Verify that there is less than or equal to 100 millivolts of ripple with an oscilloscope.

TABLE 6: SPECIFIC FAULT TROUBLESHOOTING

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
000001	FL01	IOC	**Instantaneous DC overcurrent (L,T)** -- Motor winding grounded. -- Mechanical Problems - binding shaft, unreleased brake, vibration, gearbox, or load problems. -- IOC parameter (address 043) set too low. -- Current limit set too high (check ILGEN and ILMMX ADR 040 and 041). -- Current feedback isolator failure - check 5PL & Base Drive Card. -- Check input at SJ2 (4TB16 or 13PL-16). -- ADR011-SJAJP set greater than zero. -- Transistor failure; try transistor test (Programmer Test 12). -- Incorrect values for parameters ILPPG, TQCPG, TQCTC (ADR 019, 020, 021).
000010	FL02	OVRSP	**Overspeed (sensed by analog tach circuitry) (L,T)** -- ADR073-SPDLM set too close to ADR068-REFSF and ADR071-REFLM - set a minimum of 5% greater than REFSF and REFLM. -- Speed overshoot occurs - speed regulator response too high, reduce gain. (Refer to TABLE 8, "Stability Troubleshooting Chart".) -- Tach feedback circuit failure -- check 7PL & tach wiring. -- Improper input at SJ1 or SJ2 (4TB15 or 4TB16).
000011	FL03	TOVTR	**DC Link Timed Overvoltage (L,T)** -- DB duty cycle signal exceeded programmed limit (DBTOT, ADR049); set DCELT (ADR058) higher/increase duty cycle time. -- Check connection of C1PL cable. -- Check N2 to N1 voltage across SM1, should not exceed 2 volts. -- Check AC line voltage, should not exceed rating +10%.
000100	FL04	FLOSS	**Magnitude of reconstructed flux below programmed percent (L,T)** -- Check FLDLS (ADR078), FLDMX (ADR076) and HVBAS (ADR033) settings. -- Check for open motor leads. -- Check for overload causing motor pull-out.

KEY: L - LATCHED N - NON-LATCHED H - REQUIRES HARD RESET
T - TRIPS DRIVE A - ANNUNCIATED ONLY

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
000101	FL05	TPOL	**Tach polarity incorrect (L,T)** -- Reversed tach connections to Process Interface card. -- Switch SW1-7 on Process Interface card set improperly - open for DC tach, closed for AC tach.
000110	FL06	JMPER	**Software jumper illegally changed while running (L,T)** -- Drive must be stopped to change parameters in ADR000 to ADR014.
000111	FL07	RESET	**Reset while running (L,T)** -- Drive powered up with RUN or JOG closed - check 6PL-12 and 6PL-13 (3TB58 and 57). -- Momentary loss of incoming power while running.
001000	FL08	TOC	**Timed overcurrent (L,T)** -- Motor mechanically overloaded - check for change in process load, worn bearings, gears or brakes. -- ADR044-TOCTH and ADR045-TOCGN set too low - check against the parameter list shipped with drive. -- Current feedback failure -- check signals at Control card, test posts IA, IB, IC. -- Motor winding failed.
001001	FL09	LNKTR	**Serial Link trip (drive told to trip via serial link) (L,T)** -- Drive received external trip signal - check for external fault. -- 12PL or 8PL incorrectly connected or damaged.
001010	FL10	LNKFL	**Serial Link fault (communications failure) (L,T)** -- Programmer card jumpers incorrect - refer to TABLE 1. -- Baud rate set incorrectly - check ADR126-BAUDR. -- 12PL or 8PL incorrectly connected or damaged. -- Intermittent connection between 3TB32 (CTSA) and 12PL-7 or 3TB41 (CTSB) and 12PL-14.
001011	FL11	FLT11	**(Unused at this time)**
001100	FL12	FLT12	**(Unused at this time)**
001101	FL13	EEXXX	**Checksum error or parameter out of range where XXX equals a 3-digit number, indicating an out-of- range parameter address or altered checksum. (L, T, H)** -- Spontaneous parameter alteration - check all parameters and verify EEROM is not write "ENABLED" during power up or power down, or "hard" reset. -- Value other than zero in ADR255-UNDFD - replace EEROM. -- Control card revision level earlier than "AD" -- change Control card.

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T - TRIPS DRIVE A - ANNUNCIATED ONLY

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
001110	FL14	ILOOP	**Current Loop: Analog Current Regulator power-up test failed (L,T)** -- Check for triangle wave present at Control card test posts. -- Check cable connections, 5PL, 1APL, 2APL and 3APL. -- Replace Base Drive card. -- Check current feedback at Control card test posts IA, IB and IC to PCOM. If in standby these voltages exceed 0.75 volts, check connections to the shunt for that phase.
001111	FL15	VFBLS	**(Unused at this time)**
010000	FL16	NOTRP	**Trip from N.O. on Application card (L,T)** -- External input at 4TB8 (13PL-8). -- Application card absent but selected by soft jumper ADR010-APPJP should be zero when no Application card is present.
010001	FL17	XSTOP	**Coast stop trip (L,T)** -- Check for input at XSTOP on 3TB1 (6PL-16). -- Check ADR004-STPJP - a 4 causes coast stop when XSTOP is selected.
010010	FL18	STALL	**Stalled overcurrent (L,T)** -- Mechanical problems - failed brake, frozen bearings, application problem, etc. -- ADR046-STLTH and ADR047-STLGN set wrong - check against "as shipped" parameter list. -- Drive running less than 1% of top speed and motor current greater than ADR046-STLTH. -- Motor winding failed.
010011	FL19	OVERW	**Overfrequency (L,T)** -- Motor frequency exceeded 1.15 times programmed limit FRQLM (ADR080) -- check maximum speed and frequency programmed. -- Check tach feedback for noise.
010100	FL20	TLOSS	**Loss of tachometer feedback (L,T)** -- Tach leads disconnected. -- Tach feedback circuit failure - 7PL disconnected or damaged. -- Process Interface card jumpers or switches incorrect. Refer to TABLE 1. -- P7-SFB ZERO on Process Interface card misadjusted - set test point SFB for zero volts at zero speed. -- Process Interface card failed - replace.

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T - TRIPS DRIVE A - ANNUNCIATED ONLY

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
010101	FL21	NCTRP	**Trip from N.C. Interlock on Application card (L,T)** -- Loss of external input on 4TB7 (13PL-7). -- Application card absent but selected by soft jumper - ADR010-APPJP should be zero when no Application card present.
010110	FL22	PHLOS	**AC phase loss (L,T)** -- Fuse blown in AC line - check FU1-3. -- Low or unbalanced AC line voltage - line-to-line or line-to-neutral more than 5% low. -- Loose AC power connections. -- AC line impedance too low -- see rules under "Reactors" in "Main Controller Elements and Their Function". -- Check for failed SM modules. -- Replace Power Supply card.
010111	FL23	OVERF	**(Unused at this time)**
011000	FL24	U3 FL	**Faulty RAM on U3 8155 (L,T)** -- Replace U3 on the Control card.
011001	FL25	DC-OV	**DC Link Overvoltage (L,T)** -- If DB option is furnished, check if DBPL cable is damaged or disconnected. -- If DB option is furnished, check DB operation. -- Reduce regeneration current limit -- ILGEN (ADR040). -- Increase deceleration time -- DCELT (ADR058). -- Check for proper connection of C1 lead between Power Supply card and DC Link if DB provided. -- Check to see if DB resistor is hooked up.
011010	FL26	ITRIP	**Inverter Fault Trip (L,T)** -- Check motor winding for shorts or grounding. -- Check for shorted transistor (Programmer TEST 12). -- Replace Base Drive Card.
011011	FL27	DC-UV	**DC Link Undervoltage (L,T)** -- Check AC line voltage -- should agree with controller rated AC voltage. -- Check for blown AC power fuse. -- Check for blown DC link fuse. -- Check for failed rectifiers or transistors. -- Check if C1PL cable is disconnected or damaged. -- Check if charge control resistor is open. -- Check to see if DB resistor is hooked up.
011100	FL28	RMODE	**Illegal selection of RUN mode (N,A)** -- Do not command DIAGNOSTIC RUN or STATIC while running or vice versa. -- Attempted to run Auto-Tune when feature not supplied.

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
011101	FL29	CNLON	**Control on input opened (L,T)** -- Open circuit between 3TB3 and 3TB4. -- 6PL incorrectly connected or damaged. -- Noise between 3TB3 and 3TB4.
011110	FL30	NOAPP	**Application card failure (L,T)** -- Application card absent but selected by soft jumper - ADR010-APPJP should be zero when no Application card present. -- 3PL incorrectly connected or damaged. -- Replace Application card.
011111	FL31	DBCHK	**Dynamic Brake Check (L,T)** -- Check if DB resistor kit is disconnected. -- Check if P1 lead, C1 lead or C1PL cable is disconnected or damaged. C1 to N2 should not exceed 20 volts (when not running). -- Check for possible DC link charge control malfunction and if N2 to N1 voltage across SM1 module exceeds 5 volts, replace SM1. -- Replace Power Supply card.
100000	FL32	TOCAN	**Timed overcurrent annunciation (N,A)** -- Motor mechanically overloaded - check for change in process load, worn bearings, gears, or brakes. -- ADR044-TOCTH and ADR045-TOCGN set too low - check against the parameter list shipped with drive. -- Motor winding failed. -- Check OPTJP (ADR012) for desired overcurrent detection.
100001	FL33	TOVAN	**DC Link Timed Overvoltage (N,A)** -- DB duty cycle exceeded 70% of DBTOT (ADR049 setting). -- AC supply voltage exceeds rating +10%. -- Deceleration time (ADR058 - DECEL) set too low.
100010	FL34	PLLSY	**Flux Reconstruction loss of sync (L,T)** -- Check for open motor leads. -- Check for motor shaft binding, unreleased brake etc., preventing motor rotation. -- Check LHVBA (ADR034) for too high a setting -- try reducing setting.
100011	FL35	O SPD	**Motor rotating when RUN commanded (N,A)** -- RUN command initiated during motor rotation. -- ADR048-RSTOP set too low. -- Test point SFB contains an offset at zero speed - adjust P7-SFB ZERO on the Process Interface card.

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T - TRIPS DRIVE A - ANNUNCIATED ONLY

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
100100	FL36	LNKER	**Uncipherable messages detected on serial link by drive or other device (N,A)** -- Excessive noise on 12PL-7 thru 14. -- 12PL incorrectly connected or damaged. -- 5V DC power supply noisy. -- Baud rate incorrectly set - ADR126-BAUDR normally set at 4800. -- Check Process Interface card jumpers JP1, JP11, JP12, JP13 and JP14. -- Serial link message protocol incorrect, message too long, checksum byte incorrect, or handshaking incorrect - refer to "Serial Link User's Guide". -- Replace Process Interface card. -- Replace Control card. -- Replace Digital Operator Station.
100101	FL37	UNDFD	**(Undefined at this time)**
100110	FL38	ELOCK	**Attempt to change software adjustments when soft locked (N,A)** -- ADR003-DGNJP must be odd to program EEROM - add one to value in DGNJP.
100111	FL39	EFAIL	**Software adjustment not stored in EEROM (N,A)** -- JP13 (or SW1) on Control card in hard lock position. -- EEROM failed - replace.
101000	FL40	MXSPD	**Speed reference greater than REFLM speed adjustment (N,A)** -- Manual Reference jumper JP1 on Control card improperly set - see TABLE 1. -- P1-REF SCALE on the Control card misadjusted - see TABLE 1. -- ADR071-REFLM set below ADR070-RPMSF. -- P1-ASR SCALE, on the Application card misadjusted (if used) - see TABLE 1. -- Process follower on Process Interface card misadjusted (if used) - see TABLE 1. -- RPM/HZ (ADR075) set improperly for frequency reference (if used). -- Top speed reference for parallel input (if used) is set higher than ADR068-REFSF and ADR070-RPMSF. -- Preset references RUNRF, R1RF, R2RF, and SLWRF (ADR 061, 064, 065, and 062) set too high (if used).
101001	FL41	KEYOF	**Keypad function disable (N,A)** -- Programmer key depressed for a function that is not enabled -- check setting of SERJP (ADR005).

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Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
101010	FL42	RLOGC	**Run logic error (N,A)** -- JOG, RUN, or DIAGNOSTIC RUN called for with XSTOP open. -- Keypad STOP called for during an external run cycle. -- STOP not called for before calling for START.
101011	FL43	U11FL	**A/D converter failure (L,T)** -- Check for noise on ACOM. -- Replace U11 on the Control card. -- Check U18, U19, and U3 if U11 ok.
101100	FL44	U4LAT	**U4 to U5 communication error (L,T)** -- Check PROM revision compatibility. -- Check for noise on power supplies or analog inputs. -- Replace Control card.
101101	FL45	SLOOP	**Speed tracking fault (L,T)** -- Drive entered current limit of CEMF limit when tracking enabled. -- ADR036-SERLM set too low - should be between 1and 7. -- Speed regulator unstable - see "Stability Troubleshooting" paragraph and Table 8 in this section.
101110	FL46	----	**SPARE**
101111	FL47	OADJ>	**A/D converter offset too low (N,A)** -- P4-OFFSET on the Control card misadjusted - turn CW. -- Check for noise on ACOM.
110000	FL48	OADJ<	**A/D converter offset too high (N,A)** -- P4-OFFSET on the Control card misadjusted - turn CCW. -- Check for noise on ACOM.
110001	FL49	NO U4	**U4 to U6 communications error (L,T)** -- Check for U1 and U6 revision compatibility. -- Replace U4, U2, and/or U1 on the Control card.
110010	FL50	U4BAD	**U4 RAM, timer or ALU is faulty (L,T)** -- Replace U4 on the Control card.
110011	FL51	U1ROM	**U1 ROM is faulty (L,T)** -- Replace U1 on the Control card.
110100	FL52	NCLSD	**Annunciate N.C. interlock on Application card (N,A)** -- Check external input on 4TB7 (13PL-7). -- Application card absent but selected by soft jumper - ADR010-APPJP should be zero if no Application card is present.

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T - TRIPS DRIVE A - ANNUNCIATED ONLY

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
110101	FL53	U4COM	**U4 communication with U5 or U6 is bad (L,T)** -- Check for PROM revision compatibility. -- Replace U5 and/or U6 or the Control card. -- Digital tach pulse rate exceeds 41000 Hertz.
110110	FL54	NO U5	**U5 to U4 communication error (L,T)** -- Check revision compatibility. -- Replace U5 on the Control card.
110111	FL55	U5BAD	**(Not used at this time)**
111000	FL56	U5ROM	**U5 ROM is failed (L,T)** -- Replace U5 on the Control card.
111001	FL57	NOPEN	**Annunciate N.O. interlocks on Application card (N,A)** -- External input on 4TB8 (13PL-8). -- Application card absent but selected by soft jumper - ADR010-APPJP should be zero if no Application card is present.
111010	FL58	U5COM	**U4 not communicating correctly with U5 (L,T)** -- Check for U1 and U5 PROM revision compatibility. -- Digital tach pulse rate exceeds 41,000 Hertz. -- Noise on tach input. -- Replace Control card.
111011	FL59	NO U6	**U6 does not communicate properly with U4 (L,T)** -- Check revision compatibility. -- Replace U6 on the Control card.
111100	FL60	U6BAD	**U6 RAM, timer or ALU is faulty (L,T)** -- Replace U6 on the Control card.
111101	FL61	U6ROM	**U6 ROM is faulty (L,T)** -- Replace U6 on the Control card.
111110	FL62	----	**SPARE**
111111	FL63	U6COM	**U4 not communicating correctly with U6 (L,T)** -- Check for U1 and U6 PROM revision compatibility. -- Digital tach pulse rate exceeds 41,000 Hertz. -- Check for noise on tach input signal. -- +15V shorted to ACOM (3TB).
000000	FL64	----	**SPARE**
000001	FL65	A1FOP	**TM1 (C1-E1) is not operating (L,T)** -- Check for open B1, E1 leads or faulty 1APL cable. -- Replace TM1 transistor module.

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
000010	FL66	A2FOP	**TM2 (C1-E1) is not operating (L,T)** -- Check for open B1, E1 leads or faulty 1BPL cable. -- Replace TM2 transistor module.
000011	FL67	A3FOP	**TM3 (C1-E1) is not operating (L,T)** -- Check for open B1, E1 leads or faulty 1CPL cable. -- Replace TM3 transistor module.
000100	FL68	A4FOP	**TM1 (C2-E2) is not operating (L,T)** -- Check for open B2, E2 leads or faulty 2APL cable. -- Replace TM1 transistor module (TM4 on 40HP or larger).
000101	FL69	A5FOP	**TM2 (C2-E2) is not operating (L,T)** -- Check for open B2, E2 leads or faulty 2BPL cable. -- Replace TM2 transistor module (TM5 on 40HP or larger).
000110	FL70	A6FOP	**TM3 (C2-E2) is not operating (L,T)** -- Check for open B2, E2 leads or faulty 2CPL cable. -- Replace TM3 transistor module (TM6 on 40HP or larger).
000111	FL71	UNDFD	**(Undefined at this time.)**
001000	FL72	MOTOP	**Motor open (all cells detected open)** -- Check for open motor contactor. -- 5PL cable disconnected or damaged. -- Replace Base Drive card. -- Check for blown DC link fuse.
001001	FL73	ALGOP	**'A' Leg Open** -- Check for open motor lead. -- Check for open base/emitter leads on TM1 (or TM4 if greater than 40HP); replace TM1 or TM4.
001010	FL74	BLGOP	**'B' Leg Open** -- Check for open motor lead. -- Check for open base/emitter leads on TM2 (or TM5 if greater than 40HP); replace TM2 or TM5.
001011	FL75	CLGOP	**'C' Leg Open** -- Check for open motor lead. -- Check for open base/emitter leads on TM3 (or TM6 if greater than 40HP); replace TM3 or TM6.
001100	FL76	UNDFD	**(Undefined at this time.)**
001101	FL77	UNDFD	**(Undefined at this time.)**
001110	FL78	UNDFD	**(Undefined at this time.)**
001111	FL79	UNDFD	**(Undefined at this time.)**

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
010000	FL80	UNDFD	**(Undefined at this time.)**
010001	FL81	A1FSH	**TM1 (C1-E1) is shorted (L,T)** -- Check TM1 transistor module and replace if shorted. -- Check for grounded motor winding.
010010	FL82	A2FSH	**TM2 (C1-E1) is shorted (L,T)** -- Check TM2 transistor module and replace if shorted. -- Check for grounded motor winding.
010011	FL83	A3FSH	**TM3(C1-E1) is shorted (L,T)** -- Check TM3 transistor module and replace if shorted. -- Check for grounded motor winding.
010100	FL84	A4FSH	**TM1 (C2-E2) is shorted (L,T)** -- Check TM1 transistor module (TM4 on 40HP or larger) and replace if shorted. -- Check for grounded motor winding.
010101	FL85	A5FSH	**TM2 (C2-E2) is shorted (L,T)** -- Check TM2 transistor module (TM5 on 40HP or larger) and replace if shorted. -- Check for grounded motor winding.
010110	FL86	A6FSH	**TM3 (C2-E2) is shorted (L,T)** -- Check TM3 transistor module (TM6 on 40HP or larger) and replace if shorted. -- Check for grounded motor winding.
010111	FL87	NOUAP	**Microapplication card failure (L,T)** -- Microapplication card absent but selected by soft jumper - ADR014-SPCJP should be set to disable Microapplication card interface (refer to Custom Software Description). -- 3PL incorrectly connected or damaged. -- U30 or U33 on Microapplication card failed - replace. -- Replace Microapplication card.
011000	FL88	UAPWD	**MUP watchdog misoperated (L,T)** -- Application card (G2) installed when Microapplication card is called for - check setting in ADR014-SPCJP per Custom Software Description. -- U30 on Microapplication card failed or jumpers set incorrectly - replace card if necessary. -- Refer to Table 5, "Noise or Random Nuisance Faults" item.
011001	FL89	MUPTO	**RUP to MUP communication failure (L,T)** -- U30 on Microapplication card failed - replace. -- Replace Microapplication card.

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
011010	FL90	NOMSG	**RUP to MUP watchdog detected no message received within time period expected (L,T)** -- Serial Link or Local Area Network transmission lines disconnected. -- 12PL damaged or disconnected. -- External device failed to send message in time period set by ADR005-SERJP.
011011	FL91	----	**SPARE**
011100	FL92	----	**SPARE**
011101	FL93	----	**SPARE**
011110	FL94	----	**SPARE**
011111	FL95	----	**SPARE**
100000	FL96	----	**SPARE**
100001	FL97	OCTFL	**Inconsistent Results on Open Circuit Test** -- Check for open motor leads or contactor. -- Check all Base Drive cables (1APL, 2APL, 1BPL, 2BPL, 1CPL, 2CPL) for damage or disconnection. -- Check for blown DC link fuse. -- Check 5PL cable for damage or disconnection. -- Replace Base Drive card.
100010	FL98	OCTMF	**Multiple Fault; ambiguous results on Open Circuit Test** -- Check for open motor leads or contactor. -- Check all Base Drive cables (1APL, 2APL, 1BPL, 2BPL, 1CPL, 2CPL) for damage or disconnection. -- Check for blown DC link fuse. -- Check 5PL cable for damage or disconnection. -- Replace Base Drive card.
100011	FL99	----	**SPARE**

KEY: L - LATCHED N - NON-LATCHED H - REQUIRES HARD RESET
T - TRIPS DRIVE A - ANNUNCIATED ONLY

TABLE 7: SPECIFIC ERROR TROUBLESHOOTING

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
--	ER01	BUART	**Faulty UART on the 8031 U.C.** -- Check JP3 and JP4 on Programmer card. -- Programmer card failed - replace.
--	ER02	BKPAD	**The keypad has shorted leads.** -- Replace ribbon cable. -- A key pressed during power up or reset. -- Programmer card failed - replace.
--	ER03	BDMEM	**The random access memory is faulty.** -- Programmer card failed - replace.
--	ER04	BPRGM	**The program memory checksum is incorrect.** -- Programmer card failed - replace.
--	ER05	BTIMR	**The timers in 8031 U.C. are faulty.** -- Programmer card failed - replace.
--	ER06	NADDR	**An address is not specified.** -- "ENTER" key pressed without specifying a parameter address.
--	ER07	2 KEYS	**Two keys were pressed together.** -- Press keys one at a time. -- Keypad failed - replace. -- Programmer card failed - replace.
--	ER08	ILBRD	**Illegal broadcast.** -- Programmer card failed - replace.
--	ER11	NOACK	**No acknowledgement received.** -- Device connected to the serial link not responding to the acknowledge requests from the Programmer card. -- Wires to the serial link disconnected. -- Driver or receiver chip failed on Programmer, Process Interface card, or external device. -- Processors locked up, use hard reset. -- If Hand-Held Programmer, may be failed U6 port - replace. -- Noise on transmission wires. -- RS232C on hand-held cable.

Diagnostic Readout	Prog. Card	Name	Description/Probable Causes/Remedies
--	ER12	N-CTS	**No "clear-to-send".** -- Serial link transmission lines disconnected. -- Check JP3 and JP4 on Programmer card. -- Noise on 5-volt power supply. -- Ground loops or unsuppressed relays. -- Programmer card failed - replace.
--	ER13	ORBUF	**Receive buffer has overflowed.** -- Message sent to the drive which exceeds the maximum number of bytes (11 data bytes). -- Baud rate set incorrectly. -- Byte protocol set incorrectly on external device - refer to Serial Link User's Guide. -- Message protocol used incorrectly - refer to "Serial Link User's Guide". -- Hardware configuration incorrect for serial link operation. -- External serial link device failed.
--	ER14	CKSUM	**Incorrect checksum received.** -- The serial link message checksum has been calculated improperly by external Serial Link device - refer to the "Serial Link User's Guide". -- Baud rates set incorrectly. -- Byte protocol set incorrectly on external device - refer to "Serial Link User's Guide". -- Message protocol used incorrectly - refer to "Serial Link User's Guide". -- Hardware configuration incorrect for serial link operation. -- External serial link device failed.
--	ER15	BDMSG	**Something other than a "start of transmission" was received when "S.O.T." was expected.** -- "Start of transmission" byte of the message sent over the serial link has been determined incorrectly - refer to the "Serial Link User's Guide". -- Baud rates set incorrectly. -- Byte protocol set incorrectly on external device (1 start bit, 8 data bits, and 1 stop bit). -- Message protocol used incorrectly - refer to "Serial Link User's Guide". -- Hardware configuration incorrect for serial link operation. -- External serial link device failed.

TROUBLESHOOTING STABILITY PROBLEMS

There are three broad categories of drive symptoms broadly classified as "Stability" problems. Technically, two of these are not actually stability problems but, due to their behavior, are included in this troubleshooting procedure. These categories are as follows:

"Stability" problems: "Hunting", oscillations in speed or current, etc. In general all problems in this category are cyclical in nature, with well-defined sinusoidal periods of oscillation. Usually these problems are electrically induced and can be tuned out with proportional and integral gain adjustments.

There are at least three regulating loops in the AC-300 drive, any one which can be mistuned, causing stability problems under the right conditions. They are as follows:

(1) Current regulator
(2) Voltage regulator
(3) Speed regulator

NOTE: *Various forms of position regulators are also available, which, due to their variety are not addressed here.*

Mechanical Oscillation Problems: These are a direct result of mechanical deficiencies but under certain conditions can be aggravated or excited by electrical frequencies. Examples of this are vibration, audible motor noise, gear chatter, etc.

Erratic Operation Problems: Intermittent speed or current fluctuations, mechanical "bumping", or other disturbances non-sinusoidal or non-periodic in nature. These may be induced by intermittent grounds, "noise", or mechanical problems.

NOTE: *A Programmer (Local or Hand-Held) is required to make parameter checks called for in Part I and adjustments as required in Part II.*

I. **Before troubleshooting the drive, first verify the following:**
   - -- All software jumpers are in accordance with drive parameter list.
   - -- All hardware jumpers are in accordance with the jumper selection chart.
   - -- All software parameters are in accordance with the parameter list.
   - -- All wiring is correct and in agreement with the drive elementaries.

II. **Use Table 8 to decide which type of problem actually exists, keeping the following pointers in mind:**
   - -- AC-300 drives are almost always stable.
   - -- Most stability problems, but not all, are corrected by adjusting speed loop parameters.
   - -- In general, when troubleshooting cascading or series regulating loops, the correct procedure is to stabilize the inner regulator first and then move outward; i.e., current regulator, then speed regulator, then position regulator (if furnished).
   - -- If at any time the drive becomes stable as a result of a step, record any parameter changes, and do not continue with the remaining procedures.

TABLE 8: AC-300 STABILITY TROUBLESHOOTING

SYMPTOM	PROCEDURES
A. Unstable when unloaded but stable when loaded, drive is stable but exhibits gearbox or motor noise, **or** drive runs erratically.	1. See Custom Software Description for gain changes in current loop regulator. 2. Perform Transistor Test 12 to verify transistors are good. 3. If instability still exists, return adjusted parameters to their original setting and skip to Symptom D.
B. Speed instability below base speed.	1. See Custom Software Description for gain changes in speed loop. 2. If no improvement, return the adjusted parameters to their original settings. 3. Adjust ADR016 (BSLIG) for operation with tach or ADR026 (CLPIG) for no tach operation using the **[INC]** and **[DEC]** functions on the keypad until stability is achieved. Press **[ENTER]** to permanently save any changes. 4. If stablility problems still exist, change value of ADR037 (SLEAD) using the **INC/DEC** functions on the keypad until stability is achieved (usually a lower value). Reenter the operator mode by pressing **[RESET] [RESET] [RESET].** 5. If there is some improvement but instability still exists - document changes and skip to Symptom D.
C. Speed stability problems above base speed.	1. See Custom Software Description for changes in stability parameters. 2. If no improvement, return the adjusted parameters to their original settings. 3. If the drive is unstable above base speed, adjust ADR018 (WSLIG) for tach operation or ADR026 (CLPIG) for no tach operation using the **[INC]** and **[DEC]** functions on the keypad until stability is achieved. Press **[ENTER]** to permanently save any changes. 4. If stablility problems still exist, change value of ADR037 (SLEAD) using the **INC/DEC** functions on the keypad until stability is achieved (usually a lower value). Reenter the operator mode by pressing **[RESET] [RESET] [RESET].** 5. If there is some improvement but instability still exists - document changes and proceed to Symptom D.

SYMPTOM	PROCEDURES
D. Drive is unstable: over a small part of the speed range, **or** when loaded but stable when unloaded, **or** such that the oscillating frequency is proportional to speed.	1. Confirm that the problem is not tach related - correct tach feedback problem (misaligned or loose coupling, belt slippage, wiring, or excessive ripple). 2. Verify that the power supplies are clean with an oscilloscope (refer to "Checking Power Supplies" paragraph in this section). 3. Run drive to top speed and push the "RESET" button on the Control card. The drive will **coast** stop. Mechanically induced problems will still be present as the motor decelerates. If problem disappears, continue to next step. 4. Determine resonant frequency "Fr" in Hz with an oscilloscope, if possible. Convert to radians per second as follows: "Wr" = 6.28 x "Fr". 5. If no improvement, return the parameters to their original settings and skip to note at bottom of chart.
E. Undirectional Stability Problems.	1. Inspect the load to make sure that nothing is wrong mechanically. See symptom **D.**

If any additional procedures are required, contact GE DRIVE PRODUCTS OPERATION, Product Service Engineering, for assistance/instructions. (814) 875-3851

NOTES:

10. TYPICAL WAVEFORMS

The following are examples of typical waveforms that can be seen on an oscilloscope. Each figure is accompanied by the name of the signal, the test point which it is derived from (in parenthesis), and the mode of operation if applicable.

All waveform examples are at **rated speed** and **rated load**. Graph divisions are **0.2 Volts per division vertical** and **50 milliseconds per division horizontal**. (For information on using an oscilloscope, consult the owner's manual.)

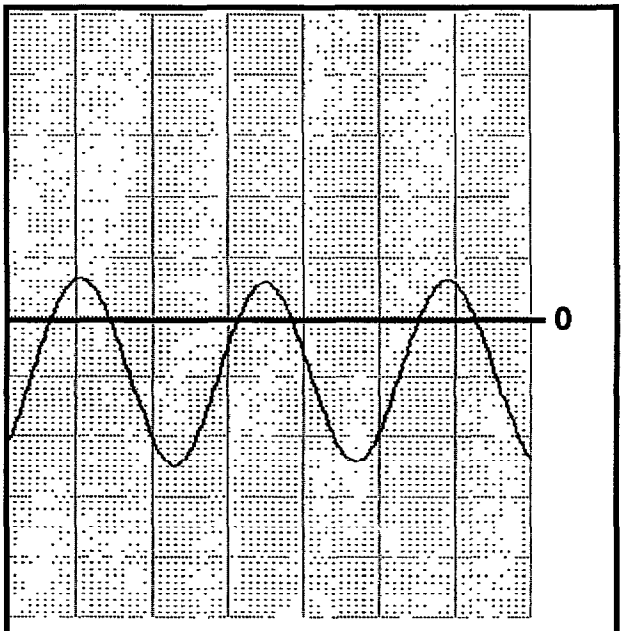


Figure 10-1. Control Card Test Point IA* (Motor Phase A Current Reference at Rated Load and Speed)

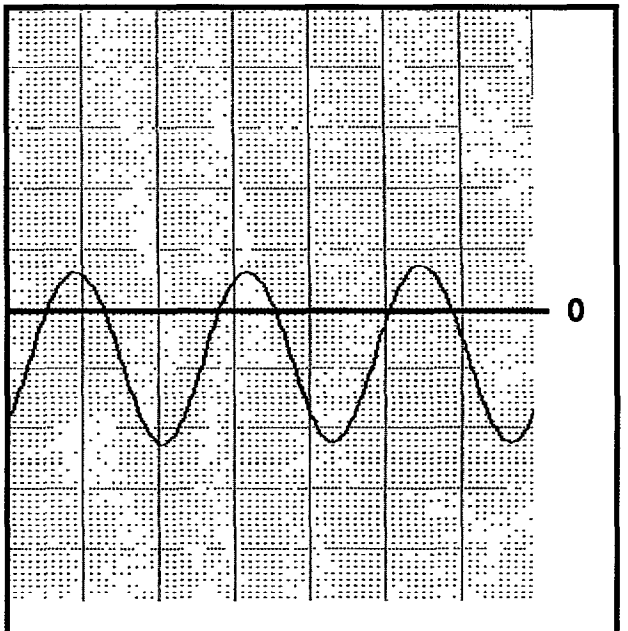


Figure 10-2. Control Card Test Point IB* (Motor Phase B Current Reference at Rated Load and Speed)

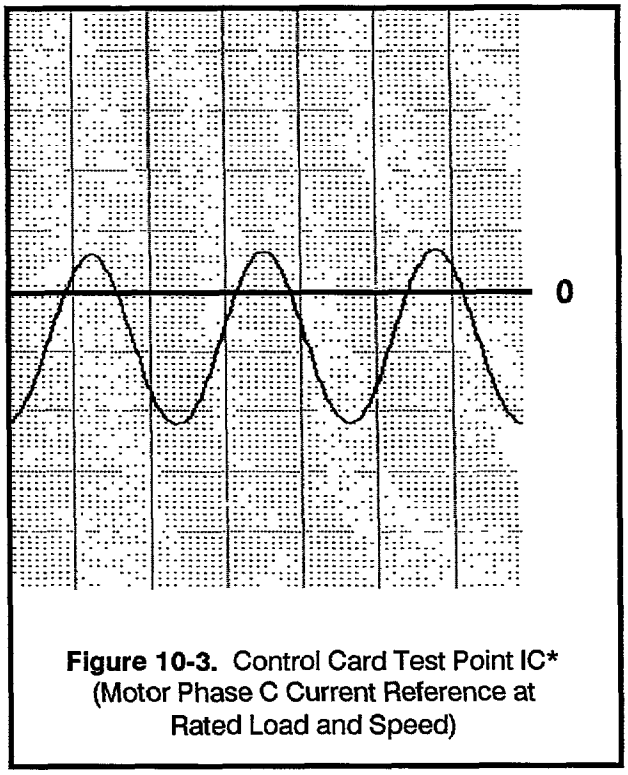


Figure 10-3. Control Card Test Point IC*
(Motor Phase C Current Reference at
Rated Load and Speed)

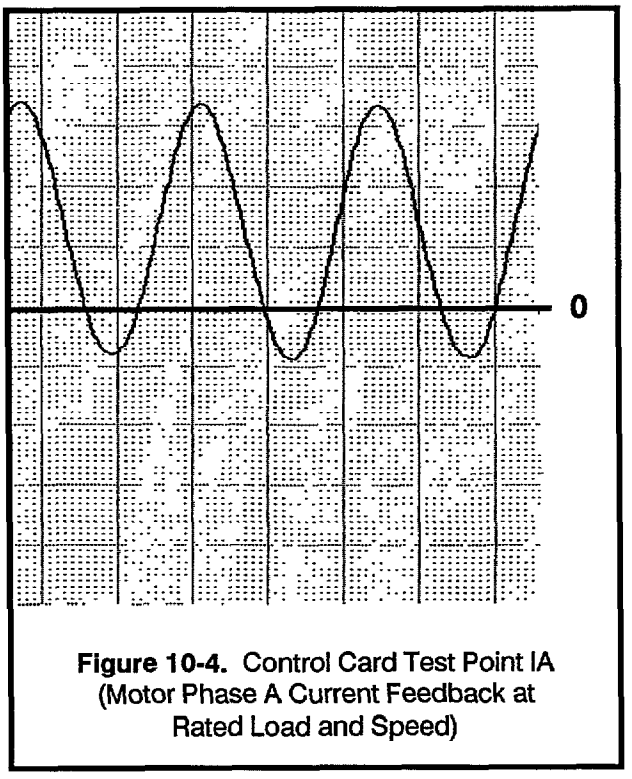


Figure 10-4. Control Card Test Point IA
(Motor Phase A Current Feedback at
Rated Load and Speed)

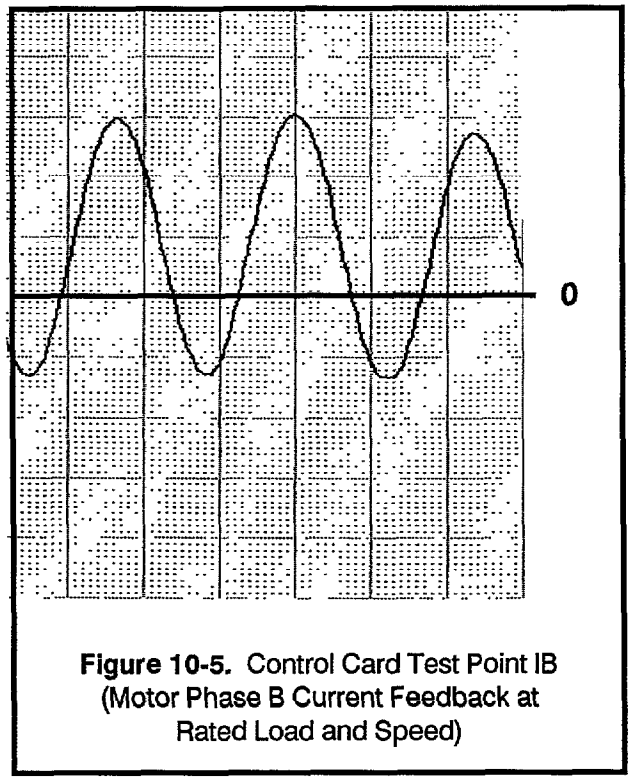


Figure 10-5. Control Card Test Point IB
(Motor Phase B Current Feedback at
Rated Load and Speed)

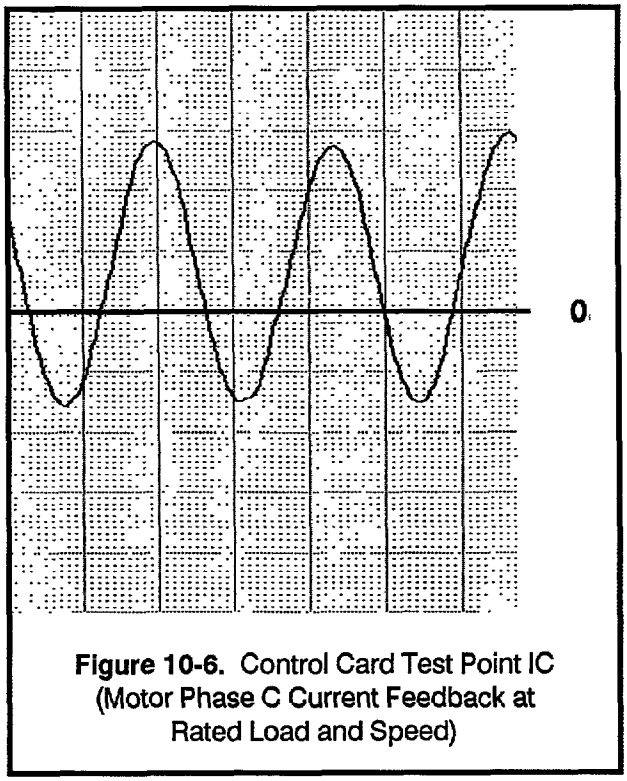


Figure 10-6. Control Card Test Point IC
(Motor Phase C Current Feedback at
Rated Load and Speed)

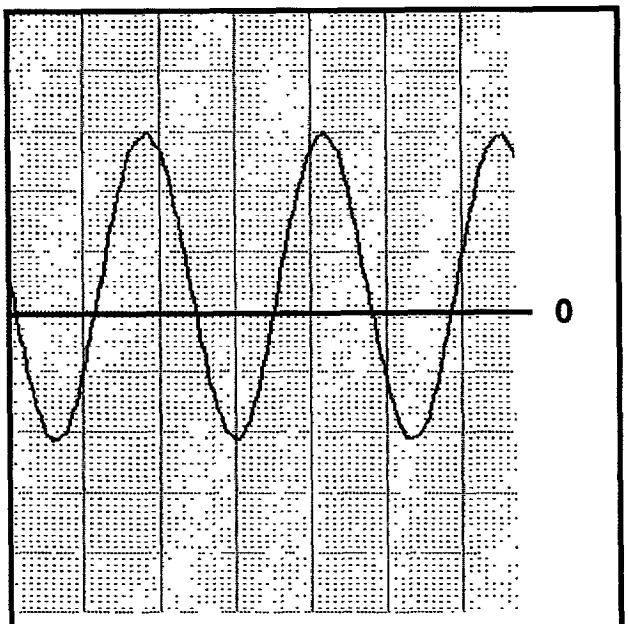


Figure 10-7. Control Card Test Point VA
(Motor Voltage Feedback at Rated Load/Speed)

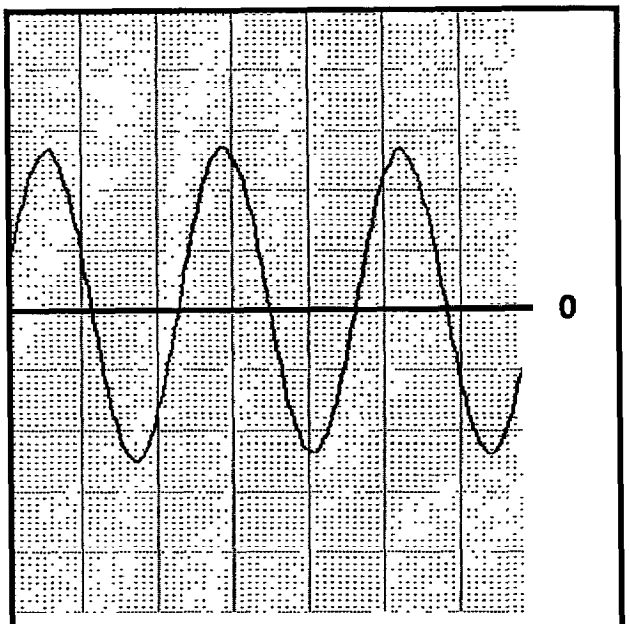


Figure 10-8. Control Card Test Point VB
(Motor Voltage Feedback at Rated Load/Speed)

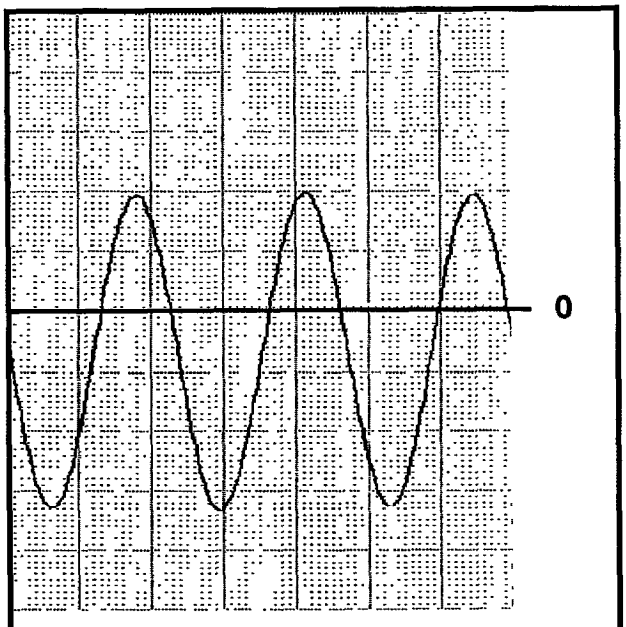


Figure 10-9. Control Card Test Point YA
(Y_A Volts/Hertz [Flux] Below Crossover)

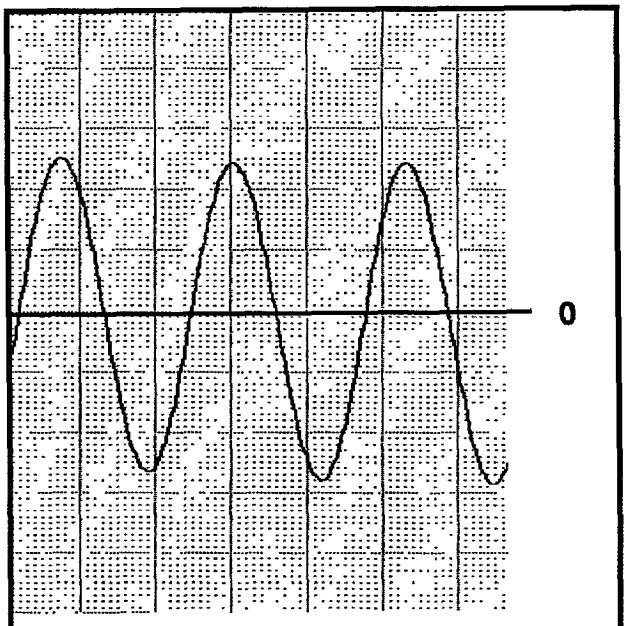


Figure 10-10. Control Card Test Point YB
(Y_B Volts/Hertz [Flux] Below Crossover)

NOTES:

11. USE OF THE OPTIONAL PROGRAMMER MODULE

The optional Programmer assembly is available in two configurations, as shown in Figures 11-1 and 11-2. The **Local** configuration is mounted on the Control card and plugged into 18PL. The **Hand-held** configuration is contained in a box and plugged directly into the version "102" Control card at 18PL. The Hand-held Programmer uses a different cable than the Local Programmer and care must be exercised when the Hand-held Programmer is plugged in.

CAUTION: The hand-held programmer cable must be oriented properly or the Control and Programmer cards will be damaged.

Orient the Hand-held Programmer cable as follows:

1. Hold the unit in front of the drive and extend the ribbon cable back to 18PL.
2. Verify that the cable is not twisted, and plug into 18PL.
   -- The colored stripe in the ribbon cable should be on the left when looking at the front of the Hand-held Programmer and should be on the left when plugged into 18PL.

Orient the Local Programmer cable as follows:

1. Hold the unit in front of the drive and extend the ribbon cable back to 18PL.
2. Verify that cable is oriented with the colored wire and unused connector pins on the left side of the ribbon cable when facing the drive, and plug into 18PL.

The optional Programmers provide an alphanumeric readout of Fault Codes (instead of the diagnostic LEDs) and operation/adjustment of the drive through three modes - **OPERATE, PARAMETER,** and **DIAGNOSTIC.**

OPERATE Mode - The "Normal" Mode, allows the red and black keypad functions to be active (the **RED** keys are dominant). This is the mode the drive will enter when power is applied.

PARAMETER Mode - Allows the user to examine and change as necessary any of 254 EEROM (E^2) Parameter addresses to "customize" the drive for a given application. In this mode all **BLACK** keys are active plus **[INC], [DEC],** and **[STOP].**

DIAGNOSTIC Mode - Allows the user to utilize any of the internal digital displays, plus run a series of diagnostic checks and self-tuning procedures. In this mode, both the **BLUE** and **BLACK** key functions are active (with **BLUE** keys dominant), plus **[INC], [DEC],** and **[STOP].**

CHANGING BETWEEN MODES

NOTE: *Throughout this section, a number or function encased in brackets [ ] indicates a programmer key pressed. The drive will normally enter the* ***OPERATE*** *mode as soon as power is applied.*

To enter the PARAMETER Mode, perform the following:

1. Confirm that the display reads **OPERATE.**
2. Press **[SET], [DRV#], [7], [7],** and **[ENTER].**
   -- The display will show **PARAMETER .**

NOTE: *In order to change parameter values, two conditions must be present:*

1. *The EEROM (E^2) enable device must be in the "ENABLE" position.*
   JP13: 1-2; position; E^2 write "PROTECT"
   2-3; E^2 write "ENABLE"
2. *Parameter address #3 (designated as ADR003-DGNJP) must contain an* ***odd*** *number (1, 3, 5, etc.). If an even number exists, add "1" to the present value and enter the new number.*

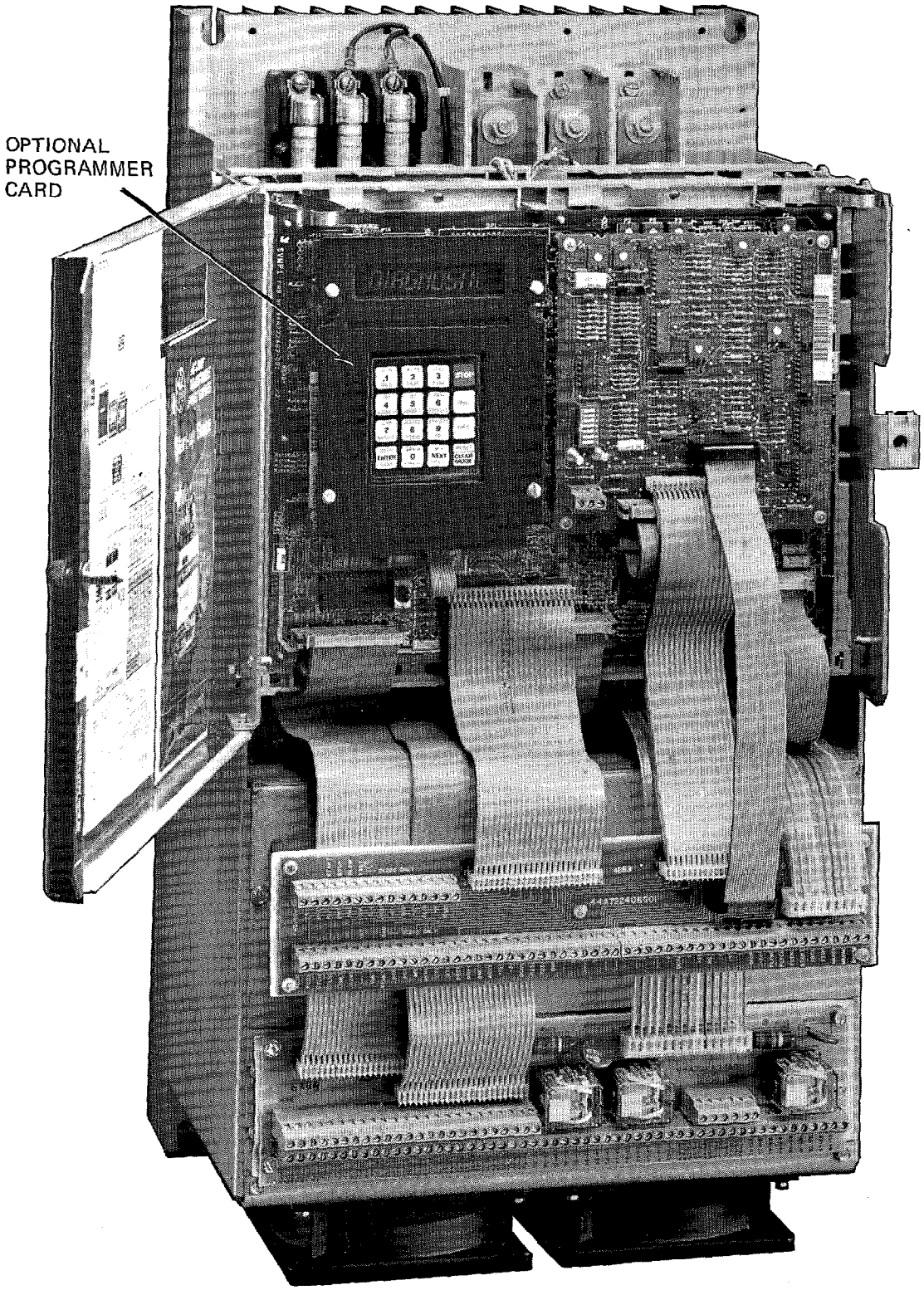


Figure 11-1. AC-300 Controller With Optional Programmer Installed

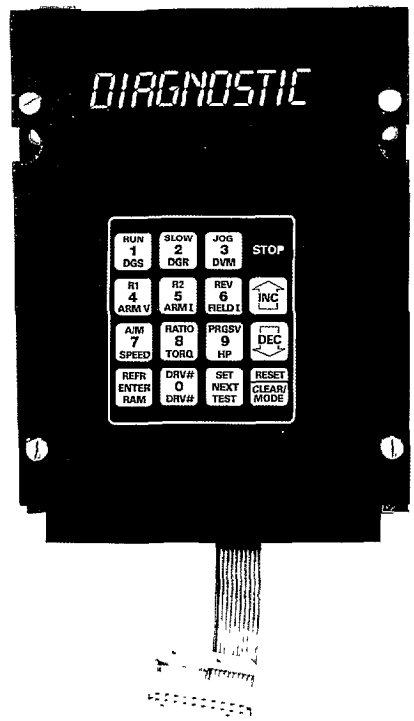


Figure 11-2. Optional Hand-Held Programmer

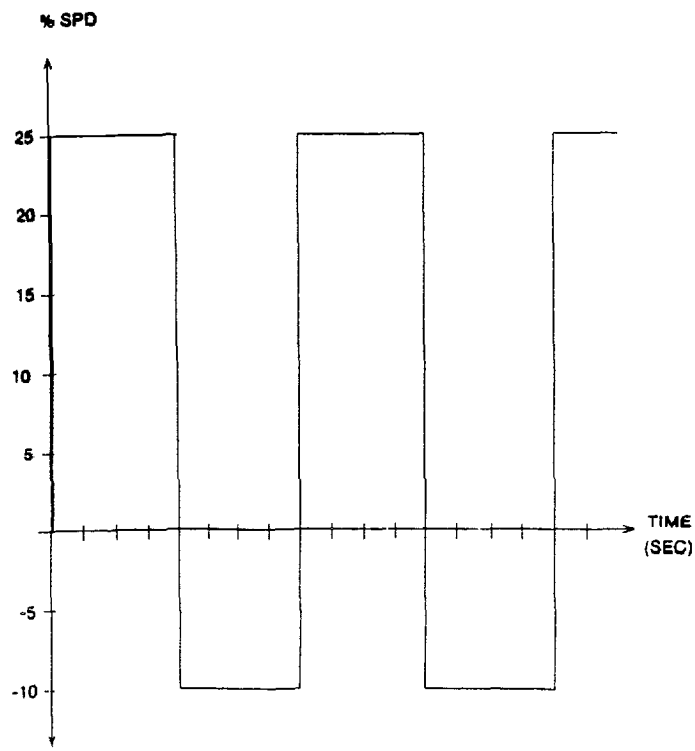


Figure 11-3. Test 25 Square Wave Reference Example

To enter the DIAGNOSTIC mode, perform the following: (The drive must first be in the **PARAMETER** mode.)

-- If the display shows **PARAMETER**, press **[CLEAR/MODE]** and the display will immediately show **DIAGNOSTIC.**

-- If a particular parameter is being displayed, press **[CLEAR/MODE]** causing **PARAMETER** to be displayed, then press **[CLEAR/MODE]** again, and the display will show **DIAGNOSTIC.**

To return to the OPERATE mode, perform the following:

-- If in the **DIAGNOSTIC** mode, press **[CLEAR/MODE]** once and the display will return to **OPERATE.**

-- If in the **PARAMETER** mode, press **[CLEAR/MODE]** either two or three times as required until the display returns to **OPERATE.**

NOTE: *The Programmer may appear to "hang up" due to programming errors. In such cases it may be necessary to initiate a "hardware reset" using the RESET switch (SW2) on the Control card.*

WARNING - POTENTIAL DRIVE ACTIVATION AND INADVERTENT EQUIPMENT MOVEMENT WHEN TESTING: Throughout the remainder of this section, do not use DIAGNOSTIC RUN, SQUARE-WAVE REFERENCE TEST, or any programmer RUN functions on any crane or elevator drive or other applications where mechanical movements cannot be adequately monitored.

OPERATE MODE - (RED Keys dominant)

In the OPERATE mode, any of the **RED** key functions, if activated, can be used as follows: (If key is not activated, the Programmer will display "FL41 KEYOF" momentarily.)

[RUN] - Run at speed set by RUNRF.

[SLOW] - Run at speed set by SLWRF.

[JOG] - Jog at speed set by JOGRF.

[STOP] - Controlled stop.

[R1] - Run at speed set by R1RF.

[R2] - Run at speed set by R2RF.

[REV] - Reverse RUN or JOG as required by first pressing **[REV]**, then either **[RUN], [SLOW], [JOG], [R1], [R2],** or **[REFR].**

[A/M] - Changes drive from AUTO to MANUAL, or vice-versa, enabling use of either ASR or MSR external speed reference inputs.

[RATIO] - Displays the drive run ratio or the percentage of speed reference at which the drive will run. Ratio is used to set a draw between multiple drives monitoring the same external speed reference. Normally set at 10,000 indicating a ratio of 1:1.

[PRGSV] - For future use.

[REFR] - Run at speed reference at which drive was last run, (includes preprogrammed keypad references or changes caused by using the **[INC]** and **[DEC]** keys, MSR, or ASR).

[DRV#] - Used to examine the drive number (in most cases "00"). Also part of access code to **PARAMETER** mode.

[SET] - Used to adjust the settings of various keypad functions; activated by pressing **[SET]**, then **[RUN], [SLOW], [JOG], [R1], [R2],** or **[RATIO]**, then the desired setting in decimal numbers, then **[ENTER].** Also, used for entering **PARAMETER** mode. The EEROM must be write "ENABLED" to make these adjustments (see "Changing Between Modes" paragraph in this section).

[INC] & [DEC] - Increases or decreases the speed of any of the keypad Run or Jog functions, or the Ratio function; simply enter the desired function and then press **[INC]**or **[DEC]** as desired.

[RESET] - Resets the drive after fault or error messages appear. The reset key is referred to as a "Software Reset".

The normal display in the **OPERATE** Mode will be **M XX% YY%** where **M** indicates MANUAL Mode using the Manual System Reference, MSR (alternatively an **A** will indicate AUTO Mode using the Auto System Reference, ASR on the optional Application card [or Microapplication card]). **XX%** indicates percent of rated speed and **YY%** indicates percent of rated load.

PARAMETER MODE - (BLACK Keys)

In the **PARAMETER** mode, any of the **BLACK** key functions plus **INC, DEC & STOP**, if activated, can be used as follows: (If key is not activated, the Programmer will display "FL41 KEYOF" momentarily.)

1. Enter the **PARAMETER** mode by pressing **[SET], [DRV#], [7], [7],** and **[ENTER]**.

2. Examine the present value of any parameter by keying in its three-digit parameter address (do not press **[ENTER]**).
   - -- The name for that address will immediately be displayed along with the present contents of that address.
   - -- Press **[CLEAR/MODE]** to clear displayed parameter (display will show **PARAMETER** again).
   - -- Key in the address number for the next parameter desired.

NOTE: *Occasionally, it may become necessary to check a series of parameters. Simply call up the lowest parameter address and press **[NEXT]** as often as required until the highest parameter number is reached.*

NOTE: *After a period of inactivity, the display may return to **'M.XX%YY%** (as displayed in the OPERATE Mode). The drive is still in the* PARAMETER *Mode (as indicated by the apostrophe before the **M**) awaiting further keypad entries. To proceed, enter the next parameter address to be examined.*

CAUTION: Do not apply or remove power from the drive or perform a "HARDWARE RESET" with the E^2 enable device in the "ENABLE" position.

3. Change parameters as shown in steps of the following Example.

NOTE: *In order to change parameter values, two conditions must be present:*

1. *The EEROM (E^2) enable device must be in the "ENABLE" position.*
   JP13: 1-2; position; E^2 write "PROTECT"
   2-3; E^2 write "ENABLE"
2. *Parameter address #3 (designated as ADR003-DGNJP) must contain an **odd** number (1, 3, 5, etc.). If an even number exists, add "1" to the present value and enter the new number.*

Example:

Examine and change the drive acceleration and deceleration rates. **Assume** the drive acceleration rate is set at 2 seconds and the deceleration rate is set at 10 seconds. Change the acceleration rate (ADR057-ACELT) from 2 seconds to 7.5 seconds. Reduce the drive deceleration rate (ADR058-DCELT) from 10 to 5.2 seconds.

1. Confirm that the E^2 enable device is in the "ENABLE" position and that ADR003-DGNJP contains an odd number.
2. Press **[CLEAR/MODE]** and **[0], [5], [7].**
   - -- The display will show **ACELT.20**, corresponding to a 2-second acceleration rate.
3. Press **[INC]** until the display value increases to **75** or press **[7], [5].**
   - -- In either case the display will now show **ACELT. 75.**
4. Press **[ENTER]** to place the new value into EEROM memory.
5. Press **[NEXT]** to command the drive to go to the next parameter, ADR058-DCELT.
   - -- The display will show **DCELT. 100**, corresponding to a 10-second deceleration rate.
6. Press **[DEC]** until the display shows **DCELT. 52**, or press **[5], [2].**
7. Press **[ENTER]** to place the new value into memory.
8. Press **[CLEAR/MODE]** to return the display to **PARAMETER.**
9. Key in the appropriate three-digit parameter address number to examine other addresses.

NOTE: *This next step is **extremely important** to prevent accidental reprogramming of the drive.*

10. **Seal** the new parameter values as follows:
    a. Remove the "1" from ADR003-DGNJP (returning it to the "as shipped" **even** number).
    b. Return E^2 enable device to the "PROTECT" position (JP13 in the 1-2 position).

4. Exit the PARAMETER Mode as follows:
   - -- Press **[CLEAR/MODE]** either once or twice to return to the DIAGNOSTIC Mode.
   - -- Press **[CLEAR/MODE]** again to return to the OPERATE Mode.

DIAGNOSTIC MODE

The Diagnostic Tests listed below will aid in the start-up, tune-up, and troubleshooting of the drive. To use these diagnostic features either a Local or Hand-held Programmer is required. Availability of certain tests may be dependent on hardware or software configurations provided.

WARNING - POTENTIAL DRIVE ACTIVATION AND INADVERTENT EQUIPMENT MOVEMENT WHEN TESTING: Throughout the remainder of this section, do not use DIAGNOSTIC RUN, SQUARE-WAVE REFERENCE TEST, or any programmer RUN functions on any crane or elevator drive or other applications where mechanical movements cannot be adequately monitored.

STANDARD DIAGNOSTICS TESTS:

Test 2	**[DGR]** - Diagnostic Run.
Test 3	**[DVM]** - Displays Digital Voltmeter.
Test 4	**[ARM V]** - Displays inverter frequency (WI).
Test 5	**[ARM I]** - Displays torque current component.
Test 6	**[FIELD I]** - Displays flux current com ponent.
Test 7	**[RPM]** - Displays motor speed.
Test 8	**[TORQ]** - Displays motor torque in 1 lb-ft.
Test 9	**[HP]** - Displays motor horsepower.
Test 10	Displays software revision of Control card microprocessors.
Test 11	Displays diagnostic variables (selectable).
Test 12	Transistor test.
Test 51	Revision date of Programmer software.
Test 52	Revision level of Programmer software.
RAM TEST	Displays single diagnostic variable (selectable).
DAC 0, 8, & 12	Digital to Analog converter outputs.
BITS 1, 2, & 3	Status bit outputs.

ADVANCED DIAGNOSTICS TESTS:

Test 20	Electronic Wire-Check.
Test 21	Diagnostic Variable Pairs.
Test 22	Diagnostic Variables through DAC Outputs.
Test 25	Square-Wave Reference.

STANDARD DIAGNOSTIC TESTS DESCRIPTIONS:

NOTE: *For Tests 2 through 9, simply enter the DIAGNOSTIC mode and press the appropriate key. To enter all other diagnostic tests, press **[TEST], [X]**, where "X" is the number of the particular test to be performed, and then press **[ENTER]**.*

WARNING - POTENTIAL DRIVE ACTIVATION AND INADVERTENT EQUIPMENT MOVEMENT WHEN TESTING: Throughout the remainder of this section, do not use DIAGNOSTIC RUN, SQUARE-WAVE REFERENCE TEST, or any programmer RUN functions on any crane or elevator drive or other applications where mechanical movements cannot be adequately monitored.

TEST 2 - Diagnostic Run (DGR)

Diagnostic run (DGR) causes the drive to run as a CEMF regulator. The Diagnostic Run reference is determined by parameter ADR003-DGNJP and may set by either MSR, the Programmer INCREASE/DECREASE keys, or the internal programmed references, ADR092 and ADR093 (TSTR1 and TSTR2).

- -- Refer to the Custom Software Description to determine which reference is used.
- -- Pressing **[STOP]** will initiate a coast stop.
- -- These features, CEMF regulation and coast stopping, will aid significantly when trying to separate mechanical problems from drive stability problems.

TEST 3 - DVM Mode (Requires Optional Process Interface or Encoder Process Interface Card; Also See Diagnostic Mode for Drive Setup)

Test 3 provides the capability of utilizing the Local or Hand-held Programmer and Process Interface card (or Encoder Process Interface card) as a digital voltmeter (DVM) for checking various analog test points (TP) on the drive cards. DC voltage in the +/-40 volt range can be measured.

-- Usage of DVM Mode requires a test probe wire (supplied in the miscellaneous parts packet shipped with the drive) routed from test point DVM on the Process Interface card (or Encoder Process Interface card) to the particular test point to be monitored.

-- The DVM may be calibrated by monitoring the +5 volt test point and adjusting P3-DVM TRIM on the Process Interface card.

-- Pressing **[TEST]** discontinues DVM Mode and displays **CLR TEST.**

TESTS 4 thru 9 - Various Displays (ARM V, ARM I, FIELD I, SPEED, TORQ, HP)

Tests 4 thru 9 are initiated by setting the drive in DIAGNOSTIC Mode and then entering the desired test number.

-- To change to a different test simply press another number from 4 thru 9.

-- To cancel the last test press **[TEST]** once; the Programmer will display **CLR TEST.**

TEST 10 - Software Revision

Test 10 will display the revision level of the microprocessors on the Control card. For example, **R20F20T20** says the regulator microprocessor (RUP) is version 2.0 software, the firing microprocessor (FUP) is version 2.0, and the tachometer microprocessor (TUP) is version 2.0. This can be verified by comparing the information with the white labels on the individual microprocessor chips located on the Control card.

TEST 11 - Diagnostic Variable Display (also see TEST 21 and TEST 22)

Test 11 can be used to display (in hexadecimal numbers) certain drive variables for the purpose of comparison. Perform steps of the following example to display/compare variables.

Example:

To compare speed reference and speed feedback, perform the following:

1. Place the drive in PARAMETER Mode.
2. Determine (from the list that follows) the RAM address of the variables to be monitored.
3. Enter the first variable RAM address number into parameter ADR122-DIAG1.
4. Enter the second variable RAM address into parameter ADR123-DIAG2.
5. Place the drive in DIAGNOSTIC Mode and select Test 11.
   -- The programmer will display **DGXXXX. YYYY**, where **XXXX** is the hexadecimal value of the RAM address entered in ADR122, and **YYYY** is the hexadecimal value of the RAM address entered in ADR123.

NOTE: *The value displayed is in 2's complement. Values from 0000 to 7FFF are positive magnitude (0 to +32767). Values from FFFF to 8000 are negative magnitude (FFFF = -1 and 8000 = -32768).*

RAM ADDRESSES FOR "ANALOG" SIGNALS

RAM ADR	DESCRIPTION	RAM ADR	DESCRIPTION
030	IALPHA Feedback	103	WI-Inverter Frequency
031	IBETA Feedback	105	WM-Rotor Frequency
053	Speed Loop Output	448	Filtered CFB
055	IREAL	463	Field Current
065	Timer Speed Reference	494	ACOM
069	Jog Reference	501	TOC Timer
071	Speed Feedback	492	HP

NOTE: *These addresses will contain a variable number, representing an "analog" signal. Refer to "RAM Variable Scaling", TABLE 4 in the Adjustment Section, for the relative hexadecimal values of the displayed variables.*

RAM ADDRESSES FOR "LOGIC" SIGNALS

RAM ADR	DESCRIPTION	RAM ADR	DESCRIPTION
137	Fault	148	Reference Enable
140	Current Limit Flag	152	Precondition
141	Freq Limit Flag	175	Start/Stop Control
147	Ready To Start Flag		

NOTE: *This list contains addresses of signals which will be either "high" (display "8080") or "low" (display "0000"). If the signal is high, then the statement is true. For example, if "137" is in ADR122, Test 11 is initiated, and a fault condition exists, then the display will show* ***DG8080.YYYY****, (YYYY value will depend on the number in parameter ADR123). If the drive is not faulted then the display will be* ***DG0000.YYYY****.*

WARNING - ELECTRICAL SHOCK HAZARD: The following test will cause current to circulate in the motor.

TEST 12 - **Transistor Test**

Test 12 will drive each transistor once. During this test, current feedback is monitored to determine if any transistors are open or shorted.

-- If problems exist, a fault code will be displayed to help isolate the nature and area of the problem. For example, "FL65 A1FOP" means that transistor TM1 did not conduct current. Transistor TM1 can be identified from the Information Label on the inside of the drive door.

-- Refer to the Troubleshooting Section for a complete list of fault codes.

TEST 51 - **Programmer Card Software Revision Date**

Test 51 provides a check of the software **date** of the Programmer microprocessor (PUP).

TEST 52 - **Programmer Card Software Revision Level**

Test 52 provides a check of the software **level** of the Programmer microprocessor (PUP).

RAM TEST - **Displays RAM Location Value**

RAM test allows a "one-shot" look at a RAM address (refer to TEST 11 list of for variables). The value displayed will be in decimal numbers. To display a RAM address, perform the following:

1. Press **[RAM]** and the Programmer will display *****RAM*****.
2. Press the keys corresponding to the desired RAM variable and the drive will display **XXX YYYY**, where **XXX** is the RAM address and **YYYY** is the RAM value.
3. Press **[CLEAR/MODE]** to clear the display.

WARNING - DRIVE MISOPERATION HAZARD: These outputs may already be utilized by the drive. Refer to the Drive Elementaries and Custom Software Description.

DAC0, DAC8, and DAC12 **(DAC8 and DAC12 Require an Optional Application or Microapplication card)**

Three selectable analog outputs, DAC0, DAC8, and DAC12, are available for test purposes and are programmed similar to Test 11. To program DAC0, perform the following:

1. Enter the PARAMETER Mode and call up ADR090-DAC0.
2. Enter the signal to be monitored from TABLE 4, "RAM Variable Scaling". (Putting that RAM Address in ADR090 will cause that signal to appear at the DAC0 test point.)

   NOTE: *DAC8 and DAC12 are monitored in a similar manner, except the signal RAM address is loaded into ADR096-DAC8 or ADR097-DAC12 and the output appears at the DAC8 or DAC12 terminal on 4TB.*

WARNING - DRIVE MISOPERATION HAZARD: These outputs may already be utilized by the drive. Refer to the Drive Elementaries and Custom Software Description before using this feature.

STATUS BITS 1, 2, and 3 **(Requires Optional Application or Microapplication card)**

ADR098-BIT#1, ADR099-BIT#2, and ADR100-BIT#3 are programmed in the same manner as DAC8 and DAC12. They are different in that they are logic outputs rather than analog outputs.

-- Signals listed in "RAM Addresses for Logic Signals" chart can be used to select addresses to enter in ADR098, ADR099, and ADR100.

-- Variables listed in "RAM Addresses for Analog Signals" chart may also be used in the bit outputs to indicate the polarity of the signal (positive or negative).

-- Outputs will appear on 4TB as "B01", "B02", and "B03" respectively.

ADVANCED DIAGNOSTIC TEST DESCRIPTIONS:

WARNING - POTENTIAL DRIVE ACTIVATION AND INADVERTENT EQUIPMENT MOVEMENT WHEN TESTING: Throughout the remainder of this section, do not use DIAGNOSTIC RUN, SQUARE-WAVE REFERENCE TEST, or any programmer RUN functions on any crane or elevator drive or other applications where mechanical movements cannot be adequately monitored.

WARNING - DRIVE MISOPERATION HAZARD: Signal level detectors and bit outputs may control external devices. Refer to the Drive Elementaries and Custom Software Description before using this feature.

TEST 20 - Electronic Wire Check

Test 20 allows for **six** different levels of wire checking and output forcing tests as follows:

20-1:	Signal Level Detector (SLD) forcing.
20-2:	Status Bit monitoring.
20-3:	Fault Code testing.
20-4:	Logic input monitoring (Application card).
20-5:	Logic input monitoring (Control card).
20-6:	Microapplication card input monitor ing (future).

Proceed as follows to perform all Electronic Wire Check tests:

1. Confirm that programmer is in DIAGNOSTIC Mode and that the EEROM is "Write Enabled" (see TABLE 1, Control card, JP13). (Change EEROM to "Write Protect" after test is completed).
2. Key in **[TEST]**, **[2]**, **[0]**, and **[ENTER]**.
   - -- This causes Test 20 to be active and the display to show **1.'.00111111**. The left-most digit indicates the test level, in this case, the **1.'.** indicates level 1 or Signal Level Detector (SLD) forcing.
3. Press the **[INC]** or **[DEC]** key to change the type of test until the desired test number is displayed in the left-most digit.
   - -- For example, if Test 20 is called for and the initial display is shown above, press **[INC]** once to run the next level (20-2), Status Bit Monitoring. The programmer display will be similar to **2.'.11111100**.
   - -- Each test is described in the following paragraphs.

NOTE: *Test 20-1 temporarily changes the value of Parameter ADR009-SLDJP. If the AC control power is removed while keys 0 thru 5 are depressed, ADR009-SLDJP may be changed and will not contain the correct value when AC power is reapplied. The value of SLDJP should be recorded prior to initiating this test.*

Test 20-1 - **Signal Level Detector (SLD) Forcing (Requires Application or Microapplication card)**

The display will show **'1.'.00111111** when Test 20-1 is initiated.

- -- SLD0 is energized if the right most digit is "1." and dropped out if it is a "0".
- -- The status of all six SLDs (SLD0 through SLD5) can be seen by scanning from right to left. For example, the second digit from the right is SLD1, the third digit from the right is SLD2, etc.
- -- To change the state of SLD0, depress **[0]** key.

NOTE: *SLDs cannot be toggled while the drive is running or the drive will fault. While the **[0]** key is depressed, SLD0 will be maintained in the opposite state it was originally. When the **[0]** key is released, SLD0 returns to its original state. Keys 1-5 operate in the same manner for SLD1 thru SLD5.*

Test 20-2 - **Status Bit Outputs (Requires Application or Microapplication card)**

Status bits B01, B02, and B03 (ADR098, ADR099, and ADR100 respectively) are accessed as described in Test 20.

- -- When Test 20-2 is initiated, B01 will appear as the right most character in the display, B02 will be shown on the left of B01, and B03 is shown on the left of B02.
- -- If Bit#1 is "high" the right most character will be "1"; if it is "low", this character will be "0".

Test 20-3 - **Fault Code Outputs (Requires Application or Microapplication card)**

Test 20-3 is used to test remote fault code lines and their effects on external equipment.

1. Access Test 20-3 as described in Test 20.
2. Enter the desired fault code by keying in **[0]**.
   - -- The programmer will display **FAULT**.
3. Key in the number of the Fault Code desired and press **[ENTER]**.
   - -- Note that Fault 63 will cause FC5 (4TB-45) to be high.

NOTE: *Test 20-3 does* ***not*** *cause the drive to fault unless a number greater than 127 is entered. This test is a way of forcing the fault code output lines on the Application card (or Microapplication card) to check interfaces with external equipment. The Diagnostic LEDs will also display the appropriate Fault Code.*

Test 20-4 - Logic Input Monitoring (Requires Application or Microapplication card)

The display will show **'4.'.XXXXXX** when Test 20-4 is initiated. The Xs will be either **1** or **0**.

- -- A **1** will be displayed if a particular input is tied to REF24 and the drive is programmed for **negative** logic (JP19 and JP20 on the Control card are in 1-2 and 3-4 position respectively, and JP5 and JP6 on the Application card [or Microapplication card] are in the 1-2 and 3-4 position respectively).
- -- From right to left the digits are MODE 0 thru MODE 7.

Test 20-5 - Logic Input Monitoring (Control card)

The Test 20-5 display is similar to the Test 20-4 display except that the left four digits do not apply. Starting at the first digit from the right, the status of four logic inputs will be displayed as "5.XXXXDCBA" where:

- **A** = Status of RUN command inverted ("0" indicates RUN is initiated).
- **B** = Status of JOG command inverted ("0" indicates JOG is initiated).
- **C** = Status of POL command inverted ("0" indicates POL is initiated).
- **D** = Status of XSTOP command ("1" indicates XSTOP initiated).

Test 20-6 Microapplication Card Input Monitoring (Microapplication card required)

Test 20-6 is reserved for future use.

TEST 21 - Diagnostic Variable Pairs

NOTE: *Test 21 will cause ADR122-DIAG1 and ADR123-DIAG2 to change.*

Test 21 performs in a similar manner to Test 11 except that several pairs of RAM addresses are already pre-programmed in memory. Test 21 is entered from the **DIAGNOSTIC** mode as follows:

1. Key in **[TEST], [2], [1]**, and **[ENTER]**.
   - -- The programmer will display **TST21**.
2. Key in **[0], [1]; [2]; [3]; or [1]; [0]** (corresponding to Tests 21-0, 21-1, 21-2, 21-3, or 21-10 respectively), and **[ENTER]**, to display the pairs of RAM addresses shown in the chart below.
   - -- As in Test 11, the display will show **DGXXXX.XXXX**. The left and right four characters are as selected from the listing.
   - -- Test 21-3 will display the variables programmed into ADR096-DAC8 and ADR097-DAC12.
   - -- When **[1], [0]** is entered (calling for Test 21-10), the display will prompt with **ADDR1**. Proceed as follows for this test:
     a. Key in the first RAM address you wish to monitor and press **[ENTER]** (typical RAM addresses are listed under Test 11). The display will then prompt with **ADDR2.**
     b. Key in the second RAM address and press **[ENTER]**. The display will now show **DGXXXX.XXXX**, where the left four digits are the value of the signal address stored in ADDR1, and the right four digits are the value of the signal address stored in ADDR2.

RAM VARIABLE PAIRS FOR TESTS 21 AND 22

TEST NUMBER	LEFT 4 DIGITS (RAM-Addr)	RIGHT 4 DIGITS (RAM-Addr)
0	Speed loop output (053)	IREAL (055)
1	Speed Reference (065)	Speed feedback (071)
2	Position Error (471)	In position flag (154)
3	DAC8 (ADR096)	DAC12 (ADR097)
10	Diag. Var. #1 (ADR122)	Diag. Var. #2 (ADR123)

TEST 22 - Diagnostic Variables Through DAC Outputs (Requires Application or Microapplication Card)

Test 22 performs in a similar manner to Test 21 with the addition that the various signal pairs are automatically sent to the DAC8 and DAC12 terminals on 4TB. Test 22 is entered from the DIAGNOSTIC Mode as follows:

1. Key in **[TEST], [2], [2],** and **[ENTER].**
   -- The programmer will display **TST22.**
2. Key in **[0], [1], [2], [3], or [1], [0]** (corresponding to Tests 22-0, 22-1, 22-2, 22-3, or 22-10 respectively), and **[ENTER],** to display the pairs of RAM addresses shown in the chart below.
   -- As in Test 21, the display will show **DGXXXX.XXXX.** The left and right four characters are as selected from the listing.
   -- Calling for Test 22-3 causes +5V to appear at the DAC8 and DAC12 terminals in order to quickly check the integrity of DAC8 and DAC12.
   -- When **[1], [0]** is entered (calling for Test 22-10), the display will prompt with **ADDR1.** Proceed as follows for this test:
      a. Key in the first RAM address you wish to monitor and press **[ENTER]** (typical RAM addresses are listed under Test 11). The display will then prompt with **ADDR2.**
      b. Key in the second RAM address and press **[ENTER].** The display will now show **DGXXXX.XXXX,** where the left four digits are the value of the signal address stored in ADDR1, and the right four digits are the value of the signal address stored in ADDR2.

NOTE: *Test 22 temporarily changes the value of ADR096-DAC8, ADR097-DAC12, ADR122-DIAG1, and ADR123-DIAG2. If AC control power is removed while conducting Test 22, then DAC8, DAC12, and DIAG2 will not contain the original value when power is reapplied. The value of DAC8, DAC12, DIAG1, and DIAG2 should be recorded prior to conducting this test.*

WARNING - POTENTIAL DRIVE ACTIVATION AND INADVERTENT EQUIPMENT MOVEMENT WHEN TESTING: Throughout the remainder of this section, do not use DIAGNOSTIC RUN, SQUARE-WAVE REFERENCE TEST, or any programmer RUN functions on any crane or elevator drive or other applications where mechanical movements cannot be adequately monitored.

TEST 25 - Square Wave Test Reference

Test 25 causes the drive to run and follow an internal drive square wave reference set up by the values in ADR092-TSTR1, ADR093-TSTR2, and ADR094-TSTT1. TSTR1 and TSTR2 are run reference levels and TSTT1 is the time duration of each half cycle. (See Example)

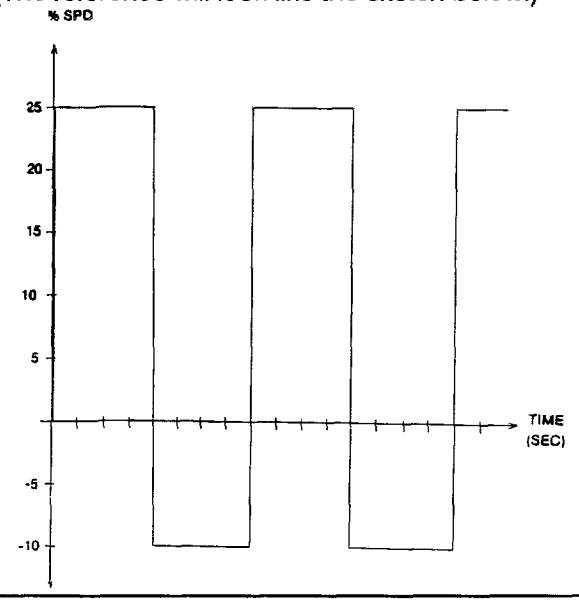
-- Run references are 0 to 127 with 100 equal to 100% speed with positive reference. Negative reference is initiated with values from 128 to 255 with the magnitude equal to 256 minus the setting (for example, 156 corresponds to -100%).
-- The time period is set with TSTT1 and will have a 50% duty cycle for powers of 2 (1, 2, 4, 8, 16, etc.). The cycle time will be 262* TSTT1 milliseconds. That is, with TSTT1 set to 8, the test will cycle between TSTR1 and TSTR2 with approximately 1 second for each reference.
-- Pressing **[STOP]** will end the test.

NOTE: *Test 25 temporarily changes the value of ADR003-DGNJP. If AC control power is removed while conducting this test, then DGNJP will not contain the correct value when AC power is reapplied. The value of DGNJP should be recorded prior to conducting this test.*

Example:
Program the square wave reference to run 25% of top speed forward for approximately 4 seconds and 10% reverse for approximately 4 seconds.

ADR092-TSTR1 = 25 (25% speed)
ADR093-TSTR2 = 246 (-10% speed)
ADR094-TSTT1 = 4000 ms/262 = 16

(The reference will look like the sketch below.)



DIAGNOSTIC MODE FOR DRIVE SETUP

Step 1 of the following procedure utilizes Diagnostic Test 21-10. All other steps utilize the DVM Mode. These procedures allow accurate tuning of all analog pots as listed below.

Control card	OFFSET	P4
Control card	REF SCALE	P1
Process Interface card*	SFB ZERO	P7
Process Interface card*	PF OFF	P5
Process Interface card*	PF GAIN	P4
Application card**	ASR SCALE	P1
Application card**	FRA SCALE	P2

**Or Encoder Process Interface card*

***Or Microapplication card*

1. Calibrate **P4-OFFSET** on the Control card and **P7-SFB ZERO** on the Process Interface card (or Encoder Process Interface card) as follows:
   a. Place the drive in DIAGNOSTIC Mode.
   b. Enable EEROM and key in **[TEST]**, **[2]**, **[1]**, **[ENTER]**, then **[1]**, **[0]** and **[ENTER]**.
      -- The drive will display **ADDR1**.
   c. Key in **[4]**, **[9]**, **[4]** and **[ENTER]**.
      -- The drive will display **ADDR2**.
   d. Key in **[0]**, **[7]**, **[1]** and **[ENTER]**.
      -- The drive will display **DG.XXXX.XXXX**.
   e. Use the left four characters and P4-OFFSET on the Control card to adjust the drive offset to zero.
   f. Use the right four characters and P7-SFB ZERO on the Process Interface card (or Encoder Process Interface card) to adjust speed feedback for zero.

2. Place the drive in the DVM Mode by pressing **[DVM]**.

3. Calibrate **P1-REF SCALE** on the Control card as follows:
   a. Connect the DVM test probe from TP DVM on Process Interface card (or Encoder Process Interface card) to TP SRS, located on the Control card.
   b. Set the input at MSR on 3TB65 to its maximum value and adjust P1-REF SCALE on the Control card for 4.0 VDC at TP SRS (also located on the Control card).

NOTE: *If 4.0 VDC cannot be obtained at SRS, then check JP1 on the Control card for proper position. If MSRs maximum voltage is 5-8 VDC, then position 2-3 is correct.*

4. Calibrate P1-TACH SCALE on the Process Interface card (or Encoder Process Interface card) as follows:
   a. Calculate the tachometer voltage at top speed from the tachometer nameplate and motor top speed.
   b. Refer to TABLE 1 for the Process Interface card (or Encoder Process Interface card) SW1 setting and determine which switch positions should be closed for desired drive top speed
      NOTE: *SW1, position 7 should be **open** for DC tachs and **closed** for AC tachs.*
   c. Start the drive and run at half speed (2.0 VDC at TP SRS on Control card if running in MANUAL Mode, or 2.0 volts at TP ASR on Application card [or Microapplication card] if running in AUTO mode).
      -- Monitor motor speed with either a direct reading mechanical tach or with a voltmeter between 1TB-1 and 1TB-3 on the Process Interface card (or Encoder Process Interface card).
   d. Adjust P1-TACH SCALE on the Process Interface card (or Encoder Process Interface card) for 50% of motor speed.
   e. Run the drive at top speed and recheck the motor speed.
   f. Readjust P1-TACH SCALE as necessary for the correct top speed.
      -- TP SFB on the Process Interface card (or Encoder Process Interface card) should now read 4.0 VDC and be equal to TP SRS voltage (TP ASR if running in AUTO Mode).

This completes the calibration of the speed feedback circuit and setup of the drive's analog pots.

5. Check the setup of the Process Follower option (if used) as follows:
   a. Verify that JP2 and JP3 on the Process Interface card (or Encoder Process Interface card) are set as required according to the following chart:

JP2	JP3	Follower Mode
1-2	1-2	5-65 MA
1-2	2-3	5-50 MA
2-3	1-2	4-20 MA
2-3	2-3	1-5 MA or 0-10V

   b. Set the programmer in DIAGNOSTIC Mode and turn on the DVM (press **[DVM]**).
   c. Connect the test probe from TP DVM on the Process Interface card (or Encoder Process Interface card) to TP PFRF (also on the card).
   d. Adjust P5-PF OFF on the Process Interface card (or Encoder Process Interface card) for zero volts on the DVM.
   e. Raise the process follower input to its maximum value and adjust P4-PFGAIN on the Process Interface card (or Encoder Process Interface card) for 7.5 VDC on the DVM.

NOTE: *It is not necessary to run the drive to calibrate the process follower, only to vary its input as described above.*

12. PARTS REPLACEMENT

CAUTION: Treat all cards with static sensitive handling techniques. Use "grounding" strap when changing cards and store cards in the anti-static bags they are shipped in.

CARD REPLACEMENT PROCEDURE

(Figures 12-1 and 12-2)

1. Carefully disconnect all cables.
2. Release all hold-down tabs and remove board.
3. Set all jumpers on the replacement board exactly as the jumpers on the failed board.
4. Install the new board verifying that all tabs snap tightly into position.
5. Reconnect all cables verifying that they are properly seated at both ends.
6. Set all pots in the same physical position as the pots on the failed card and fine tune as required per instructions in TABLE 1 of the Adjustments section.
7. If the Control card is changed, remove the EEROM from the failed card and install in the new card. If the EEROM from the failed card is suspected to be damaged, it will be necessary to reprogram a new EEROM to the custom parameter list. See Figure 12-3 for typical EEROM removal tool and grounding techniques.

REPLACEMENT OF TRANSISTOR MODULES (Figures 12-4 and 12-5)

1. Lower the card carriers to expose the rear of the unit.
2. Disconnect the wires from the damaged module and remove the bus bar and mounting screws.
3. Clean the mounting surface of old lubricant and dirt.

CAUTION: When mounting transistor module on a heatsink, certain precautions should be taken to prevent any damage against a sudden torque. If a sudden torque is ("one-sided tightening") applied at only one mounting terminal, the ceramic insulation plate or silicon chip inside the module may get damaged.

CAUTION: The mounting screws are to be fastened in a certain order (the examples of which are shown below and in Figure 12-5). Also, care must be taken to achieve maximum contact area (i.e., minimum contact thermal resistance) for the best heat dissipation. Application of thermal compound on the contact surface improves its thermal conductivity.

4. Apply a thin film of General Electric GL322 Versilube® Plus lubricant (Part No. 104X214AB059) to both contacting surfaces and install the module. Tighten screws to the torque specifications shown in the following chart per the methods shown in Figure 12-5.

   "A" Method: Two-point mounting type
   Temporary tightening sequence: 1 - 2
   Final tightening sequence: 2 - 1

   "B" Method: Four-point mounting type
   Temporary tightening sequence: 1 - 2 - 3 - 4
   Final tightening sequence: 4 - 3 - 2 - 1

TORQUE SPECIFICATIONS

TRANSISTOR MODULE PART NUMBER	TORQUE METHOD	TORQUE REQUIREMENTS MOUNTING SCREWS*	TERMINAL SCREWS
104X170CB010	A	26 inch pounds	17 inch pounds
104X170CB011	B	26 inch pounds	26 inch pounds
104X170CB012	B	26 inch pounds	26 inch pounds
104X170CB013	A	26 inch pounds	17 inch pounds
104X170CB014	A	26 inch pounds	17 inch pounds
104X170CB015	A	26 inch pounds	17 inch pounds
104X170CB026	B	26 inch pounds	EC @ 95 inch pounds BBE @ 12 inch pounds
104X170CB027	B	26 inch pounds	EC @ 95 inch pounds BBE @ 12 inch pounds
104X170CB028	A	17 inch pounds	--

** Tighten mounting screws one at a time (progressing diagonally opposite corners on 4-screw mounting) until snug. Finally tighten to 26 inch pounds in opposite order (beginning with last screw snugged on 4-screw mounting).*

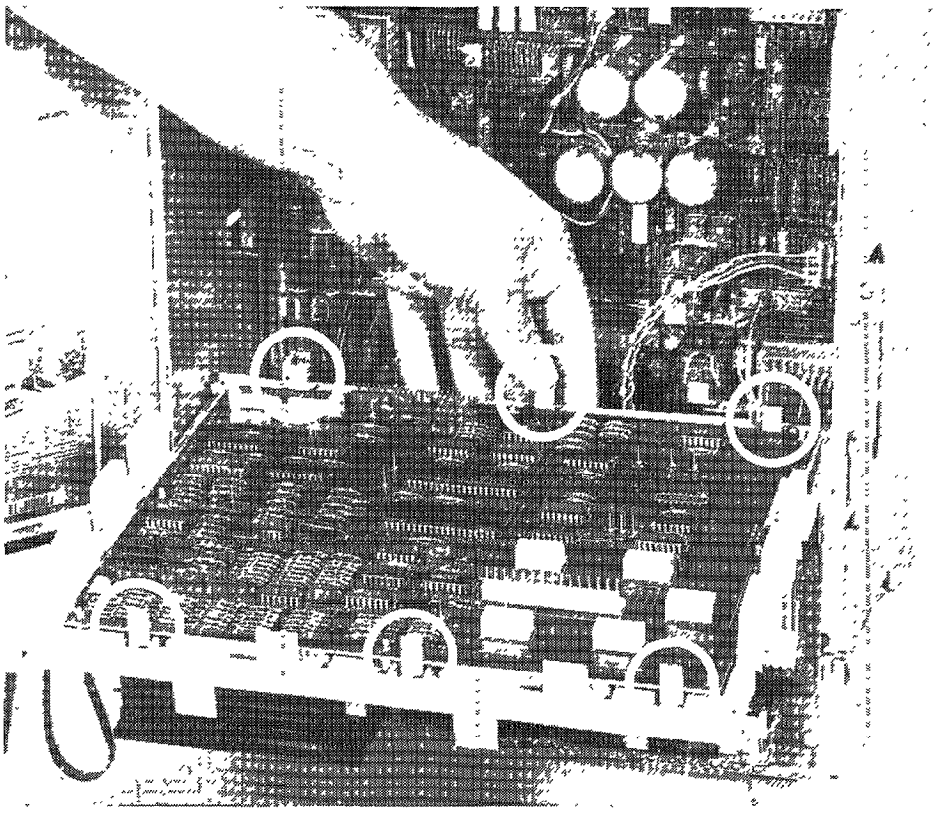


Figure 12-1. Six Card Release Tabs

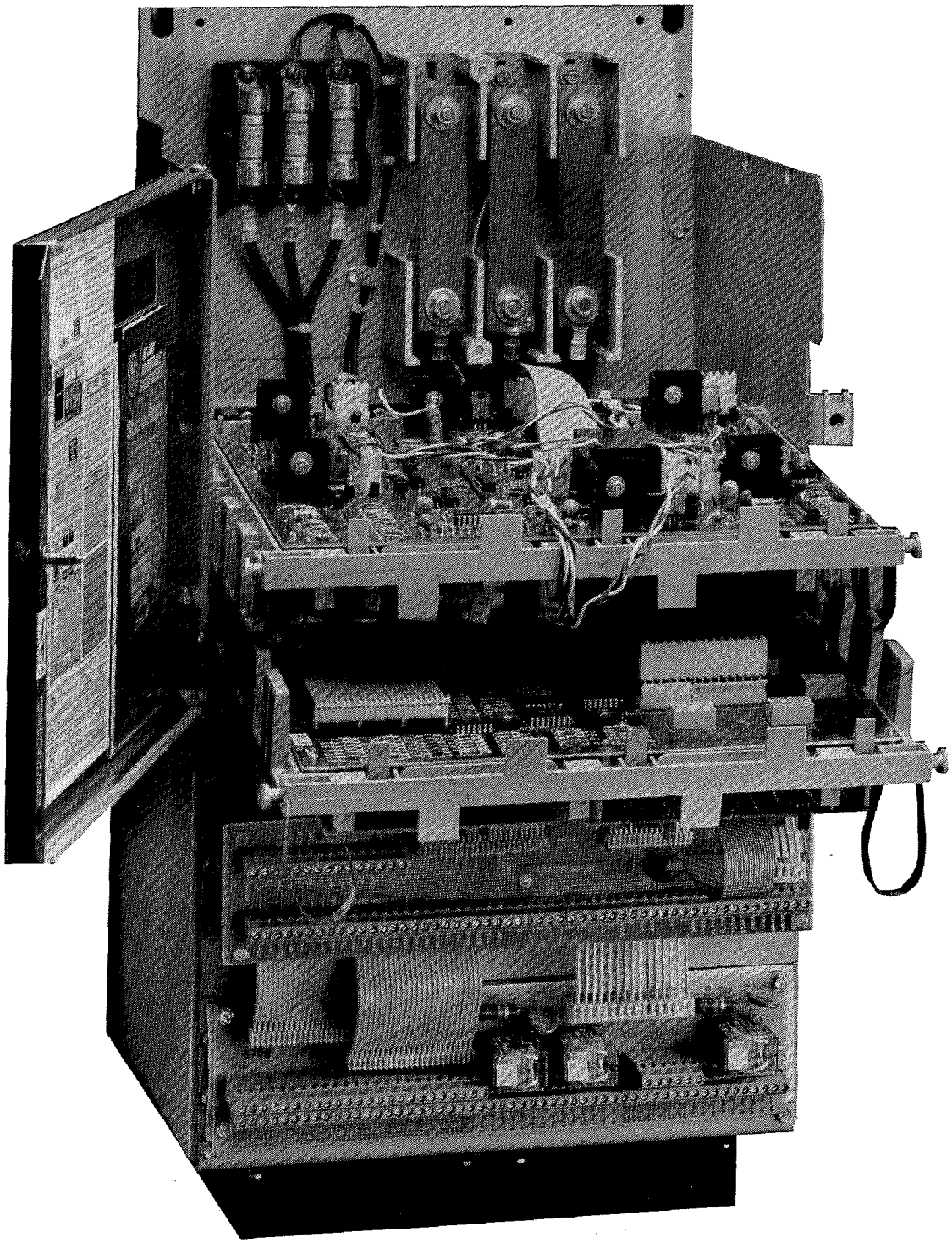


Figure 12-2. Controller With Front And Rear Card Carriers Lowered

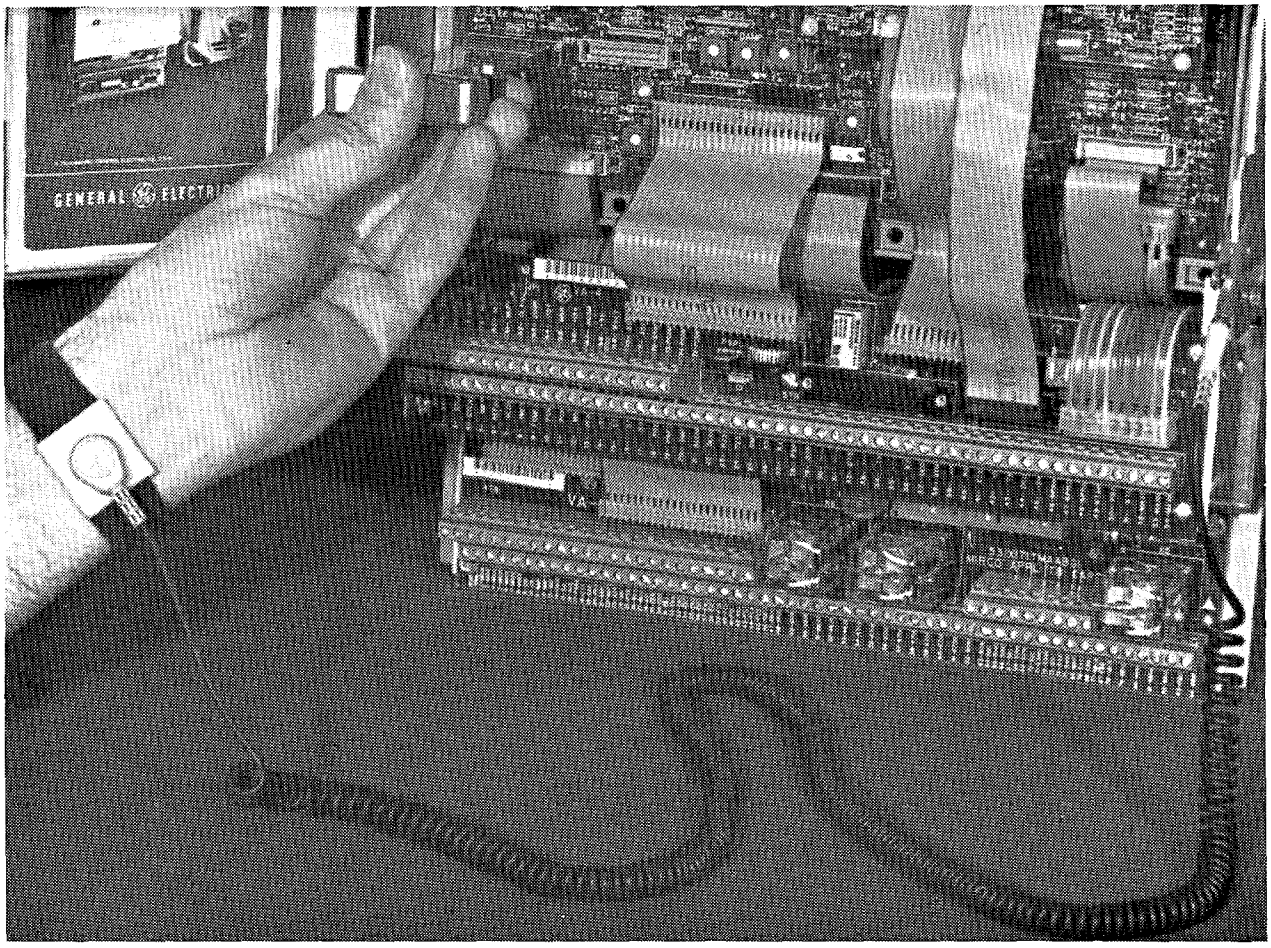


Figure 12-3. Typical EEROM Removal And/Or Replacement Equipment

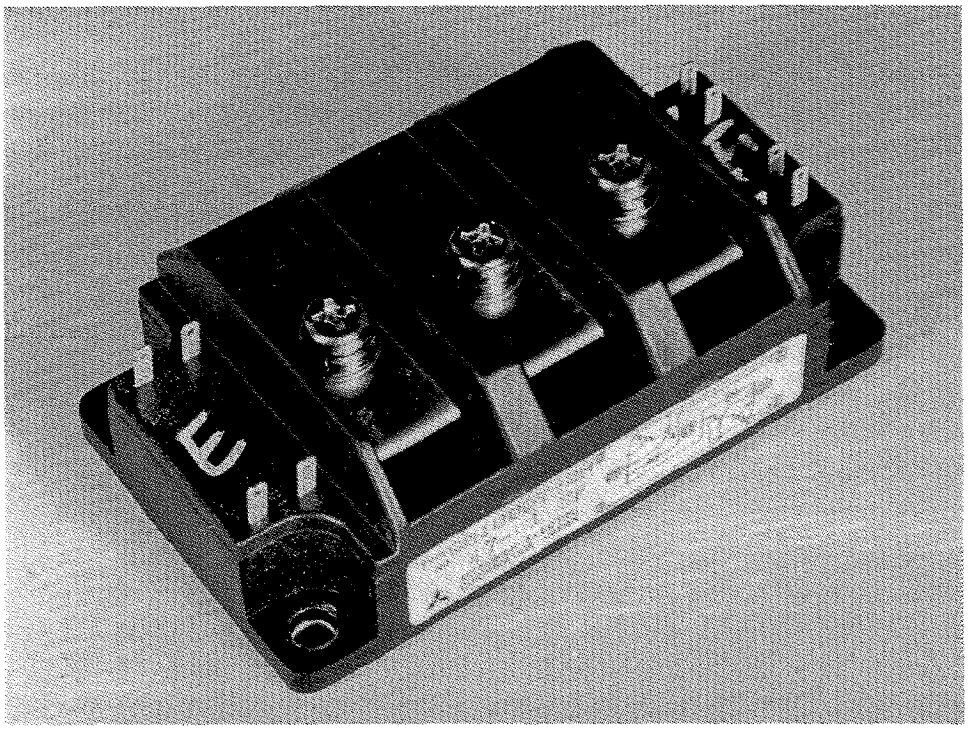


Figure 12-4. AC-300 Drive Controller Transistor Module

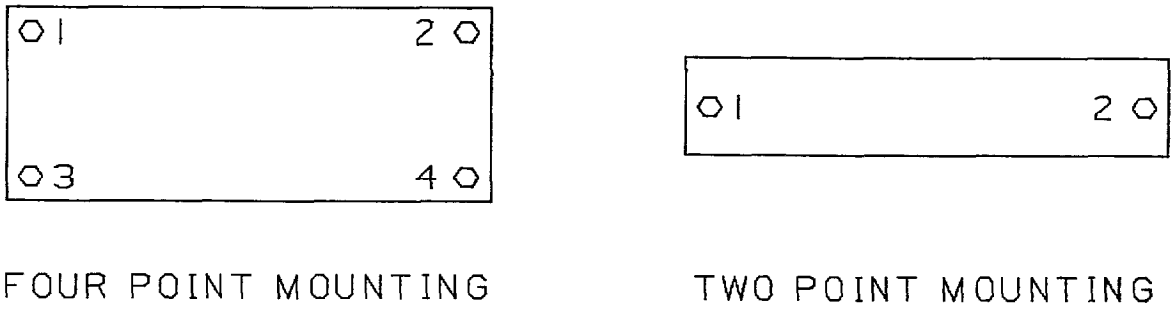


Figure 12-5. Transistor Module Tightening Sequences

NOTES:

PRINTED CIRCUIT CARDS -- STANDARD	CATALOG NUMBER (Root)
Control card (with microprocessors)	531X102CCHC_M_
Base Drive card	531X146BDHA_G1 (up to 150 HP, 460 VAC) 531X146BDHA_G2 (up to 75 HP, 230 VAC) 531X146BDHA_G3 (200-300 HP, 460 VAC) 531X146BDHA_G4 (100-150 HP, 230 VAC)
Power Supply card	531X112PSAA_G1* 531X112PSAA_G2**
Base Drive Amplifier (150-300 KVA)	531X194BDAA_G1
Snubber	531X125SNHA_G1
Snubber	531X125SNHA_G2
Snubber	531X129SNCA_G1
Snubber (150-300 KVA)	531X149SNCA_G1
Snubber (DB)	531X190RCSA_G4
Snubber (DC Link)	531X190RCSA_G2
PRINTED CIRCUIT CARD -- OPTIONAL	
Programmer card	531X135PRGA_M1
Application card***	531X139APMA_G2
Microapplication***	531X139APMA_M_
Encoder Process Interface card	531X134EPRB_G1
Process Interface card	531X133PRUA_G1
Terminal Board card - 3TB	531X170TBSA_G1
Terminal Board Relay card - 4TB***	531X171TMAA_G1
Programmer card Assembly	36B605548A_G01
Hand-Held Programmer Assembly	6VHHP10A1

**Up to 150 HP, 460 VAC; up to 75 HP, 230 VAC.*
***200-300 HP, 460 VAC; 100-150 HP, 230 VAC.*
****This card replaces several versions of the card that were previously used. All new cards are furnished with substitution matrix and set-up instructions.*

SUBASSEMBLIES AND COMPONENTS

Subassembly/Component	Catalog Number
Control Power Transformer (230 VAC)	36B605538ARG01
Control Power Transformer (460 VAC)	36B605538ARG02
Fan (small, 4-1/2")	104X215CA003
Fan (large, 7")	104X215CA004
Blower Motor	104X215BA023
Blower Motor Fuses	104X109AD039
Filter Capacitor	104X122AA371
Fuses (Control Power)	
3 Amps (FU1,2)	104X109AD004
0.5 or	104X109AD029
1 Amps (FU3)	104X109AD034
3.2 Amps (FU4 & 5)	104X109AD038

13. SPARE AND RENEWAL PARTS

Renewal parts information for this equipment is included in the **CUSTOM INSTRUCTION BOOK** provided for the drive. The renewal parts list is generated specifically for the equipment furnished on each order and provides information on parts relative to the particular rating and options incorporated in the drive at the time of shipment. If the renewal parts information is missing, a replacement list may be obtained by contacting your local **GE Sales Office** or **GE DRIVE PRODUCTS OPERATION Authorized Distributor** with the drive **MODEL and REQUISITION numbers.** Parts lists may also be obtained direct from **GE DRIVE PRODUCTS OPERATION Technical Services, (814) 875-2075.**

It is recommended that the parts list be reviewed and a set of spare parts be maintained to minimize down time should repairs be required. Renewal parts may be ordered by contacting your local **GE Sales Office** or an **Authorized GE Parts Distributor in your area.**

PART NUMBER IDENTIFICATION

The parts listed are divided into four categories, each with its own part number structure:

Order-Specific Assemblies
Common Assemblies
Component Parts
Circuit Cards

Order-Specific Assemblies are generally major assemblies consisting of a group of common assemblies that make up the specific controller provided. Other items which are obtained specifically for the order may also be provided with a similar part number structure.

Example:

7VXYZ999CI01

7V	--	assembly provided by GE DRIVE PRODUCTS OPERATION
XYZ999	--	identification number (original order)
CI	--	assembly type (CI = Core Inverter)
01	--	unit identification number

Common Assemblies are subassemblies, consisting of a group of component parts, which are in common use in GE DRIVES products and are not specific to a particular order. The part number consists of the number "36" followed by an alpha character and may be up to 14 characters in total length.

Example:

36B605538ARG01 is the part number for a Control Power Transformer (230VAC).

Component Parts are the basic components used in the assemblies and generally represent the lowest discrete level. Component part numbers consist of a combination of alphanumeric characters, defining the class of component and the specific item.

Example:

104X109AD038

104X	--	component part
109	--	component type (fuse)
AD038	--	specific item (500 V, 3.2A FNQ)

Circuit Cards are identified by a catalog number giving the product class, card function and revision levels. It should be noted that some cards contain software which is designated by the suffix "Mn" where "n" is the software number. These cards are also available less software with suffix "Gn" designation. The factory may substitute later versions of cards based on availability and design enhancements.

Example:

531X102CCHCDM4

531X	--	card for GE DRIVE PROUCTS OPERATION microprocessor based controller
102CCH	--	Control card for AC-300 controller
CDM4	--	revision level, including software

As an aid in expediting card identification, a partial listing of the cards, functions, and associated root part numbers are shown in the following charts.

INTERCONNECTION CABLES

Connector	Catalog
1PL	36A358208AAG02
2PL	104X154CB111
3PL	36A358208AAG04
4PL-3	36A358208AAG07
5PL	36A358208AAG01
6PL	36A358208AAG08
7PL	36A358208AAG05
8PL-10	36A358208ABG01
9PL	36A358208CXG01
10PL-3	36A358208AAG09
11PL	36A358208AAG11
12PL	36A358208AAG10
13PL	36A358208AAG12
14PL	36A358208EYG01
16PL (6 foot)	36A358208AAG18
16PL (8 foot)	36A358208AAG19
16PL (10 foot)	36A358208ARG01
16PL (15 foot)	36A358208ARG02
18PL	36A358208ADG01

NOTE: *Connectors used may vary with options provided.*

AC-300 SPARE AND RENEWAL PARTS
(HP Related, 460 VAC)

Part Description	Cat. # 104X-	Horsepower 2-15	20-30	40-60	75-125
AC Line Fuse	109AG-	004	007	009	017
DC Link Fuse	109AD-	091	092	094	094
DB Fuse	109AD-	--	--	--	094
Power Transistor	170CB-	010	011	012	026
DB Transistor	170CB-	013	014	015	026
Base Drive Transistor	170CB-	--	--	--	---
Diode Module	125DC-	014	015	016	039
Charge Control Module	125DC-	017	018	019	040
Shunts	See seperate table, SHUNTS (460 VAC)				
AC Line Reactor	220FA-	001	002	003	004
Inverting Transistor Modules	170CB-	010	011	012	026
DB Transistor Modules	170CB-	013	014	015	026
Base Drive Amplifier Transistor	170CB	--	--	--	---
DB Resistor Kit[1] *	36C774360-	ARG01	AXG01	AYG01	AYG01[3]
DB Resistor Kit[2] *	36C774360-	BBG01	BAG01	--	---
DB Contactor*	36C774545-	AAG01	AAG01	AAG01	AAG02[4]

(1) Standard 5% duty cycle.
(2) Heavy duty 20% duty cycle.
(3) Quantity 2 required for std. duty; Quantity 1 required for light duty.
(4) AAG01 for light duty.
* *G01 - For separate mounting (enclosed).*
G11 - For panel mounting (open).
G21 - For case roof mounting.

AC-300 SPARE AND RENEWAL PARTS (HP Related, 460 VAC) (continued)

Part Description	Cat. # 104X-	Horsepower 150*	150**	200	250	300
AC Line Fuse	109AG-	019	019	021	025	027
DC Link Fuse	109AD-	110	107	107	107	107
DB Fuse	109AD-	094	094	094	094	094
Power Transistor	170CB-	027	027	027	027	027
DB Transistor	170CB-	027	027	027	027	027
Base Drive Transistor	170CB-	---	028	028	028	028
Diode Module	125DC-	039	051	051	051	051
Charge Control Module	125DC-	040	052	052	052	052
Shunts	See seperate table, SHUNTS (460 VAC)					
AC Line Reactor	220FA-	005	005	005	006	006
DB Resistor Kit***	**36C774360-**	AYG03	AYG03	AYG03	AYG03	AYG03

"Q" Frame Size
***"T" Frame Size*
****Quantity of two required; light duty cycle only.*

SHUNTS (460 VAC)

Cat. #	Horsepower 2	3	5	7.5	10	15	20	25	30	40
36B605625AA-	G10	--	G11	G01	G02	G03	G04	G05	G06	G07

Cat. #	Horsepower 50	60	75	100	125	150	200	250	300
36B605625AA-	G08	G09	--	--	--	--	--	--	--
104X149-	--	--	CA007	CC029	CA008	CA009	CA021	CA010	CA014

SHUNTS (230 VAC)

Cat. #	Horsepower 1	2	3	5	7.5	10	15	20	25	30
36B605625AA-	G10	G11	G01	G02	G03	G04	G06	G07	G08	G09

Cat. #	Horsepower 40	50	60	75	100	125	150
104X149-	CA007	CC029	CA008	CA009	CA021	CA010	CA014

AC-300 SPARE AND RENEWAL PARTS (HP Related, 230 VAC)

Part Description	Cat. # 104X-	Horsepower 1-7.5	10-15	20-30	40-60
AC Line Fuse	109AG-	004	007	009	017
DC Link Fuse	109AD-	091	092	093	094
DB Fuse	109AD-	---	---	---	094
Power Transistor	170CB-	010	011	012	026
DB Transistor	170CB-	013	014	015	026
Base Drive Transistor	170CB-	---	---	---	---
Diode Module	125DC-	014	015	016	039
Charge Control Module	125DC-	017	018	019	040
Shunts	See seperate table, SHUNTS (230 VAC)				
AC Line Reactor	220FA-	001	002	003	004
DB Resistor Kit[1]	36C774360-	ADG01	AEG01	AFG01	AYG01[3]
DB Resistor Kit[2]	36C774360-	AXG01	AYG01	---	---
DB Contactor	36C774545AA	G01	G01	---	---

(1) Standard 5% duty cycle.
(2) Heavy duty 20% duty cycle.
(3) Quantity of two required.

AC-300 SPARE AND RENEWAL PARTS (HP Related, 230 VAC) (continued)

Part Description	Cat. # 104X-	Horsepower 75*	75**	100	125	150
AC Line Fuse	109AG-	019	019	021	025	027
DC Link Fuse	109AD-	110	107	107	107	107
DB Fuse	109AD-	094	094	094	094	094
Power Transistor	170CB-	027	027	027	027	027
DB Transistor	170CB-	027	027	027	027	027
Base Drive Transistor	170CB-	---	028	028	028	028
Diode Module	125DC-	039	051	051	051	051
Charge Control Module	125DC-	040	052	052	052	052
Shunts	See seperate table, SHUNTS (230 VAC)					
AC Line Reactor	220FA-	005	005	005	006	006
DB Resistor Kit***		---	---	---	---	---
DB Contactor***		---	---	---	---	---

**"Q" Frame Size*
***"T" Frame Size*
****To be available at a future date; consult factory for availability/application.*

14. WARRANTY PARTS AND SERVICE

1. Consult the **GE DRIVE PRODUCTS OPERATION Terms and Conditions** for details of the product warranty, including warranty period and parts or service coverage.

2. Warranty replacement parts or service assistance may be obtained direct from the factory during regular working hours **(8 a.m. - 5 p.m.; M-F)** by calling:

GE DRIVE PRODUCTS OPERATION
PRODUCT SERVICE ENGINEERING
ERIE, PA.
(814) 875-3851

3. An **EMERGENCY HOT-LINE** has been established by **GE DRIVE PRODUCTS OPERATION** to provide a 24 hour telephone answering service to our customers.

GE DRIVE PRODUCTS OPERATION
EMERGENCY HOT-LINE
(814) 455-3219

For a **PROMPT RESPONSE THE NEXT WORKING DAY** the answering service operator will record the required information and your **NAME, COMPANY and PHONE NUMBER.** In cases of emergency, where assistance must be obtained immediately, the operator may be requested to contact a Customer Service Representative (for parts) or a Product Service Engineer (for technical assistance) at home.

4. So that we may serve you more expediently, please have the following information ready whenever contacting the factory:

DRIVE SERIAL NUMBER & MODEL NUMBER
PART NUMBER & DESCRIPTION

5. **GE DRIVE PRODUCTS OPERATION** may request that failed parts replaced under warranty be returned to the factory. A RETURN AUTHORIZATION tag or number is to be obtained and included with any parts being returned. When returning parts, pack them carefully to prevent damage in transit and ship to:

GENERAL ELECTRIC COMPANY
DRIVE PRODUCTS OPERATION
1100 LAWRENCE PARKWAY
ERIE, PA 16531
ATTENTION: PARTS REPAIR AND EXCHANGE

NOTES:

15. RECOMMENDED POWER STUD WIRING, TERMINALS, AND CRIMPING TOOLS

RECOMMENDED POWER STUD WIRING AND CRIMPING TOOLS

ITEM	AMP CRIMPING TOOL CAT. NO.	TYPE	FOR WIRE SIZE	TERMINAL TYPE
1	49592	Hand	16--14 AWG	Solistrand*
2	49935	Hand	12--10 AWG	Solistrand
3	69062	Hand-hydraulic with self- contained dies	8--2 AWG	Solistrand

	AMP ELECTRO-HYDRAULIC POWER UNIT 115VAC (CAT. NO.) Crimp Tool	Head	Die		
4	69120-1	69065	46322-2	1-1/0 AWG	Solistrand
5	69120-1	69065	46323-2	2/0 AWG	Solistrand
6	69120-1	69065	46324-2	3/0 AWG	Solistrand
7	69120-1	69065	46325-2	4/0 AWG	Solistrand
8	69120-1	69065	46326-2	250 MCM	AM Power
9	69120-1	69065	46327-2	400 MCM	AM Power

**Trademark of AMP Incorporated.*

RECOMMENDED POWER STUD WIRING AND TERMINALS

Drive HP	230 VAC Stud Dia. AC	Power Wire (a, b) Qty.	AWG	MCM	Terminal - AMP Solistrand GE Cat.# 104X161AA	AMP Cat.#	Hole Dia.	Crimping Tool - See tool list:
5	.25	1	12		009	33458	.25	Item 2
7.5	.25	1	10		009	33458	.25	Item 2
10	.25	1	8		012	33461	.25	Item 3
15	.25	1	8		012	33461	.25	Item 3
20	.25	1	6		015	33465	.25	Item 3
25	.25	1	4		059	31811	.25	Item 3
30	.25	1	2		057	320383	.25	Item 3
40	.25	1	1		074	321866	.25	Item 4
50	.25	1	2/0		075	36921	.25	Item 5
60	.25	1	3/0		025	36925	.38	Item 6
75	.38	1	4/0		068	321878	.38	Item 7
100	.38	2	2/0		023	36923	.38	Item 5
125	.38	2	3/0		025	36925	.38	Item 6
150	.38	3	4/0		068	321878	.38	Item 7

NOTES:

(a) *Wire size from NEC Table 310-17 and UL/CSA standards. Copper wire rated 75°C and 1.25 times rated current. These are minimum wire sizes. Consult the NEC Codes for other wire ratings and conform to all local ordinances.*

(b) *Quantity of wires and terminals in parallel per stud.*

RECOMMENDED POWER STUD WIRING AND TERMINALS

Drive HP	460 VAC Stud Dia. AC	Power Wire (a, b) Qty.	AWG	MCM	Terminal - AMP Solistrand GE Cat.# 104X161AA	AMP Cat.#	Hole Dia.	Crimping Tool - See tool list:
7.5	.25	1	14		006	34124	.25	Item 1
10	25	1	12		009	33458	.25	Item 2
15	.25	1	10		009	33458	.25	Item 2
20	.25	1	8		012	33461	.25	Item 3
25	.25	1	8		012	33461	.25	Item 3
30	.25	1	8		012	33461	.25	Item 3
40	.25	1	6		015	33465	.25	Item 3
50	.25	1	4		059	31811	.25	Item 3
60	.25	1	2		057	320383	.25	Item 3
75	.25	1	1		074	321866	.25	Item 4
100	.25	1	2/0		075	36921	.25	Item 5
125	.25	1	3/0		025	36925	.38	Item 6
150	.38	1	4/0		068	321878	.38	Item 7
200	.38	2	2/0		023	36923	.38	Item 5
250	.38	2	3/0		025	36925	.38	Item 6
300	.38	3	4/0		068	321878	.38	Item 7

NOTES:

(a) *Wire size from NEC Table 310-17 and UL/CSA standards. Copper wire rated 75°C and 1.25 times rated current. These are minimum wire sizes. Consult the NEC Codes for other wire ratings and conform to all local ordinances.*

(b) *Quantity of wires and terminals in parallel per stud.*

NOTES:

16. GLOSSARY OF TERMS

A/D	--	Analog to Digital Converter.
ACOM	--	Analog common.
CFU1,2,3	--	Control Fuse.
CPT	--	Control Power Transformer.
CPTFU(1, 2)	--	Control Power Transformer Fuse 1 or 2.
D/A	--	Digital to Analog Converter.
DB	--	Dynamic Brake.
DCOM	--	Digital Common.
DM1, 2, 3, 4, 5, 6	--	Diode Module in Rectifier Bridge.
EEROM	--	Electrically Erasable Read-Only Memory, also called EE or E^2 (E squared). Contains the controller Parameters.
EE Enable Device	--	Device used to "Enable" or "Protect" EEROM programming. A hardware jumper (JP13) located on the Control card.
EN	--	Enable Signal.
FU	--	Fuse.
"Hard" Reset	--	Hardware Reset, resets the drive cards. Used to clear some faults and allow certain parameter changes. Activated by depressing SW2 on the Control card or connecting 3TB69 to 3TB66.
IMET	--	Load Current Meter Output.
IOC	--	Instantaneous Overcurrent.
JOGR	--	Jog Reference.
JP	--	Jumper.
K1	--	Inverter start Relay located on Power Supply card.
K2	--	"RUN" (or "MAX") relay located on the Power Supply card.
K3	--	"FAULT" Relay located on the Power Supply card.
LED	--	Light Emitting Diode.
L(1, 2, 3)	--	Input 3 phase - Line 1, 2, or 3.
MOV	--	Metal Oxide Varistor.
MSR	--	Manual System Reference.
MUP1, 2, 3	--	Multiple Use Point on Control card.
N1	--	Negative Converter Output.
N2	--	Negative DC Link Output.
Parameters	--	Adjustable software settings used to program and tune the controller regulators.
PCN, P	--	Process Interface Input (- and +).
PCOMX	--	Power Supply Common Point.
PI	--	Process Interface card.
P1	--	Positive Converter Output.
PSC	--	Power Supply card..
PWM	--	Pulse Width Modulated.
RAM	--	Random Access Memory.

RS232C	--	Serial Link Communication Standard.
RS422	--	Serial Link Communication Standard.
R1	--	Capacitor Bank Charge Control Resistor.
R2 (N, P)	--	Capacitor Bank Discharge Resistor.
R(3)	--	DB Resistor.
SH 1, 2, 3	--	Current Feedback Shunt
SLD	--	Signal Level Detector, SLD0 thru SLD5, (RX1 or K1 thru RX5 or K5) located on optional Application card and SLD5 on optional 3TB.
"Soft" Reset	--	Software Reset initiated by pressing the RESET key on the optional Programmer.
SRS	--	System Reference test point on Control card.
THSW	--	Thermal Switch.
TM1, 2, 3, 4 , 5, 6	--	Inverter Transistor Module.
TM7	--	DB Transistor Module.
TP	--	Test Point.
T1, 2, 3	--	Output 3 Phase - Line 1, 2, or 3.
UV	--	Undervoltage.
VCO	--	Voltage Controlled Oscillator.
WMET	--	Frequency Meter Output.
(1, 2,....)PL	--	Ribbon Cable Harness.
1, 2C	--	DC Link Filter Capacitor.



GE Drive Systems

GEK-85781C (7/91)

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